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TECHNICAL SUPPLEMENT TO NEURAL BELLMAN OPERATORS

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Complete proofs of the retained own-future NBO construction, original-law qualification and information results are followed by proofs of the active-region Bellman compiler, centered tail-switch bound, completed-incumbent and recursive improvement, gain-budgeted stopping and risk-distorted composition. A separate Brownian Markov recursive-utility result proves its own well-posedness and residual certificate. The current addition proves the original Bellman and observation foundations, constrained gradients, stopped boundaries, tail-uniform information bounds, implementable original-law tail defects, bounded preliminary work, packed evaluation and atom-safe coupled comparisons. The present addition proves the original action comparison modulus, paired information bounds, constrained tail-defect transfer, safe actual-policy changes, independent all-action validation and simultaneous population transfer. Finite executable checks supplement these continuum arguments. Statements introduced in this supplement are numbered S.*; references to main-text statements retain their main-text numbers.

1. PROOFS OF THE INHERITED NBO RESULTS

These proofs retain the original statements and arguments. Their placement in the supplement changes the reading order, not their hypotheses.

1.1. *Proof of the result 1*

PROOF: Backward comparison and cash invariance give

$$V_t^* \geq f_t - \sum_{k=t}^{T-1} B_{t,k} u_k - B_{t,T} u_T, \quad J_t^\pi \leq f_t + \sum_{k=t}^{T-1} B_{t,k} v_k - B_{t,T} \ell_T,$$

where $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$. The terminal claim is the stipulated enclosure; applying the Bellman or fixed-policy operator proves each inductive step. Subtraction gives (3).

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A shift $f_t \mapsto f_t + c_t^0$ adds $c_t^0 - \beta_t c_{t+1}^0$ to u_t and subtracts it from v_t. The terminal width is unchanged. *Q.E.D.*

1.2. *Proof of the result 2*

PROOF: The selected node label lies between the node Bellman infimum minus e_t and that infimum plus $D_t k_t + e_t$. The Lipschitz property of the infimum implies that every cone lies above it minus e_t. At a nearest node its cone exceeds the infimum by at most $D_t k_t + e_t + 2L_t h_t$. At the attaining original index, the two feasible pairs and (4) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \le y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t \zeta_t.$$

This proves (6). The terminal residual belongs to $[-e_T, e_T + 2L_g h_T]$; Proposition 1 gives the loss bound. A finite minimum of equally Lipschitz cones is equally Lipschitz. The identities $|z| = \mathrm{ReLU}(z) + \mathrm{ReLU}(-z)$ and $\min(u, v) = u - \mathrm{ReLU}(u - v)$ give the exact network. Finite lexicographic comparison and Borel repair give measurability. Effective refinement of the finitely many positively weighted terms yields the termination assertion. *Q.E.D.*

1.3. *Proof of the result A.11*

PROOF: The triangle inequality bounds $f(x)$ above by each corner expression. Choose the lexicographically first attaining site $i_*(x)$. In every coordinate choose the cell endpoint between x_j and $x_{i_*,j}$. At this corner distances add exactly. An owner of strictly smaller transformed value would improve the objective at x; a tied smaller owner would contradict the original tie rule. Thus this corner has owner $i_*(x)$, proving both identities. Along a coordinate line, forward and backward relaxations respectively minimize over sites on either side, always comparing value–original-index pairs. Iterating over coordinates gives the separable distance transform and its stated work. Rational arithmetic and unchanged repair complete the claim. *Q.E.D.*

1.4. *Proof of the result A.72*

PROOF: The state cost has gradient infinity norm at most $8/d$, Hessian norm at most $22/d$, and diagonal second derivatives at most $5/d + 16/d^2$. For the terminal

cost the corresponding bounds are $10/d$, $26/d$, and $9/d + 16/d^2$. They follow by differentiating the quadratic terms and the continuously differentiable squared shortage. The bound on the cycle Laplacian is four. The shortage Hessian, where it exists, is $16\mathbf{1}\mathbf{1}^\top/d^2$; the same second-difference bound holds across its kink.

For feasible (x, a) and (y, b), put $w = x - y$. The difference between the transition at their convex combination and the corresponding combination of transitions is $\theta(1-\theta)w_j w_{j+1}/16$ in coordinate j. Convexity and the G_{t+1} Lipschitz bound therefore give a convexity defect of at most

$$\frac{\beta G_{t+1}}{16}\theta(1-\theta)\sum_j |w_j w_{j+1}| \le \frac{\beta G_{t+1}}{16}\theta(1-\theta)\|w\|_2^2.$$

The state cost supplies curvature $4/d$, while the two-control action cost has Hessian eigenvalues at least $7/4$. Thus the state curvature of Q_t^* is at least $4/d - \beta G_{t+1}/8$, which is at least $107/(128d)$ when $G_{t+1} \le 27/d$. Partial minimization over the affine capacity graph preserves the state convexity inequality. The minimizing action is unique, including at a binding capacity constraint.

To propagate Lipschitz continuity, write $a = c(x)u$, where u lies in the fixed unit simplex. Transfer the same u from x to y. Each action-cost partial derivative has absolute value at most $13/16$, and $\sum_j u_j \le 1$. The action-cost transfer contributes at most $13\|x - y\|_1/(128d)$. The transition Jacobian at a fixed action has column sums at most $11/16$; the capacity transfer adds at most $3/64$. Comparing a minimizing normalized action in both directions gives the stated recursion for G_t. Its right side maps $27/d$ below itself.

For a fixed u, the transition as a function of x has Jacobian norm at most $11/16 + 1/16 = 3/4$. Its second-order composition term is bounded by $G_{t+1}/8$. The action-cost Hessian has norm at most $21/4$, and the capacity gradient has squared norm $1/(64d)$. These facts give the recursion for M_t. The infimum over u preserves this common semiconcavity bound: after subtracting the same quadratic, each fixed-u function is concave, and an infimum of concave functions is concave. The right side maps $54/d$ below itself. Convexity together with semiconcavity yields the claimed differentiability and gradient regularity; this conclusion does not require a differentiable optimizing-action map.

Along one state coordinate, the fixed-u transition has zero second derivative. The squared norm of its first derivative is at most $85/256 + 23/(256d)$: the two original nonzero entries are bounded by $9/16$ and $1/8$, and the capacity-induced entries are bounded by $1/(16d)$. Combining this with the diagonal stage-cost bound gives D_t. Finally, positive interpolation along one coordinate has error between zero and $D_t h^2/8$. Applying the coordinate interpolations successively and adding the bounds proves (A.147). *Q.E.D.*

1.5. *Proof of the result A.73*

PROOF: Joint convexity, including the feasible graph, gives

$$Q_t^*(x, \bar{\pi}_t(x)) \le \sum_v w_v(x) Q_t^*(v, a_t(v)) \le I_n q_t^{\mathrm{sel},+}(x).$$

The lower interpolation bound gives $V_t^*(x) \ge I_n l_t(x) - \kappa_t$. Subtraction proves the uniform optimal-advantage bound e_t. The partial action derivatives of Q_t^* are bounded by Λ_t, so feasible implementation error adds at most $\Lambda_t b_t$. Adding the future policy loss and proceeding backwards proves the recursion. No convexity of $J^{\widehat{\pi}}$ is used. *Q.E.D.*

1.6. *Proof of the result A.62*

PROOF: Apply concavity to $f(a) - Ha^2/2$ and substitute the two lower endpoints. The displayed quadratic has derivative $D - K + 2Ks$. Minimizing it on the unit interval gives the three cases. Its coefficient of K is $-s(1-s)$, which proves the last assertion. *Q.E.D.*

1.7. *Proof of the result A.63*

PROOF: At $r = 1$, analytic endpoint enclosures and the minorant lemma give the assertion. Suppose it holds for $r - 1$. Each child interval encloses the true optimal continuation. If the numerical next state is an interval, expanding the queried center value by G_{r-1} times the maximum ℓ^1 displacement encloses all its possible values. Conditional-mean lower bounds and the semiconcavity remainder then give valid endpoint brackets for the true Q_r. The minorants cover the entire action interval.

A downward capacity endpoint can omit only an interval of length equal to the certified capacity-rounding difference; subtracting Λ_r times that difference accounts for its possible improvement. The minimum upper endpoint is the cost of a feasible first action followed by the optimal future. This proves (i) by induction. Retaining earlier valid lower bounds preserves every conclusion.

For termination, an interval of length h has minorant at least its smaller lower endpoint minus $H_r h^2/8$. Its smaller upper endpoint is an available global upper bound. With the endpoint-width guard, an interval of length at most $h_\delta = \sqrt{3\delta/H_r}$ cannot be responsible for a gap exceeding δ. Every refined parent is therefore longer than h_δ. Its children are longer than $h_\delta/5$. Apart from the at most two initial fragments made by a prediction, disjoint terminal fragments consequently have total number at most $5c(x)/h_\delta$, up to endpoint counting. Equation (A.129) is a conservative point budget. Bounded action-quantum rounding of a proposed quarter-to-three-quarter split preserves the one-fifth property whenever the quantum is smaller than $h_\delta/20$. Otherwise precision must be increased; a fixed-precision run is not allowed to ignore this requirement.

For (iii), let $\bar{x}$ be the acquired state. Its downward displacement is at most $d2^{-b}$ in ℓ^1, and $c(\bar{x}) \le c(x)$ preserves feasibility. At a fixed action the stage cost is $8/d$-Lipschitz and the transition has column sums at most $11/16$. Thus replacing $\bar{x}$ by x changes Q_r by at most $(8 + 11\beta d G_{r-1}/16)2^{-b}$. The corresponding value change is at most $dG_r 2^{-b}$. Add these two allowances to (i). Substituting the policy's own future yields the usual one-step loss recursion; backward induction gives (A.130). No policy-value oracle and no population claim about fitting are used. *Q.E.D.*

1.8. *Proof of the result A.53*

PROOF: A routed action is merely another feasible partition point. The endpoint and minorant induction in Theorem A.63 therefore applies without change. After each evaluation retain the smallest upper endpoint and the maximum of the old and newly valid global lower endpoints. Their difference cannot increase. The stopping test uses this difference, not a fitted value or a prediction error.

Put $h_\delta = \sqrt{3\delta / H_r}$. A partition interval of length at most h_δ has minimum minorant at least its smaller lower endpoint minus $3\delta/8$. The associated upper endpoint, with the stipulated width and capacity allowance, is within δ of that minorant. Such an interval cannot be responsible for an unresolved global gap. Every split parent therefore has length greater than h_δ. The one-fifth condition makes each of its children longer than $h_\delta/5$. Starting with one interval, disjoint terminal fragments consequently number at most $1 + 5c(x)/h_\delta$. Counting endpoints and allowing for rounding of this numerical bound gives the displayed conservative budget. Floating failure is handled by the separately charged exact-rational procedure, not by this real-arithmetic count.

The first assertion about predictor evaluation follows from the order of the stopping test and the routing rule. For the last assertion, use backward induction on the remaining horizon, with the common analytic terminal kernel as base. At each nonterminal query, declined predictions leave the same conventional split, lower bound, witness and tie rule. Induction over partition refinements gives the same next query and returned endpoints; the induction over dates then gives the same implemented policy. Model-evaluation work can still differ, so transcript identity is not a zero-cost fitting claim. *Q.E.D.*

1.9. *Proof of Proposition A.54*

PROOF: Every retained lower endpoint is at least its true endpoint value minus w, and hence at least $f(a^*) - w$. On an interval of length at most h, the parabolic minorant is therefore at least $f(a^*) - w - Hh^2/8$. Subtracting the omitted-capacity allowance gives $L \geq f(a^*) - w - Hh^2/8 - \Lambda\Delta$; retaining an older, larger valid lower bound only improves this inequality. The upper endpoint at $\widehat{a}$ is at most $f(\widehat{a}) + w$. Semiconcavity and the one-sided first-order condition give $f(\widehat{a}) - f(a^*) \leq \lambda e + He^2/2$. Subtract the lower bound to prove (A.115). At an interior optimum the derivative is zero. The example $f(a) = a$ on $[0, c]$ shows that the boundary linear term is needed. Finally, retaining bounds makes $U' \leq U$ and $L' \geq L$; expansion proves (A.116). *Q.E.D.*

1.10. *Proof of the result A.50*

PROOF: For each drawn bin sequence, conditional integration over its continuous realizations places the exact path expectation between the interval endpoints. Uniform mixing over the equal bins recovers the original continuous law. The lower endpoint expectation is thus no larger than D_i, and the upper endpoint expectation is no smaller. Apply the bounded empirical-Bernstein inequality of Maurer and Pontil (2009) to the endpoint samples and take the union bound, with outward arithmetic included in the mean, variance and radius. Conditional validity for every $\mathcal{H}$ implies unconditional validity by integration. On this one simultaneous event, all ensuing deterministic comparisons of the covered quantities hold together. *Q.E.D.*

1.11. *Proof of the result A.51*

PROOF: Subtract the gain intervals from k and compare the resulting interval endpoints with zero. Taking the image of $[l, u]$ under the absolute-value map gives (A.111). The assertions hold simultaneously for every nonnegative fee on the original coverage event. *Q.E.D.*

1.12. *Proof of the result A.52*

PROOF: Subtract the common J_{b_0}. The selected implementation has net difference at most $U(\lambda)$, and the best catalogue implementation has net difference at least $L(\lambda)$. This proves the regret bound pointwise on the one simultaneous event and hence for every λ. A minimum of finitely many affine functions changes its active affine expression only at an intersection. For $\lambda_2 > \lambda_1$, sum the two minimizing inequalities to obtain $(\lambda_2 - \lambda_1)(s_{\widehat{b}(\lambda_2)} - s_{\widehat{b}(\lambda_1)}) \leq 0$. Since s_b increases strictly with b, the precision comparative static follows. *Q.E.D.*

2. PROOFS FOR REFERENCE-CENTERED RISK QUALIFICATION

2.1. *Centering and the original Bellman optimum*

For any bounded z, substituting $f_{t+1} + z$ in (8) and subtracting the finite number $\rho(f_{t+1})$ proves (10). Each dual objective at f_{t+1} is at most its supremum, giving $k^f \geq 0$. Approximating that supremum gives $\inf_q k^f(q) = 0$ without an attainment

assumption. Since each q is a probability, adding a constant b to z adds b to $\mathcal{E}^f(z)$. If $z \le w$, every dual objective at z is no larger than its counterpart at w, proving monotonicity. The inequalities $w - \|z-w\|_\infty \le z \le w + \|z-w\|_\infty$ prove nonexpansiveness.

PROOF OF THEOREM 3: Let $d_t^\pi = V_t^\pi - f_t$ and $d_t^* = V_t^* - f_t$. The original recursions and (10) give the exact identities

$$d_t^\pi(x) = r_t(x, \pi_t(x)) + \beta_t \mathcal{E}^f_{t,x,\pi_t(x)}(d_{t+1}^\pi),$$
$$d_t^*(x) = \inf_{a \in A_t(x)} \{r_t(x,a) + \beta_t \mathcal{E}^f_{t,x,a}(d_{t+1}^*)\}. \tag{1}$$

Both have terminal value $g - f_T$. Suppose $d_{t+1}^\pi \le U_{t+1}$. Monotonicity, nonnegative β_t, the upper residual inequality and (11) imply $d_t^\pi \le U_t$. Similarly $L_{t+1} \le d_{t+1}^*$ implies $L_t \le d_t^*$. Backward induction establishes both statements. Feasibility of the supplied complete policy implies $V_t^* \le V_t^\pi$, giving (12). The argument takes place at each fixed parameter vector; applying it to every vector satisfying the premises proves the class statement without assembling incompatible continuations.

Exact equality recursions coincide with (1), proving the exactness assertion by induction. For the scalar comparison, set $a_t = \ell_t + b_t^-$ and $b_t = u_t + \gamma_t + b_t^{\pi,+}$ from the signed assumptions. Every original action has residual at least a_t, since its numerical action value is at least the numerical Bellman infimum. The implemented action has residual at most b_t. Define $\underline{h}_T = d^-$, $\overline{h}_T = d^+$, and

$$\underline{h}_t = a_t + \beta_t \underline{h}_{t+1}, \qquad \overline{h}_t = b_t + \beta_t \overline{h}_{t+1}.$$

Since $\mathcal{E}^f(b) = b$ for constants, these are feasible lower and upper certificates. Exact backward evaluation is order preserving, so $L_t \ge \underline{h}_t$ and $U_t \le \overline{h}_t$ for exact residuals and terminal discrepancies. Their date-zero difference is at most (A.35).

For two policy supersolutions, their pointwise minimum remains a supersolution: the operator at the minimum of their successor bounds is at most its value at either bound. The corresponding pointwise maximum of two optimum subsolutions remains a subsolution by the reverse argument. Finite combinations therefore

preserve an already valid scalar guarantee. No claim about improved bounds from an uncontrolled numerical optimizer is needed. *Q.E.D.*

2.2. *Risk occupations and certificate orientation*

PROOF OF PROPOSITION A.25: For fixed π, (1) is the Bellman equation for a finite-horizon maximizer selecting a risk index j after observing the current state. Its reward is $r_{ts} - \beta_t k_{tsj}$, transition vector is q_{tsj}, and terminal reward is $g - f_T$. Finite maxima have measurable maximizers. Backward induction establishes a Markov maximizing choice at each date and state.

Any nonanticipating choice rule, including a history-dependent one, induces occupation masses satisfying (A.50). Conversely, for a feasible flow, choose j at (t,s) with probabilities y_{tsj}/m_{ts} when $m_{ts} > 0$; at zero-mass states choose an arbitrary feasible index. Induction on t recovers the declared masses. The corresponding expected sum of rewards is exactly the objective in (A.51). Thus the flow maximum equals the centered fixed-policy value. This representation is under the adversarial risk kernels and must not be called the physical occupation law.

Multiply each dual inequality by $B_t y_{tsj}$ and sum. The flow equalities cancel intermediate potential terms, leaving the primal objective at most $\mu^\mathsf{T} U_0$. The argument allows a zero discount: it uses $B_{t+1} = B_t \beta_t$ but never divides by either factor. Feasible primal and dual witnesses with equal objectives certify the maximum. For an optimum subsolution, a separate feasible risk witness for every original action supplies a lower bound on that action's supremum; a minimization over a restricted sampled action set does not supply the required all-action inequality. *Q.E.D.*

2.3. *Kernel exclusion and its sharp boundary*

For any $z \in \mathcal{D}$, the difference between the affine pieces indexed by q and q_0 is

$$(q - q_0)^\mathsf{T} z - k^f(q) + k^f(q_0).$$

A strict negative upper bound excludes q uniformly. This proves (A.52). Maximizing coordinate by coordinate gives the box formula. For a span ball, the coefficients sum to zero; adding a constant to z changes nothing. Normalize $\min_i z_i = 0$ and

assign s to the positive coefficients and zero to the negative ones. The maximum is $s\sum_i (q_i - q_{0i})_+ = s\|q - q_0\|_1/2$.

As a sharp example, take two equally likely outcomes and average value at risk at level $1/2$. Then $\rho(v) = \max(v_1, v_2)$ and the dual vertices are e_1, e_2. At $f = (0,1)$ their centered penalties are $1, 0$. For $z = (2,0)$, the centered value is 1. Keeping only the maximizing kernel e_2 from the reference gives 0, which is invalid as an upper bound; dropping penalties gives 2, which is conservative but not exact. On the span ball of radius $1/2$, the first kernel is safely excluded. At radius 1, $z = (1,0)$ makes the two pieces equal, so the strict exclusion must retain the boundary. This example proves both the necessity of allowing the risk kernel to change and sharpness of the strict threshold convention.

2.4. *Entropic duality without changing the objective*

Let p be a probability measure, f, z bounded, and $\lambda > 0$. Boundedness makes $Z_f = \int e^{\lambda f} dp$ strictly positive and finite, and makes p_f equivalent to p on its support. For $q \ll p$ with finite relative entropy,

$$\lambda^{-1}\mathrm{KL}(q\|p_f) = \lambda^{-1}\mathrm{KL}(q\|p) - \int f\, dq + \lambda^{-1}\log Z_f.$$

If q places mass outside the support of p, its penalty is infinite. For any bounded v, define $p_v = e^{\lambda v} p / Z_v$. Nonnegativity of $\mathrm{KL}(q\|p_v)$ gives

$$\int v\, dq - \lambda^{-1}\mathrm{KL}(q\|p) \le \lambda^{-1}\log Z_v,$$

with equality at $q = p_v$. This proves the variational representation, its centered penalty, and (A.53). It also shows exactly why using the fixed tilt p_f as an expectation is not generally exact: the centered evaluation remains entropic under p_f, rather than linear. For $\lambda \downarrow 0$, bounded convergence gives the expectation limit. Exponential and logarithmic evaluations in a numerical certificate require outward arithmetic; the rational average-value-at-risk experiments are not evidence of exact floating-point entropic computation.

2.5. *Occupation-sensitive precision and a strict improvement*

For fixed penalties and transitions, write the flow objective as $A(y)$ and the additional nonnegative residual contribution as $E(y) = \sum_{t,s} B_t e_{ts} m_{ts}$. If $H = \max_y A(y)$ and $H_e = \max_y \{A(y) + E(y)\}$, then

$$0 \leq H_e - H \leq \max_y E(y).$$

The upper inequality follows from $A(y) \leq H$ for every feasible y; the lower follows from $E(y) \geq 0$. Since each occupation layer has total mass one, $\max_y E(y) \leq \sum_t B_t \max_s e_{ts}$. This establishes (A.54). The argument assumes the penalties are held fixed while residual uncertainty is enlarged. Altering the underlying stored continuation changes the penalties and requires requalification.

For a strict horizon example, let two states be absorbing under both actions, let action zero cost zero and action one cost one in both states, let $g = 0$, and let all discounts equal one. Every normalized monetary risk evaluation of these deterministic transitions agrees with evaluation at the absorbing state. Put $f_t(s) = (T - t)s$ for $s \in \{0, 1\}$ and choose action zero everywhere. Each date's implemented and minimum residuals are $(0, -1)$; the signed scalar account charges one at each date, yielding T. The exact state-dependent lower and upper recursions both equal $-(T - t)s$, giving zero policy loss at every state. There are two feasible actions and every action is checked. No state-dependent bias was cancelled without propagation: its entire state dependence was carried through the original transition at every date. This is consistent with, and sharpens rather than contradicts, the statewise-bias warning in Section AJ.

Finally, the finite backward recursion evaluates each stored transition support once for each retained risk piece at each date and state-action pair. Summing these support sizes yields the stated arithmetic count. A direct descendant recursion repeatedly evaluates the same subproblem; removing that repetition does not establish a polynomial bound in the number of continuous states, risk pieces, acquisition cells or rational bit length. The comparison is a representation-conditioned resource statement, and the full construction and support-qualification work remains separately charged.

3. CONTINUOUS INVESTMENT: PRIMITIVE PROOFS

3.1. *Curvature, the action constraint, and state moduli*

PROOF OF PROPOSITION 4: For $x \in X$, $0 \le a \le 1/4$ and $|z| \le 1/32$, the transition remains in X. Indeed, the state part is increasing in both of its arguments, its maximum is $1/16 + 1/2 + 1/8$, and $b_j a \le 1/8$. Its minimum after the shock is at least $1/32$, and its maximum is at most $27/32$. Thus every derivative bound below is used on its stated domain.

Both S and g are convex and continuously differentiable; their squared hinges have positive semidefinite second derivative on each smooth region. Direct differentiation gives

$$|\partial_j g| \le \frac{10}{d}, \qquad |\partial_j S| \le \frac{15}{2d}, \qquad \nabla^2 S \succeq \frac{4}{d} I.$$

The first bound adds $5/d$ from the target term, $1/d$ from the cyclic difference term, and $4/d$ from the hinge. Only $-x_j x_{j+1}/16$ in F_j has nonzero second derivative. For a joint direction (v, w), the possibly negative part of the second derivative of $g(F)$ is

$$-\frac{1}{8} \sum_j \partial_j g(F) v_j v_{j+1} \;\ge\; -\frac{5}{4d} \|v\|_2^2,$$

since $\sum_j |v_j v_{j+1}| \le \|v\|_2^2$. The remaining term is the quadratic form of the positive semidefinite Hessian of g applied to the full state-action Jacobian. In particular, it includes the mixed state-action terms and is not discarded selectively. Since $h''(a) = 2 + 48a^2 \ge 2$, every fixed-shock objective satisfies (15). Integrating against a fixed feasible risk density preserves this bound. Taking the supremum over those densities preserves the same strong convexity. The bound extends across the hinge by continuity of the first derivative, or by integrating the almost-everywhere second-derivative inequality along a segment.

The graph $\{(x, a) : x \in X, 0 \le a \le A(x)\}$ is convex because A is affine. Partial minimization of the jointly convex objective over this graph proves convexity of V_1. Compactness and continuity supply an attained minimum. The positive capacity permits constraint qualification in the action variable.

For a fixed feasible density, the supporting strong-convexity inequality gives, for every (y, u),

$$Q_1(y,u) \geq W + p_x^\top (y - x_0) + p_a(u - a_0) + (u - a_0)^2.$$

For feasible u, adding $\lambda(u - A(y)) - \nu u \leq 0$ to the right side preserves a lower bound. Minimize this augmented quadratic over all real u, an enlargement of the feasible interval. Completing the square gives (17). This proof uses neither equality in the risk supremum nor exact stationarity. Interval implementations bound the complete expression outwards, including its residual square and its slope.

For completeness we bound true value subgradients, rather than the potentially large slopes of arbitrary poor lower witnesses. A transition Jacobian column has nonnegative entries summing to at most $11/16$. Therefore any state subgradient of a risk-maximizing one-date objective has coordinate magnitude at most

$$\frac{1}{d}\left(\frac{15}{2} + \beta\frac{55}{8}\right).$$

At an upper-bound action, the multiplier can be chosen with $0 \leq \lambda \leq \beta(10/d)\sum_j b_j = 15\beta/4$: the nonnegative derivative of h cannot increase the negative derivative which this multiplier offsets. At a lower-bound action take $\lambda = 0$. At an interior minimizer both action-bound multipliers vanish. Envelope subgradients thus have coordinates bounded by

$$\frac{1}{d}\left\{\frac{15}{2} + \beta\left(\frac{55}{8} + \frac{15}{32}\right)\right\} < \frac{15}{d}.$$

Where the risk function is nonsmooth, the same bound holds for convex combinations of active risk gradients and valid KKT multipliers. Integrating bounded subgradients along a segment, with the one-sided limit at the boundary of X, proves (16). *Q.E.D.*

3.2. *Exact one-date risk and outward construction*

Put $m = F(x,a,0)$, $R = 1/32$ and $D = 10d^{-1}\sum_j m_j s_j$. Since $\sum_j s_j = 0$, the hinge in $g(m+sz)$ is constant in z. The target and cyclic terms give exactly

$$g(m+sz) = g(m) + Dz + 5z^2. \tag{2}$$

Let $L = 2R\alpha$ and $h = \max\{-R, \min\{R-L, -D/10 - L/2\}\}$. The interval $[h, h+L]$ is a lowest-value interval of length L for the convex quadratic. The maximizing AVaR density is $(1-\alpha)^{-1}$ on its complement and zero on that interval. The two tail moments are

$$M_1 = -\frac{L(2h+L)}{2(2R-L)}, \qquad M_2 = \frac{2R^3 - L(3h^2+3hL+L^2)}{3(2R-L)}.$$

Hence the risk equals $g(m) + DM_1 + 5M_2$. This also follows from the threshold representation of AVaR, with the threshold chosen at the boundary of the retained lower interval. Endpoint minima and equal boundary values are included by the clipping rule.

For a lower witness, any dyadic $\widehat{h} \in [-R, R-L]$ defines an exactly feasible excluded interval and a feasible tail density. Differentiating its objective holds $\widehat{h}$ fixed. This distinction is essential: a derivative of a changing approximate tail need not be the derivative of a supporting objective. Upper values use an outward enclosure of the maximizing h, while lower witnesses use the separately retained feasible $\widehat{h}$. The lower witness therefore remains valid when the two differ.

At a fixed initial state, $F(x,u,z)$ is affine in u and z. A convex continuation has a global affine minorant at every successor anchor. For a density constant on a bin, integration of this affine minorant equals its value at the midpoint. Its action dependence is affine. Adding the strong-convexity minorant of h gives (18). A fitted continuation is used only where its nonnegative convex readout makes this argument valid; no analogous assertion is made for an unrestricted nonconvex fit.

For a value upper bound, convexity places a fitted continuation below the larger endpoint value on a bin. For V_1, the midpoint upper value plus $15R/n$ bounds the whole bin by (16). If $l_i \le v(z) \le u_i$ on each bin, monotonicity of the original risk

gives

$$\rho_\alpha\left(\sum_i l_i 1_{B_i}\right) \le \rho_\alpha(v) \le \rho_\alpha\left(\sum_i u_i 1_{B_i}\right).$$

The step-function risks are finite upper-tail means because the bin probabilities are exactly equal and $n(1-\alpha)$ is an integer. Their computation does not assume that v is constant on the bins.

3.3. *Complete policy and centered comparison*

PROOF OF THEOREM 5: Every fixed density is feasible for the original risk supremum. Every continuation support is below V_1 at every successor. Accordingly its all-action quadratic minimum is below V_0, irrespective of which feasible proposal supplied the anchor. The binwise upper calculation dominates the value of the entire terminal controller; monotonicity and cash invariance charge at most $\beta\varepsilon_1$ for its terminal optimality allowance. It follows that $\underline{V}_0 \le V_0 \le J_0^{\widehat{\pi}} \le \overline{J}_0^{\widehat{\pi}}$. Subtracting proves (19).

For the centered statement take any lower and upper continuation functions ℓ, u furnished by the preceding supports and bounds. Their differences from f give admissible error functions. The exact centered identity cancels the common $\rho_\alpha(f)$ before any numerical enclosure. The preceding lower witness is a minorant of the resulting all-action expression; the preceding policy upper envelope is a majorant of the resulting policy expression. They are sufficient subsolution and supersolution inequalities in the general theorem. Computing them jointly is an implementation of that comparison, not a new objective. Loose global residual constants remain a valid alternative, but need not replace these sharper local witnesses. *Q.E.D.*

Whole-cell residual interpolation. On a state cell with vertices v_i, choose its multilinear convex weights $\omega_i(y)$. If f and V_1 are convex, their vertex upper enclosures interpolate to upper bounds. A fixed affine support L_V of V_1 satisfies

$$V_1(y) - f(y) \ge L_V(y) - \sum_i \omega_i(y)\overline{f}(v_i) = \sum_i \omega_i(y)\{L_V(v_i) - \overline{f}(v_i)\}.$$

Its right side is at least the smallest vertex difference. Maximize that lower bound over the stored supports. Reversing V_1 and f, using upper values of V_1 and lower supports of f, gives the upper residual bound. The finite maximum or minimum over anchors and cells is taken after these whole-cell inequalities. No interpolation equality for the unknown function is asserted.

Strict tail exclusion. Suppose $u_i < \tau$ and the union H of $k = n(1-\alpha)$ other bins has $v(z) \geq \tau$ throughout. It has probability at least $1-\alpha$. If a feasible maximizing density assigned positive mass to B_i, some capacity $q < (1-\alpha)^{-1}$ would remain on H. Moving equal mass from B_i to H strictly increases its risk objective, a contradiction. Equality yields only weak improvement and cannot support exclusion. Bin bounds used to construct H must come from the unpruned calculation. The argument covers a continuum of densities without enumerating all of them.

3.4. *Statistical unit and numerical allowance*

For two complete policies that share the terminal controller and differ only in their first action, the stage-action difference is at most $5/64$. Since $\sum_j b_j = 3d/8$ and the largest feasible action difference is $1/4$, (16) gives a future-value difference of at most $45/32$. Terminal-controller errors take values in $[0, \varepsilon_1]$, so their difference is at most ε_1, not twice that allowance. Monotonicity and cash invariance give the deterministic paired support

$$|J^A(x) - J^B(x)| \leq B := 5/64 + \beta(45/32 + \varepsilon_1).$$

This support is established before observing a workload. It is not replaced by an observed sample range.

For n independently sampled initial states and paired cost intervals $[D_i^-, D_i^+]$, intersect each interval with $[-B, B]$. For any fixed center c computed from these intervals,

$$\frac{1}{n-1}\sum_i (D_i - \overline{D})^2 \leq \frac{1}{n-1}\sum_i \max\{|D_i^- - c|, |D_i^+ - c|\}^2 =: \bar{s}^2.$$

Applying the two-sided empirical Bernstein inequality (Maurer and Pontil, 2009) to each of twelve fixed contrasts, followed by a union bound, is conservative with

$\log(960) < 7$. A simultaneous sampling radius is

$$r_n = \sqrt{14\bar{s}^2/n} + \frac{98B}{3(n-1)}.$$

The resulting interval is $[n^{-1}\sum_i D_i^- - r_n, n^{-1}\sum_i D_i^+ + r_n]$, evaluated outwards. The calibration selector is fixed independently of these rows, and all pairwise contrasts are retained. Innovation bins, fitting labels, repeated sorting trials and repeated executions of an unchanged stream are not additional independent observations.

3.5. *Proof of the horizon-uniform construction*

PROOF OF PROPOSITION A.26: The primitive state cost has ℓ^1 Lipschitz modulus at most $8/d$. The transition's state Jacobian has nonnegative entries with every column sum at most $11/16$. Each control exposure column has ℓ^1 norm $3d/8$. On the outer action rectangle $[0,1/4]^m$, each component of the action-cost gradient is at most $13/16$. The scalar and vector transitions remain in the cube on that rectangle: for the vector case their range is contained in $[1/64,59/64]^d$. These bounds therefore remain applicable during the state and action comparisons even when an intermediate action is not feasible at an intermediate capacity.

Capacity intervals and capacity simplexes have Hausdorff modulus $1/(8d)$ in the action ℓ^1 norm. In the vector case, when capacity decreases, scale an action radially to the smaller capacity; the required total action displacement is no larger than the capacity change. Monotonicity and cash invariance make AVaR nonexpansive in the supremum norm. It follows that any continuation with state modulus L gives a Bellman value modulus at most

$$\frac{8}{d} + \beta\frac{11}{16}L + \frac{1}{8d}\left(\frac{13}{16} + \beta\frac{3d}{8}L\right) = \frac{1037}{128d} + \beta\frac{47}{64}L.$$

Substitution of $L = 27/d$ gives $\{1037/128 + 705 \cdot 27/1024\}/d < 27/d$. The terminal cost has modulus $10/d$. This proves the uniform value modulus by backward induction. The same calculation applies to an arbitrary Lipschitz fitted continuation; convexity of that fit is not required. Consequently every own-future cone may be constructed with the common primitive modulus $27/d$.

For convexity of the original values, assume inductively that V_{t+1} is convex and its coordinate subgradients have absolute value at most $27/d$. For a smooth such function, the negative contribution from the bilinear state transition is at worst $27\|v\|_2^2/(8d)$; the Jacobian–Hessian–Jacobian term is positive semidefinite in the full state-action direction. Adding stage curvature yields

$$\frac{1}{d}\left(4-\beta\frac{27}{8}\right)\|v\|_2^2+\frac{7}{4}\|w\|_2^2=\frac{107}{128d}\|v\|_2^2+\frac{7}{4}\|w\|_2^2.$$

The action bound includes the vector cost interaction, whose off-diagonal Hessian entries equal $1/4$. For a nonsmooth convex continuation, mollify on a neighborhood of the transition image, which is a fixed positive distance from the cube boundary. Mollification preserves convexity and coordinate Lipschitz bounds there. The preceding uniform curvature inequality passes to the limit. Integration under any fixed feasible innovation density and the supremum over those densities preserve it. Partial minimization over the convex capacity graph proves convexity of V_t. This establishes the induction without assuming differentiability of the resulting risk-averse values.

The tensor state cover has ℓ^1 covering radius at most $d/(2n_x)$. Hence the two-sided cone term is at most $27/n_x$, including the terminal cone formed from exact terminal labels. The action objective modulus is bounded by

$$D_a\le\frac{13}{16}+\beta\frac{3d}{8}\frac{27}{d}=\frac{1319}{128}.$$

The scalar feasible net has radius no larger than $A(x)/(2m)$. The vector simplex lattice has ℓ^1 radius at most $A(x)/m\le 1/(4m)$. To verify the latter, scale by $m/A(x)$, take the coordinate floors, and, when they do not exhaust the integer capacity, compare the two points obtained by rounding one coordinate up. One of their ℓ^1 errors is at most one. If the floors exhaust capacity, the point is already a lattice point. Thus the common action-net contribution is at most $1319/(512m)$.

On one innovation rectangle, the maximal ℓ^1 displacement from its midpoint under the vector transition is $d(1/(32n_z)+1/(64n_z))$. A $27/d$-Lipschitz continuation therefore varies by at most $81/(64n_z)$. Applying the original risk's nonexpansiveness and then discounting bounds the objective integration error by

$e \le 1215/(1024 n_z)$. The scalar case is covered by the same bound. Exact AVaR of the midpoint step function is computed by sorting the n_z^s equally weighted values; $4 \mid n_z$ ensures integral tail counts for both risk levels. The innovation law remains continuous: the bound on e is the explicit difference between its risk and the step-function risk.

The cone proof also applies to the present risk recursion. Indeed, for any bounded u, v and scalar c, monotonicity and cash invariance give $\|\rho(u) - \rho(v)\|_\infty \le \|u - v\|_\infty$ and $\rho(u + c) = \rho(u) + c$. Feasible-action minima preserve order and this bound. Thus label, action-net and continuation-cover errors propagate by the same discounted subsolution and supersolution induction; no linear expectation or frozen maximizing density is used in that induction. The graph and action moduli established above verify its primitive hypotheses for every own-future continuation. Theorem 2, with exact feasible repair and no additional label-arithmetic error, now bounds each nonterminal residual sum by

$$b := 27/n_x + 1319/(512m) + 1215/(512 n_z).$$

The terminal cone contributes at most $27/n_x$. Thus the finite-horizon bound is $\sum_{t<T} \beta^t b + \beta^T 27/n_x \le b/(1-\beta)$, which is (A.55). The displayed sufficient resolutions divide the approximation allowance into three equal discounted shares. With the same uniform moduli, the sensor, repair and action-rounding coefficients in (A.25) are bounded independently of the horizon; their discounted sum is at most sixteen times the one-date allowance. Finite sensor and action precision can therefore fill a separately reserved positive share of the target. *Q.E.D.*

Work, storage, and exact-number growth. Every date uses S targets, at most A_m feasible actions per target and n_z^s midpoint queries per action. A dense cone query scans S labels in $O(dS)$ rational operations, giving the stated dense count. The original-owner transform costs $O(dS)$ per date; cell location and the exact off-grid owner query give the compiled count. Sorting costs $O(TSA_m n_z^s \log(n_z^s))$ comparisons and is charged in addition, not absorbed into an unmeasured primitive oracle. Stored labels and witnesses occupy $O(TSd)$ rational words, with their bit lengths charged separately.

All fixed meshes, costs, transition coefficients, risk weights and cone moduli in this construction are rational. Their finitely many primitive denominators divide a fixed integer D whose logarithm is $O(\log(dn_x mn_z))$. At a new date, a stored label is combined with fixed-denominator primitive values, cone distances, tail averages and the rational discount. A common-denominator implementation therefore increases the denominator exponent by only a fixed amount per date. Maxima, minima, comparisons and exact feasible repairs do not increase it exponentially. Bounded discounted stage costs and the fixed cone modulus also bound label magnitudes uniformly in the horizon. The resulting numerator and denominator bit lengths are $O((T+1)\log D)$. Arithmetic on these integers is not assigned constant bit cost. The statement is a fixed-accuracy constructive horizon bound with an explicit state and innovation curse, not a measured arithmetic crossover or a full implementation benchmark.

4. DECISION-SCALE QUALIFICATION: PROOFS

PROOF OF PROPOSITION A.17: For every real threshold η, the function $\phi_\eta = (v-\eta)_+$ is convex and K-Lipschitz. Conditional Jensen and the chord inequality give $M_n\phi_\eta \le P\phi_\eta \le E_n\phi_\eta$. Apply the AVaR threshold formula $\inf_\eta\{\eta + (1-\alpha)^{-1}\mu\phi_\eta\}$ for each of the three probability measures. Taking infima preserves the two orders and proves the enclosure. In particular, no interchange of an action infimum and a risk supremum occurs.

Here is a uniform bound which remains valid at kinks. Let $h = 2R/n$. Choose one-sided slopes of ϕ_η at each bin's outer endpoints. The slope increases across each bin; its total increase over the whole interval is at most $2K$. The difference between the endpoint average and midpoint value on bin i is no greater than $h/4$ times the slope increase across that bin. This follows by integrating the left and right slopes over the two half-bins. Choose the left endpoint right derivative and right endpoint left derivative; jumps at shared endpoints may be omitted, which only decreases the sum of within-bin slope increases. Therefore

$$0 \le E_n\phi_\eta - M_n\phi_\eta \le \frac{h}{4n}(2K) = \frac{KR}{n^2}.$$

For completeness, the half-bin expression equals one half the integral of the difference between the right and left slopes at equal distances from the midpoint, so its upper bound also follows directly without assuming a classical second derivative. The preceding uniform bound passes through the two infima and gives (A.39). Monotonicity and cash invariance add $e_- + e_+$ for endpoint error. They also charge the actual terminal controller's one-sided error only on the upper side.

For a product law, apply the one-coordinate midpoint–endpoint comparison successively, holding the other coordinates fixed. Products of positive averaging operators preserve inequalities and telescope. Each partially averaged positive-part function remains convex and has the same Lipschitz modulus in each unintegrated coordinate. Summing the one-coordinate error bounds and then taking the threshold infimum proves the product statement. This argument uses neither a discrete economic transition assumption nor differentiability of v. *Q.E.D.*

PROOF OF THEOREM A.19: Every service's regret belongs to $[0, \tau]$ at every possible initial state. The difference of two complete-policy values therefore belongs to $[-\tau, \tau]$, because the common original optimal value cancels. Each interval intersection retains the true paired difference. A difference of two intervals of width at most w has width at most $2w$. Intersecting with a valid support cannot increase that width.

Hoeffding's inequality for a variable of range length 2τ gives

$$\Pr\{|\overline{D} - \mathbb{E}D| > r\} \leq 2\exp\{-Nr^2/(2\tau^2)\}.$$

A union bound across the H fixed contrasts gives the stated coverage. Pairwise contrasts need not be mutually independent; only the initial states within each contrast must be independent, conditional on the frozen preparation. Replacing the unobserved sample mean by its numerical lower and upper bounds yields (A.40). Their difference is at most $2(r + w)$. This is an application of the established concentration inequality (Hoeffding, 1963), not a claim to a new concentration principle.

For the chosen rational parameters, $Nr^2/(2\tau^2) = 8$. The inequality $24e^{-8} < 1/20$ can be checked entirely with rational arithmetic: the partial sum $\sum_{j=0}^{11} 8^j/j!$ is greater

than 480. Equation (A.41) is an exact rational comparison. In numerical classification the margin is rounded inward and the reported confidence endpoints outward. No observed range or variance is used to choose the radius. *Q.E.D.*

Totalization of the admitted service. The first-date objective with the original V_1 has action Lipschitz constant less than 7: the primitive action derivative is at most $3/4$, and the discounted continuation contribution is at most $(15/16)(15/d)(3d/8)$. The feasible interval has length at most $1/4$. A uniform net with $m = 1024$ intervals therefore gives an all-action lower value by subtracting $7/(8m)$ from the smallest grid cost lower bound. The grid point with the smallest upper value supplies a feasible policy upper value.

The exact fallback uses $n = 256$ innovation bins and the preceding convex-order rule. Exact one-date rational supports have gaps at most 2^{-30}; the terminal controller contributes 2^{-20}. The resulting quadrature width, including these allowances, is below $1/8192$ for both risk levels. Adding $7/(8m)$ leaves the original-optimum bound below $1/256$. Downward rounding of the first action to 40 bits costs at most 72^{-40} on each side and fits strictly inside both remaining allowances. All values, risk weights, feasible actions and endpoint computations in this final construction are rational, and the terminal strong-convexity bisection reaches its stated gap in finitely many steps. Thus the fallback defines a finite valid action at every dyadic initial observation in the target distribution. For arbitrary real initial states, the separate acquisition transfer remains required; the confirmation theorem here concerns the declared dyadic initial law.

The floating implementation may reach the two targets much sooner. That does not permit removing the exact fallback from the policy definition, treating rejected proposals as missing observations, or asserting that the fallback was exercised when the actual counter is zero. Hardware or resource failure aborts the experiment and is recorded as a failure, not a change in its conditional sampling law.

PROOF OF PROPOSITION A.20: Subtract the two complete acquisition-and-deployment accounts. Using the upper policy, acquisition and fee bounds and the lower computational saving bounds above (A.42) by $m(d^+ - \lambda s^-) + \lambda k^+ + F^+$. Nonpositivity gives (A.43). Solving this scalar inequality proves the integer thresh-

old and its sign qualifications. The separate quality inequality depends on policy cost rather than price and cannot be deduced by increasing λ. A common audit cost cancels only when it is actually common and purchased in both alternatives; otherwise its incremental charge remains in k^+ or F^+. *Q.E.D.*

PROOF OF COROLLARY A.18: Write $h = 2R/n$ and $k = n(1-\alpha)$. The two roots of $v_n - \eta_n$, where $\eta_n = K\alpha R$, are $c_n - \alpha R$ and $c_n + \alpha R$. They lie strictly inside the domain and at two distinct bin midpoints, since αn is an even integer. Under M_n the mass strictly above η_n is $(k-1)/n$ and the mass at η_n is $2/n$. Under E_n there is no mass at η_n, and the mass above it is exactly k/n. Hence η_n minimizes the AVaR threshold objective under both measures, by its left and right slope conditions.

The hinge $\phi = (v_n - \eta_n)_+$ is affine on every bin except the two whose midpoints are its roots. On either exceptional bin, the endpoint average minus the midpoint value is $Kh/4$. Each bin has weight $1/n$. The two threshold minima therefore differ by

$$(1-\alpha)^{-1}(E_n\phi - M_n\phi) = \frac{1}{1-\alpha}\frac{2}{n}\frac{Kh}{4} = \frac{KR}{(1-\alpha)n^2}.$$

All equalities use the endpoint mixture's half weights at the outer boundaries. This explicit sequence establishes sharpness of the stated enclosure on its function class; it does not assert that the investment value itself equals v_n. *Q.E.D.*

5. ACTION-ALIGNED CERTIFICATES AND SELECTION

5.1. *Proof of Theorem 6*

Taking the infimum of the second inequality in (20) and subtracting it from the first gives the gap bound. Substitution of $b = a$ in that inequality shows $L \le Q(a) \le U$, hence $w \ge 0$. The minimum in (21) is at most zero because a is feasible, so $\Gamma \ge 0$.

For $z = b - a$ and the supplied $n \in N_A(a)$,

$$s^\mathsf{T} z = (s - \widehat{s} + \widehat{s} + n)^\mathsf{T} z - n^\mathsf{T} z \ge -(d + r)\|z\|_2.$$

The inequality uses $n^{\mathsf{T}} z \le 0$ and the triangle and Cauchy–Schwarz inequalities. Completing the square yields

$$-(d+r)\|z\|_2 + \frac{\mu}{2}\|z\|_2^2 \ge -\frac{(d+r)^2}{2\mu}.$$

Taking the minimum over the original feasible set proves (22). In one dimension, completion of the square centers the minimization at $a - s/\mu$, whose projection onto the interval gives the stated minimizer. This argument includes $a = 0$ and $a = A_0$ and does not differentiate a discontinuous actor. A lower numerical bound on the minimum only enlarges Γ, preserving validity. *Q.E.D.*

5.2. *Proof of the risk-witness lift*

Strong convexity of the stage cost gives

$$c(b) \ge c(a) + \nabla c(a)^{\mathsf{T}}(b-a) + \frac{\mu}{2}\|b-a\|_2^2.$$

For the common feasible risk witness, the dual representation and the continuation support imply

$$\rho(V \circ F(b)) \ge \int V(F(b,\omega))\, q(d\omega) - \psi(q) \ge \int \ell\, dq - \psi(q) + \left(\int G^{\mathsf{T}} v\, dq\right)^{\mathsf{T}} (b-a).$$

Adding the inequalities gives (23). The same q must be feasible for every b in the comparison. If an upper penalty endpoint replaces $\psi(q)$, the intercept decreases and remains valid. Errors in a vector integral must be retained as a set of possible slopes or transferred uniformly to the intercept using the bounded action displacement. Choosing a favorable slope from an interval without this transfer is invalid. The complete policy upper U must separately dominate the full risk envelope, not just the selected lower witness. *Q.E.D.*

5.3. *Dated action losses and own-future continuations*

Suppose first that a feasible complete policy has, at every date and state,

$$Q_t(V^*_{t+1}; x, \pi_t(x)) - V^*_t(x) \le \gamma_t$$

with nonnegative constants γ_t. If $V^{\pi}_{t+1} \le V^{*}_{t+1} + e_{t+1}$, monotonicity and cash invariance give

$$V^{\pi}_{t} \le Q_t(V^{*}_{t+1}; x, \pi_t(x)) + \beta_t e_{t+1} \le V^{*}_{t} + \gamma_t + \beta_t e_{t+1}.$$

Backward induction with $e_T = 0$ proves $V^{\pi}_0 - V^{*}_0 \le \sum_{t<T} B_t \gamma_t$. A certified terminal-controller allowance is inserted at its actual date. No product of tail multipliers is used to propagate a constant.

This statement does not assume that an NBO receives V^{*}_{t+1} as a free training target. A support built only for its own attained f_{t+1} bounds $Q_t(f_{t+1})$, not $Q_t(V^{*}_{t+1})$. It must therefore enter the one-sided or centered residual sandwich with a certified continuation discrepancy. The investment construction instead supplies genuine original-V_1 supports and a separately bounded terminal controller. These are two legitimate implementations of the same original-optimum obligation; a small own-future greedy gap alone is not that obligation.

5.4. *Proof of Theorem A.12*

Conditional on acquisition, the menu and all its functions are fixed. For any one h and any loss $Z_h \in [0,1]$, Hoeffding's inequality gives

$$\Pr\left(\left| n^{-1} \sum_i Z_h(X_i) - \mathbb{E} Z_h(X) \right| > e \mid \mathcal{A} \right) \le 2e^{-2ne^2}.$$

A union bound over the M candidates, at $e = e_n$, gives a common event of conditional probability at least $1 - \delta$. On that event, for every h,

$$\mathbb{E} Z_{\widehat{h}} \le \widehat{Z}_{\widehat{h}} + e_n \le \widehat{Z}_h + e_n \le \mathbb{E} Z_h + 2e_n.$$

Use $Z_h = R_h$ to obtain (A.28). Use $Z_h = \ell_h$ and the pointwise inequality $R_h \le \ell_h$ to obtain the surrogate assertion. These are separate choices of loss; claiming both events simultaneously requires a corresponding union allocation. The work implication follows by expectation of its verified pointwise cap. For direct work minimization use $Z_h = W_h / W_{\max}$ and multiply the bound by $W_{\max}$. Bounds exceeding the trivial support limit may be intersected with it. Validation and acquisition costs do

not disappear from total workload cost because they are absent from the per-context random variable. *Q.E.D.*

A minimal counterexample explains why a global residual objective cannot replace this loss. Let the action set be $[-1,1]$, the true objective $Q(a)=a^2$, and consider two continuation perturbations added to this objective. The first is $e_0(a)=a^2$, with oscillation one and unchanged minimizer zero. The second is $e_1(a)=\eta a$, with oscillation $2\eta<1$ for $0<\eta<1/2$, but its minimizer is $-\eta/2$ and its true regret is $\eta^2/4>0$. Thus even strict oscillation reduction can worsen the delivered action. This example is an explanatory construction, not a diagnosis inferred for each retained investment cell.

6. NONUNIFORM AND RANDOMIZED AVAR INFORMATION

6.1. *Proof of the nonuniform lower bound*

Let q, L, t be as in Theorem A.8 and answer every query with

$$f_0(z)=\frac{K}{2L}(z-q)_+^2.$$

The function is convex, nondecreasing and K-Lipschitz. Collect the queried points strictly inside (q,R), insert the two endpoints, and write the resulting nonempty interval lengths as $h_1,\dots,h_s$, with $s\le N+1$. Let $f_1=f_0$ to the left of q and equal its chord on each of those intervals. Successive chord slopes are nondecreasing and belong to $[0,K]$, including the junction at q. Therefore $f_1\in\mathcal{C}_{K,R}$ and is nondecreasing. The functions agree at every query, and induction on the adaptive transcript proves that a deterministic procedure has the same output on both.

Because the functions are continuous and nondecreasing and P has positive density, their upper-tail risks are $t^{-1}\int_q^R f(z)p(z)\,dz$. Flat portions cause no difficulty: the quantile formula integrates the appropriate fraction of their mass. On an interval of length h, the unweighted integral of the chord error is $Kh^3/(12L)$. Since $p\ge m$,

$$\mathrm{AVaR}_\alpha^P(f_1)-\mathrm{AVaR}_\alpha^P(f_0)\ge\frac{mK}{12Lt}\sum_{i=1}^{s}h_i^3\ge\frac{mKL^2}{12t(N+1)^2}.$$

The last inequality follows from $\sum_i h_i=L$ and convexity of the cube. A sound common interval contains both values. Finally $t=\int_q^R p\le M_pL$, so $L\ge t/M_p$, giving

the second lower bound. The worst-case error of a point-valued deterministic procedure is at least half the first width. *Q.E.D.*

6.2. *Proof of the nonuniform upper bound*

Write the uniform cell length as $h = 2R/n$. For a cell $[a, b]$, let its mass be p_i and its conditional mean be μ_i. For any convex function g,

$$p_i g(\mu_i) \le \int_a^b g(z)p(z)\,dz \le p_i \left\{ \frac{b-\mu_i}{h} g(a) + \frac{\mu_i - a}{h} g(b) \right\}.$$

For $g_\eta = (f-\eta)_+$, these are valid for every threshold η; g_η is convex and K-Lipschitz. If $a < \mu_i < b$, the difference of the right and left expressions is

$$p_i \frac{(\mu_i - a)(b-\mu_i)}{h} \left\{ \frac{g_\eta(b) - g_\eta(\mu_i)}{b-\mu_i} - \frac{g_\eta(\mu_i) - g_\eta(a)}{\mu_i - a} \right\}.$$

The factor before the braces is at most $p_i h/4 \le M_p h^2/4$. The slope difference is bounded by the variation of the one-sided derivative within the cell. Summing these nonnegative variations over the disjoint cell interiors is at most the total derivative variation, which is at most $2K$. Boundary means follow by continuity (and do not occur with strictly positive density). Hence the summed upper-minus-lower expectation is at most $M_p K h^2/2 = 2M_p K R^2/n^2$, uniformly in η.

Add η, divide each expectation by t, and minimize over η in the AVaR variational representation. The numerical lower objective lies below the original one and the numerical upper objective above it. Their difference is uniformly bounded by (A.15); taking infima preserves both enclosure and this width bound. The numerical rules require the n mean values and the $n+1$ endpoint values of f, with repetitions shared. Known moments specify their nonnegative weights, summing to one. An approximate moment rule requires a separately proved perturbation allowance and is not covered by an uncharged exact-law oracle. *Q.E.D.*

6.3. *Proof for pathwise-sound random intervals*

Use the same fixed f_0 as above, before drawing a random tape. Outside the single null set in the soundness definition, fixing the tape gives a deterministic adaptive

procedure that is sound for all inputs. The preceding chord construction therefore forces the stated width on its transcript from f_0. The chord competitor may vary with the tape; f_0 does not. Taking expectation gives the same lower bound on expected width. A procedure whose bad-tape set depends on the input need not be sound for that tape's chord competitor, which is why the argument proves no such assertion for ordinary $1-\delta$ coverage. *Q.E.D.*

6.4. *Proof of the randomized point-estimation lower bound*

For the uniform law set $q = -R + 2R\alpha$, $L = 2R(1-\alpha)$ and $s = z - q$. Partition the tail into $2N$ equal intervals of length $h = L/(2N)$. Let $\psi(u) = u^2(1-u)^2$ on $[0,1]$, and extend it by zero outside this interval. For independent signs $\theta_j \in \{-1,1\}$, define on the tail

$$f_\theta(q+s) = \frac{Ks^2}{4L} + \frac{Kh^2}{8L}\sum_{j=1}^{2N}\theta_j\psi\left(\frac{s-(j-1)h}{h}\right),$$

and put $f_\theta = 0$ to the left of q. We verify that this entire finite family belongs to the announced class. Both ψ and ψ' vanish at zero and one, so the functions and first derivatives match at every cell boundary and at q. On each cell, $\psi''(u) = 2 - 12u + 12u^2 \in [-1,2]$. Consequently

$$f_\theta'' \geq \frac{K}{2L} - \frac{K}{8L}2 = \frac{K}{4L} > 0.$$

The derivative starts at zero and ends at $K/2$, since bump derivatives vanish at both tail endpoints. It is therefore everywhere in $[0, K/2]$. The full function is convex, nondecreasing and K-Lipschitz; second derivatives at junctions are not needed.

Since $\int_0^1 \psi(u)\,du = 1/30$, the original AVaR of this monotone function equals a known baseline plus

$$b_N\sum_{j=1}^{2N}\theta_j, \qquad b_N = \frac{Kh^3}{240L^2} = \frac{KL}{1920N^3}.$$

A point query inside one open cell depends only on that cell's sign; a boundary query or query outside the tail reveals none. Conditional on the algorithm's random tape and its entire adaptive transcript, the signs in unqueried cell interiors remain

independent and equally likely. This follows inductively: the next location is measurable with respect to already revealed information and the tape, not unqueried signs. At most N of the $2N$ signs have been queried. The conditional variance of the target risk is therefore at least Nb_N^2.

For any estimator measurable with respect to the transcript and independent algorithmic randomization, conditional mean-square error is no smaller than that conditional variance. Averaging over signs and randomization gives Bayes risk at least Nb_N^2. The largest risk over this fixed finite family is at least its average; hence the worst-case root mean-square error is at least

$$\sqrt{N}b_N = \frac{KL}{1920N^{5/2}} = \frac{KR(1-\alpha)}{960N^{5/2}}.$$

This argument uses a fixed distribution over inputs rather than a tape-dependent adversarial input, and therefore applies to ordinary randomized point estimation. It supplies a lower bound only; an optimal upper algorithm for the whole convex class is not asserted. *Q.E.D.*

6.5. *Proof of the dated allocation identity*

For positive a_j, c_j, put $H = \sum_j a_j^{1/3} c_j^{2/3}$. Hölder's inequality gives

$$H = \sum_j (a_j/n_j^2)^{1/3} (c_j n_j)^{2/3} \le \left(\sum_j a_j/n_j^2 \right)^{1/3} \left(\sum_j c_j n_j \right)^{2/3}.$$

Thus feasibility implies $\sum_j c_j n_j \ge H^{3/2}/\sqrt{\varepsilon}$. Equality is attained by $n_j = (a_j/c_j)^{1/3}\sqrt{H/\varepsilon}$, for which the tolerance constraint binds. Upward integer rounding decreases each allowance and adds less than c_j per date. This argument assumes the stated linear per-resolution work law on a fixed graph; it does not substitute for pricing descendants or multidimensional tensor queries. *Q.E.D.*

7.1. *Proof of Theorem A.10*

At fixed x, only prices between $h(x)$ and $v(x)$ induce a different action from the original optimum. At such a price, the regret is $|p - v(x)|$. Both thresholds lie in $[0,3]$, so integrating the triangle under the uniform price density gives

$$\frac{1}{3}\int_{\min(h,v)}^{\max(h,v)} |p - v|\, dp = \frac{(h-v)^2}{6}.$$

Integration over x proves the unconditional identity, with no approximation or simulated policy observation.

Write $f = v - 1 = (x_1 + x_2 - 1)_+$. Under independent uniform coordinates,

$$\mathbb{E}f = \frac{1}{6}, \quad \mathbb{E}[f \mid x_1] = \frac{x_1^2}{2}, \quad \mathbb{E}[f \mid x_2] = \frac{x_2^2}{2}, \quad \mathbb{E}f^2 = \frac{1}{12}.$$

The additive projection of f is $g = (x_1^2 + x_2^2)/2 - 1/6$. Its residual has conditional mean zero given either coordinate, hence is orthogonal to every square-integrable additive function. This proves optimality over the whole additive class, not merely over additive quadratics. The constant and centered coordinate terms are mutually orthogonal, so

$$\|g\|_2^2 = \frac{1}{36} + 2\left(\frac{1}{20} - \frac{1}{36}\right) = \frac{13}{180}, \qquad \|f - g\|_2^2 = \frac{1}{12} - \frac{13}{180} = \frac{1}{90}.$$

The shifted optimum $1 + g$ has range $[5/6, 11/6]$, satisfying the price-support restriction. Equation (A.19) gives $1/540$.

For the quadratic projection use the orthogonal shifted Legendre polynomials $P_0 = 1$, $P_1(u) = 2u - 1$, $P_2(u) = 6u^2 - 6u + 1$, with squared norms $1/(2j+1)$. Direct integration on the upper triangle gives

$$\Pi_2 v = \frac{7}{6} + \frac{1}{4}(P_1(x_1) + P_1(x_2)) + \frac{1}{12}(P_2(x_1) + P_2(x_2)) + \frac{3}{10}P_1(x_1)P_1(x_2).$$

To verify these coefficients without assuming the answer, the monomial normal equations use basis $(1, x_1, x_2, x_1^2, x_1x_2, x_2^2)$, Gram entries $G_{(i,j),(k,l)} = 1/\{(i + k +$

$1)(j+l+1)\}$, and right side

$$b_{ij} = \frac{1}{(i+1)(j+1)} + \int_0^1 \int_{1-x_1}^1 x_1^i x_2^j (x_1 + x_2 - 1)\, dx_2\, dx_1.$$

The Gram matrix is positive definite because a nonzero polynomial cannot vanish almost everywhere on the square. Substituting the coefficients in (A.22) gives $G\theta = b$ exactly; the complete rational matrix, vector and zero residual are supplied in the executable evidence. Orthogonality alternatively gives

$$\|\Pi_2 f\|_2^2 = \frac{1}{36} + 2\frac{(1/4)^2}{3} + 2\frac{(1/12)^2}{5} + \frac{(3/10)^2}{9} = \frac{37}{450}, \quad \|f - \Pi_2 f\|_2^2 = \frac{1}{900}.$$

The Hessian of (A.22) has eigenvalues $11/5$ and $-1/5$. Its interior critical point $(3/11, 3/11)$ is a saddle. Checking each edge gives the minimum $143/150$ at $(3/5, 0)$ or $(0, 3/5)$ and the maximum $32/15$ at $(1,1)$. Thus it also lies in $[0,3]$, and the constrained admissibility requirement does not change the projection optimum. Dividing by six gives $1/5400$.

The original v is the displayed affine constant plus a single ReLU, and has zero error. Its native hinge, two-triangle piecewise-affine encoding, and dictionary containing that hinge are exactly the same function, so they induce the same policy at every state and price. No representational renaming changes either conclusion. *Q.E.D.*

For the worst-state conditional criterion, (A.19) gives $\sup_x \mathbb{E}_p[J^{b_h} - J^*] = \|h - v\|_\infty^2/6$. The sharp attained approximation errors in Proposition A.22 therefore yield $1/96$ and $1/1536$. The explicit attaining additive and quadratic functions shifted by one stay inside $[0,3]$. The population and worst-state optimization problems are distinct, so their minimizers need not coincide. The exact purchase equation follows by subtracting the respective policy and complete computation costs and solving the strict linear inequality. *Q.E.D.*

8. ACTIVE-REGION BELLMAN COMPILER: COMPLETE PROOF

PROOF OF THEOREM A.1: Let $v_j = r_j + b_j a$. Since d is even, $\sum_j s_j = 0$, $s_{j+1} = -s_j$, $\sum_j b_j = 3d/8$, and $\sum_j s_j b_j = d/8$. Expansion of the original terminal cost gives

$$g(v + sz) = D_x(a) + 2(H_x - 3a/4)_+^2 + k(a)z + 5z^2,$$

where

$$D_x(a) = \frac{4}{d}\sum_j (v_j - 5/8)^2 + \frac{1}{4d}\sum_j (v_j - v_{j+1})^2, \qquad k(a) = k_x + 5a/4.$$

The first sum contributes $4z^2$ and the cyclic-difference sum contributes z^2. Their linear coefficients are respectively $8d^{-1}\sum_j s_j v_j$ and $2d^{-1}\sum_j s_j v_j$. The positive-part term has no innovation argument because the innovation has zero coordinate mean. Also

$$D'_x(a) = C_x + \left\{\frac{8}{d}\sum_j b_j^2 + \frac{1}{2d}\sum_j (b_j - b_{j+1})^2\right\} a = C_x + \frac{41}{32}a,$$

since the two sums in braces equal $5/4$ and $1/32$. These identities derive the coefficients from the economic primitives rather than fitting them to data.

For the tail integral complete the square: $5z^2 + kz = 5(z + k/10)^2 - k^2/20$. When $|k| \le 10R(1-\alpha)$, the lower-cost interval of probability α is $[-k/10 - R\alpha, -k/10 + R\alpha]$, contained in the innovation support. Its conditional mean cost is $-k^2/20 + 5R^2\alpha^2/3$. The unconditional mean is $5R^2/3$. Subtracting the lower interval's contribution and dividing by $t = 1 - \alpha$ gives the first line of (A.1). When $k \ge 10Rt$, the upper-cost set of probability t is $[R(2\alpha - 1), R]$. Indeed, the lower interval touches the left boundary at the transition, and the convex quadratic thereafter ranks this right interval above its complement. Its first and second moments are αR and $R^2(4\alpha^2 - 2\alpha + 1)/3$. This yields the second expression. Reflection $z \mapsto -z$ treats negative k. Endpoint equality affects a set of probability zero. Substitution at $k = \pm 10Rt$ verifies agreement of values and derivatives, and therefore

$$\mathcal{R}'_\alpha(k) = \mathrm{clip}\{\alpha k/(10t), -\alpha R, \alpha R\}.$$

Cash invariance permits the deterministic $D_x(a)$ and hinge terms to be taken outside risk. Differentiating the resulting $S(x) + h(a) + \beta$ times terminal risk yields (A.2). In a region with hinge indicator I_h and central-tail indicator I_c, its linear coefficient is

$$p = 2 + \beta\left\{\frac{41}{32} + \frac{9}{4}I_h + \frac{25}{16}\frac{\alpha}{10t}I_c\right\} \ge 2.$$

The three listed breakpoints exhaust changes in those indicators. Sorting them with the two feasible endpoints, clipping to the original interval and discarding duplicates leaves at most four nonempty regions. The derivative is continuous, and on each open region its slope is $48a^2 + p \geq 2$. Integrating this piecewise inequality shows strict monotonicity on the entire interval. The intermediate value theorem and the constrained endpoint conditions prove existence and uniqueness, including a zero derivative exactly at a breakpoint or boundary.

For the interior root, write $a = -2u \sinh y$ with $u^2 = p/48$. The identity $\sinh(3y) = 3\sinh y + 4\sinh^3 y$ gives $16a^3 + pa = -32u^3 \sinh(3y)$. Setting $y = \operatorname{arsinh}(q/(32u^3))/3$ proves the formula. A scan of a fixed number of regions follows $O(d)$ coefficient sums. Transcendental accuracy and finite-number arithmetic remain separately charged, as claimed. *Q.E.D.*

VALUE GRADIENTS, BOUNDARIES AND IMPLEMENTATION AUTHORITY: The expanded Q_1 is continuously differentiable in (x, a); its positive-part square and the tail function are C^1 at their junctions. Set $a = A(x)u$ with $u \in [0, 1]$. The transformed objective has a unique minimizing u because $A(x) > 0$ and the action objective is strictly convex. Continuity of this minimizer follows directly by taking convergent subsequences in the compact interval and passing the minimizing inequality to the limit; uniqueness makes every subsequential limit the same. For a state increment h, compare values using the old optimizer for an upper bound and the new optimizer for a lower bound. Differentiability of the objective and continuity of the optimizer squeeze the value increment to $[\nabla_x Q_1 + u\partial_a Q_1 \nabla A]^\top h + o(\|h\|)$. At an interior optimum $\partial_a Q_1 = 0$; at zero $u = 0$; and at capacity $u = 1$ with $\partial_a Q_1 \leq 0$. This proves (A.3), its continuity and its one-sided boundary version. Inside a smooth region with strict interiority, differentiating the zero action derivative and using $\partial_{aa} Q_1 \geq 2$ proves the implicit actor derivative. The strict endpoint regions have the stated actor formulas. At junctions the value argument still holds, but no actor derivative is required.

The implementation does not claim exact evaluation of this symbolic root. It rounds a floating proposal to the inherited action quantum, verifies feasibility and evaluates the same original feasible risk witness as in Proposition 4. Its lower bound

pays the full multiplier residual and interval uncertainty, and its upper bound encloses the original objective. If its gap fails the inherited threshold, the old certified oracle is called and charged. Each branch therefore returns an original-value bracket and a valid global state support. The context manager changes bindings for optimal-continuation calls only and restores them on exceptions. It does not replace the actual terminal controller or its error allowance in complete-policy evaluation. Thus all original-policy enclosures remain those of Theorem 5; an unverified cubic evaluation never enters them as an exact value. This proves certificate invariance independently of any claim about the speed or bitwise identity of the two optimal-continuation oracles. *Q.E.D.*

9. TAIL-SWITCH ACCURACY AND ITS ECONOMIC MEANING

PROOF OF THEOREM A.2: We first justify the derivatives rather than differentiating a sorting permutation. For an outcome Y_a, write $\mathrm{AVaR}_\alpha(Y_a) = \min_\eta\{\eta + t^{-1}\mathbb{E}(Y_a - \eta)_+\}$. The unique quantile minimizes this expression. At the anchor, absence of mass at that quantile and dominated action derivatives give the action derivative of the minimized expression, with the quantile held fixed, as $t^{-1}\mathbb{E}[\mathbf{1}_{\{Y_a > \eta_Y\}}\nabla_a Y_a]$. To verify the envelope step, local L^1 continuity bounds the relevant minimizing quantiles in a compact interval. Any convergent subsequence of minimizers has to minimize the limiting continuous objective; uniqueness forces convergence to the anchor quantile. Comparing the two minimized objectives at each other's minimizers then gives the derivative formula by dominated convergence. Equivalently the unique tail density gives the same supporting derivative. The domination and uniqueness hypotheses in the statement are used at precisely these steps.

Let $\widetilde{H} = H_a + c$ and denote its quantile by $\widetilde{\eta}$. The almost-sure inequalities $\widetilde{H} - \epsilon \le Z_a \le \widetilde{H} + \epsilon$ imply $|\eta - \widetilde{\eta}| \le \epsilon$ by the monotonicity and translation rule of the generalized quantile. Put $E = \{Z_a > \eta\}$ and $F = \{\widetilde{H} > \widetilde{\eta}\}$. Both have probability t. If $Z_a > \eta + 2\epsilon$, then $\widetilde{H} > \eta + \epsilon \ge \widetilde{\eta}$, so the point lies in both upper tails. If $Z_a < \eta - 2\epsilon$, then $\widetilde{H} < \eta - \epsilon \le \widetilde{\eta}$, so it lies in neither. Hence $E \triangle F \subseteq B_\epsilon$ up to null sets. Since $P(E) = P(F) = t$, its mass is at most $2\min(t, 1-t)$ as well as $P(B_\epsilon)$.

Using the derivative formula and adding and subtracting the surrogate gradient on E gives

$$t(\nabla\rho(Z_a) - \nabla\rho(H_a)) = \mathbb{E}[\mathbf{1}_E(\nabla Z_a - \nabla H_a)] + \mathbb{E}[(\mathbf{1}_E - \mathbf{1}_F)(\nabla H_a - v_0)].$$

The subtraction of v_0 is legitimate because $P(E) - P(F) = 0$. The first term has norm at most $t\delta$; the second is at most $G_\epsilon m_\epsilon$. This proves (A.4). The two margin consequences follow by substituting $P(B_\epsilon) \le C(2\epsilon)^\kappa$ and, in the stronger case, $G_\epsilon \le 2L\epsilon$. If the band has zero probability, the second expectation vanishes regardless of its representative values.

The common stage cost cancels in the original versus surrogate action-gradient difference, leaving a bound $D = \beta(\delta + G_\epsilon m_\epsilon / t)$. For every feasible b, strong convexity gives

$$Q(b) \ge Q(a) + \nabla Q(a)^\mathsf{T}(b - a) + \frac{\mu}{2}\|b - a\|^2.$$

By the normal-cone convention, $n^\mathsf{T}(b - a) \le 0$. Thus the right side is at least $Q(a) - (r_a + D)\|b - a\| + \mu\|b - a\|^2/2$. Completing the square and minimizing over the nonnegative distance proves (A.5), including boundary actions. *Q.E.D.*

EXACT COUNTEREXAMPLE: For z uniform on $[-1, 1]$, its upper half-tail mean is $1/2$ and its lower half-tail mean is $-1/2$. Consequently $\mathrm{AVaR}_{1/2}(kz) = |k|/2$. On the stated action interval the true objective is $a^2 + (a+1)/2$ and the surrogate is $a^2 + (1-a)/2$. Each is strictly convex, with derivatives $2a + 1/2$ and $2a - 1/2$, respectively. Their feasible minimizers are $-1/4$ and $1/4$. Completing the first square gives a loss $(1/4 + 1/4)^2 = 1/4$. Direct integration of the two affine tails gives the same values. Nevertheless $\partial_a Z_a = \partial_a H_a = z$. The different maximizing densities, not a pointwise derivative error, produce the wrong decision. *Q.E.D.*

10. COMPLETE-INCUMBENT AND RECURSIVE IMPROVEMENT

PROOF OF PROPOSITION A.4: The interval premises imply $J - J' \ge l - u' > 0$ at every accepted gate. If a trial is refused, the complete incumbent action and future policy are retained exactly. Induction over the finite trace gives weak decrease from the completed starting policy, with a strict decrease whenever a gate is accepted.

The added qualification predicate holds for the starting policy and for each accepted replacement, so induction also preserves both original targets. This reasoning uses actual policy costs, not their regret upper bounds. It supplies no comparison to an unexecuted baseline or to a controller with a different future policy. *Q.E.D.*

PROOF OF THEOREM A.5: For $\|v - w\|_\infty \le h$, order preservation and cash invariance imply $\rho(w) - h \le \rho(v) \le \rho(w) + h$. It follows that $\|T_\pi v - T_\pi w\|_\infty \le \beta\|v - w\|_\infty$; taking infima over the same actions proves the identical bound for T. The space of bounded measurable functions with the supremum norm is complete, and the assumed preservation properties let Banach's fixed-point argument apply. Iteration is Cauchy by a geometric bound on successive differences, its uniform limit is a fixed point by continuity, and contraction proves uniqueness. The fixed points are the declared recursive values: finite-horizon iterates with any bounded terminal function converge to them, with terminal differences vanishing as β^n. Each fixed policy dominates the Bellman fixed point by order comparison. Conversely a measurable η-greedy selector for V^* has $J^\pi \le V^* + \eta/(1-\beta)$ by contraction and comparison. Letting η decrease to zero identifies V^* as the original infimum value. The same finite-horizon recursion treats adapted policies under the stipulated Markov conditional criterion.

The inequality $T_{\pi'} J^\pi \le J^\pi$ and monotonicity show that $T_{\pi'}^n J^\pi$ decreases to $J^{\pi'}$. In particular $D = J^\pi - J^{\pi'} \ge 0$. For any two continuations v, w at the same $(x, \pi'(x))$, the common dual set and penalty give

$$\inf_q \int (v - w)\, dq \le \rho(v) - \rho(w) \le \sup_q \int (v - w)\, dq.$$

For example, every affine dual value at v is at most its value at w plus the rightmost supremum, and taking the supremum proves the upper inequality; exchange v, w for the lower one. No optimizing density needs to be attained, and no $\inf_a \sup_q$ order is exchanged. Subtracting the two policy equations now gives

$$d + \beta P^- D \le D \le e + \beta P^+ D.$$

The maps $P^\pm$ are monotone, preserve constants, and are nonexpansive; hence the two gain maps are contractions. Iterating each comparison at D and passing to its

fixed point proves $D^- \le D \le D^+$. Iterating the lower gain map from zero proves $D^- \ge 0$. These are robust discounted gain accounts; identifying them with physical occupation measures would discard the original risk envelope.

For the successive policies, safe improvement gives $J^{\pi_{k+1}} \le J^{\pi_k}$ and, more precisely, $J^{\pi_{k+1}} \le T_{\pi_{k+1}} J^{\pi_k}$. The additional greedy inequality and contraction of T imply

$$0 \le J^{\pi_{k+1}} - V^* \le TJ^{\pi_k} - TV^* + \varepsilon_k \le \beta \|J^{\pi_k} - V^*\|_\infty + \varepsilon_k.$$

Unrolling yields $\beta^k \|J^{\pi_0} - V^*\| + \sum_{j<k} \beta^{k-1-j} \varepsilon_j$. If $\varepsilon_j \to 0$, split this sum into a finite prefix, which vanishes geometrically, and a tail bounded by $\sup_{j \ge m} \varepsilon_j / (1-\beta)$. Let m then increase to infinity. Finally $\|J^\pi - V^*\| \le \|J^\pi - TJ^\pi\| + \beta \|J^\pi - V^*\|$ proves the stopping statement. All these premises are uniform; checking a finite catalogue does not supply them. *Q.E.D.*

PAID INTERVAL RESOLUTION AND THE STOPPING DOMINANCE RULE: Containment and the width bounds give $l_n \ge J - A/n^2$ and $u'_n \le J' + A'/n^2$, proving (A.9). For a positive gap let $s = \sqrt{(A + A')/\Delta}$. The first dyadic resolution strictly above s is at most $2s$ if $s \ge 1$, and is one if $s < 1$. The sum of all dyadic resolutions through n is $2n - 1$. The stated work bound follows. When intervals overlap, choose any common point as both true costs; this is an admissible pair under the interval information alone, with no strict improvement. Thus sound strict separation cannot be asserted for every consistent pair. A zero gap remains undecidable for a strict gate, even with a valid non-strict policy comparison.

For the stopping rule, every feasible replacement has cost at least $V^*(x) \ge b$, while the incumbent costs at most u. Hence the largest possible cost saving is at most $u - b$. Subtracting an unavoidable expense of at least that amount leaves nonpositive net gain. This argument is pointwise in the declared economic units. It does not price unmeasured information, calibrate a workload, or determine the optimum of a different exploration problem. *Q.E.D.*

11. REMAINING-GAIN STOPPING AND RECURSIVE COMPOSITION

PROOF OF THEOREM A.6: Admissibility of ν and the original-optimum lower bound imply $J^\nu \ge V^* \ge b$. Since $J \le u$, we have $J - J^\nu \le u - b \le e$. The positive part

preserves this inequality because $e \geq 0$. The predicate $e \leq \theta$ proves (A.10). Stopping does not change the incumbent action, future policy, observation rule or certificate, and therefore does not change either original qualification property. The argument does not assume that the competitor uses the same future controller; that stronger scope is supplied by $b \leq V^*$.

At a stopped state apply (A.10) with the feasible policy returned by the full procedure as ν. The shortened procedure retains the stopped incumbent, so its loss relative to that comparator is at most the recorded stop ceiling. No identity of the later full trace is needed. At an un-stopped state the assumed trace agreement gives equal returned actions and the same future policy, hence zero contrast. This proves (A.11). Multiplication by nonnegative catalogue weights and summation proves the finite mean statement. When two returned actions and future policies are identical, they define the same original cost, so zero contrast is exact even when the numerical enclosures themselves have nonzero width.

To see the certificate-interface sharpness, take a one-step action problem with $A = [0,1]$ and $Q(a) = b + (u-b)a^2$, where $u > b$. Let the incumbent be $a = 1$ and provide any valid interval $[l,u]$ containing $Q(1) = u$. Its original optimum is exactly b at zero and its available improvement is exactly $u - b$. The objective is strongly convex with modulus $2(u-b) > 0$. Consequently no rule using only this interval and lower-bound information can uniformly promise a smaller available improvement over this problem class. When $u = b$, nonnegativity makes the zero bound exact. A supplied additional curvature modulus, geometry restriction, observation or formula may exclude this construction or sharpen the bound; no claim is made against that richer information interface. *Q.E.D.*

PROOF OF PROPOSITION A.7: Monotonicity and the probability-law dual representation imply that $P^+_{\pi_s}$ is monotone, preserves constants, and is nonexpansive in the supremum norm. The map $v \mapsto e + \beta P^+_{\pi_s} v$ is therefore a β-contraction on the complete space of bounded measurable functions. The stated preservation assumption allows iteration in that space. The geometric bound on successive differences proves a unique bounded fixed point E. Starting the iteration from zero and using $e \geq 0$ shows $E \geq 0$.

Write $D = J^{\pi_s} - J^{\pi_f}$. Subtracting the original policy equations and inserting the comparator's continuation gives

$$D = T_{\pi_s} J^{\pi_f} - J^{\pi_f} + \beta\{\rho_{x,\pi_s(x)}(J^{\pi_s}(F)) - \rho_{x,\pi_s(x)}(J^{\pi_f}(F))\}.$$

For every fixed original action, the two risks use the same dual set and penalty. For any v, w, each affine dual value at v is at most its value at w plus $\sup_q \int (v - w)\, dq$. Taking the supremum yields $\rho(v) - \rho(w) \leq \sup_q \int (v - w)\, dq$. Apply this to the displayed equation, use the residual bound (A.12), and then use $D \leq D_+$ and monotonicity. It follows that

$$D \leq e + \beta P^+_{\pi_s} D_+.$$

The right side is nonnegative, because e and D_+ are nonnegative and all dual laws are probability laws. Taking positive parts thus yields $D_+ \leq e + \beta P^+_{\pi_s} D_+$. Iterating the monotone gain map at D_+ preserves this comparison. Contraction makes its iterates converge uniformly to E, proving (A.13).

If $e \leq \theta$, the constant function $\theta/(1-\beta)$ is a supersolution: cash preservation gives $e + \beta P^+(\theta/(1-\beta)) \leq \theta/(1-\beta)$. Monotone iteration proves the uniform upper bound. For finite dates the same inequality is applied backward, starting from $D_{T,+} \leq E_T$ and obtaining $D_{t,+} \leq e_t + \beta_t P^+_t E_{t+1} = E_t$. Repeated application to constant bounds produces the dated discounted sum. The maximizing law at a state can differ from its physical innovation law and from the law maximizing a different continuation. Replacing these supremum operators by observed physical-state frequencies is therefore not licensed by the proof. *Q.E.D.*

DOMINANCE OF STOPPING AT AN UNAFFORDABLE IMPROVEMENT CHARGE:

For any feasible replacement, the first proof gives $J - J^\nu \leq e$. Subtracting a charge $c \geq e$ leaves a nonpositive net benefit. This is a bound on the contemplated decision-service operation with its declared unavoidable cost. It does not assign a value to unmeasured human research, guarantee a workload, or solve a distinct search problem in which computations disclose new models or objectives. Those would require their own information and cost specifications. *Q.E.D.*

12. A CONTROLLED RECURSIVE-UTILITY BSDE VERIFICATION BRIDGE

The connection between recursive utility and backward stochastic differential equations has a classical foundation (Pardoux and Peng, 1990, El Karoui et al., 1997). For an NDU/FBSDE application, the preference-state dynamics, information filtration and recursive objective must first be specified. The result below supplies a precise Brownian Markov subcase with an augmented but decoupled forward state. It does not identify a general coupled, general-martingale NDU system with the investment model's nested AVaR recursion. The two-date investment experiment supplies no observations for this different continuous-time criterion.

Fix a finite horizon T, deterministic initial state $X_0 = x$, a Brownian natural filtration with its usual augmentation, and an admissible progressively measurable action process a_s valued in a compact set. The augmented economic and preference state solves

$$dX_s = b(s, X_s, a_s)\, ds + \sigma(s, X_s, a_s)\, dW_s.$$

Assume bounded coefficients, uniform Lipschitz continuity in the state, and the usual measurability. A chosen feedback policy must additionally define an admissible, well-posed state process; arbitrary discontinuous feedback is not licensed by these assumptions on open-loop controls. The state may include preference coordinates, but there is no dependence of b, σ on the backward pair in this subcase. Let g and $f(s, x, 0, 0, a)$ be bounded. The scalar cost driver satisfies

$$|f(s, x, y, z, a) - f(s, x, y', z', a)| \le L_y |y - y'| + L_z \|z - z'\|.$$

The original policy criterion is Y_0^a, where

$$Y_s^a = g(X_T) + \int_s^T f(v, X_v, Y_v^a, Z_v^a, a_v)\, dv - \int_s^T Z_v^a\, dW_v. \tag{3}$$

Converting a utility into a cost requires the corresponding sign changes in both the driver and terminal condition. Merely changing the name of the output does not identify different utility functionals.

For a bounded $u \in C^{1,2}$ with the bounded derivatives needed below, set

$$z_u(s, x, a) = \sigma(s, x, a)^{\mathsf{T}} \nabla u(s, x),$$

$$\mathcal{H}^a u = \partial_s u + b(s,x,a)^\mathsf{T} \nabla u + \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\mathsf{T} D^2 u) + f(s,x,u,z_u,a).$$

The whole-space formulation introduces no artificial boundary. A bounded-domain version requires an invariant domain or a specified stopped process with its stopping-boundary error included in the terminal account. Deleting that term is not justified by a small interior residual.

PROPOSITION S.1—Original-recursion residual certificate: *For each admissible action process the forward equation and* (3) *have a unique square-integrable adapted solution. Suppose uniformly over the declared state-action domain*

$$\mathcal{H}^a u \geq -e_-(s) \quad \textit{for every admissible } a, \qquad \mathcal{H}^\pi u \leq e_+(s),$$
$$-\eta_- \leq g - u(T,\cdot) \leq \eta_+,$$

where $e_\pm$ are nonnegative integrable deterministic bounds and $\eta_\pm \geq 0$. Then the implemented admissible policy satisfies

$$0 \leq Y_0^\pi - \inf_a Y_0^a \leq e^{L_y T}(\eta_- + \eta_+) + \int_0^T e^{L_y s}\{e_-(s) + e_+(s)\}\, ds. \tag{4}$$

BSDE solvability alone does not imply the existence of an optimizing control. The infimum is sufficient for this certificate; compactness and continuity of the control-to-solution map are separate optimal-control assumptions.

PROOF: For the forward process, the Brownian isometry and maximal inequality bound squared maximal differences of successive Picard iterates on a short interval by a fixed Lipschitz constant times the interval length and the preceding squared difference. Choose the interval so that this coefficient is smaller than one. Completeness yields a strong adapted fixed point, and the identical difference estimate proves uniqueness. Concatenating finitely many intervals covers $[0,T]$. Bounded coefficients give finite second moments. The argument uses the stipulated admissible control process; it does not establish well-posedness for a feedback whose closed-loop coefficients violate those conditions.

Fix this forward process. For a trial pair (y,z) in the weighted square-integrable space, Brownian martingale representation of $g(X_T) + \int_0^T f(s,X_s,y_s,z_s,a_s)\, ds$ de-

fines the linear backward solution (Y, Z) with that prescribed driver. For differences between the outputs of two trial pairs, Itô's formula to $e^{ks}|\Delta Y_s|^2$, the common terminal value and expectation give

$$\mathbb{E}\int_0^T e^{ks}\{k|\Delta Y_s|^2 + \|\Delta Z_s\|^2\}\,ds \\ \le 2\mathbb{E}\int_0^T e^{ks}|\Delta Y_s|(L_y|\Delta y_s| + L_z\|\Delta z_s\|)\,ds.$$

Young's inequality bounds the integrand on the right by $(k/2)|\Delta Y_s|^2 + (4L_y^2/k)|\Delta y_s|^2 + (4L_z^2/k)\|\Delta z_s\|^2$. In the squared norm $\mathbb{E}\int_0^T e^{ks}\{(k/2)|Y_s|^2 + \|Z_s\|^2\}\,ds$, the map is a contraction for sufficiently large k, with squared-norm factor at most $\max\{8L_y^2/k^2, 4L_z^2/k\} < 1$. This complete weighted space has a norm equivalent to the usual finite-horizon one. Banach iteration gives existence and uniqueness. Applying Brownian maximal estimates to the backward equation gives $\mathbb{E}\sup_{s\le T}|Y_s|^2 < \infty$. No general-martingale representation or fully coupled forward equation has been inserted into this argument.

For verification, fix any admissible action process and apply Itô's formula to $u(s, X_s)$. Put $\Delta Y = Y^a - u(s, X_s)$ and $\Delta Z = Z^a - z_u(s, X_s, a_s)$. Subtracting the equations yields a linear BSDE with terminal error $g(X_T) - u(T, X_T)$ and driver

$$A_s\Delta Y_s + \vartheta_s^{\mathsf{T}}\Delta Z_s + \mathcal{H}^{a_s}u(s, X_s), \qquad |A_s| \le L_y, \quad \|\vartheta_s\| \le L_z.$$

Measurable divided differences first change y at fixed z, then change z; set the relevant coefficient to zero when a denominator vanishes. The Lipschitz bounds imply the displayed coefficient bounds. Bounded ϑ satisfies Novikov's condition. Under the measure whose density is $\exp\{\int_0^T \vartheta\,dW - \frac{1}{2}\int_0^T \|\vartheta\|^2 ds\}$, $W - \int \vartheta ds$ is Brownian. Multiplication by $\exp(\int_0^s A_v dv)$ and expectation give

$$\Delta Y_0 = \mathbb{E}^{\vartheta}\Big[e^{\int_0^T A_v dv}(g - u(T))(X_T) \\ + \int_0^T e^{\int_0^s A_v dv}\mathcal{H}^{a_s}u(s, X_s)\,ds\Big].$$

Here is the integrability step. The bounded drift has an exponential density with all finite moments. Cauchy–Schwarz transfers the $L^2(P)$ supremum of ΔY and

the $L^2(P)$ square-root quadratic variation of ΔZ to $L^1(P^\vartheta)$. After multiplication by the bounded finite-horizon factor, the localized stochastic integral is therefore an $H^1(P^\vartheta)$ martingale. The terminal and residual terms are dominated by their bounded or integrable deterministic envelopes. The localized identity consequently passes to expectation and the limit, rather than assuming the stochastic integral has mean zero without an integrability check.

The multiplier is positive and at most $e^{L_y s}$. The all-action lower residual and terminal lower error give

$$Y_0^a \geq u(0,x) - e^{L_y T}\eta_- - \int_0^T e^{L_y s} e_-(s)\, ds$$

for every admissible control. The policy-specific upper assumptions give the analogous upper bound for Y_0^π. Take the infimum in the first inequality and subtract. This proves (4); admissibility of π proves nonnegative loss. *Q.E.D.*

Research interface, not objective substitution. An NBO critic and actor for this declared problem must be checked against the original driver, the full graph $z_u = \sigma^\top \nabla u$, the all-action versus implemented-action residuals, and every terminal or stopping-boundary discrepancy. The proposition does not prove uniform approximation by a neural architecture, convergence of a training optimizer, discretization consistency, or a contraction for a fully coupled NDU FBSDE. A latent-preference application would first establish its Markov augmentation, information filtration, driver regularity and admissible policy class. The same economic error accounting can then be developed for that problem; investment-catalogue results cannot be transferred to it by renaming a value function.

13. WELL-POSED ECONOMIC OPERATORS AND IMPLEMENTED POLICIES

This section supplies sufficient conditions and proofs for the existence, measurability, uniqueness, and stability obligations used in the paper. These are standard dynamic-programming foundations, stated explicitly rather than claimed as a new numerical contribution (Ruszczyński, 2010, 2014). They apply directly to the original investment primitives below. A different recursive utility, information filtration,

or noncompact state model requires its own corresponding hypotheses; it does not inherit these properties by being described as a Bellman problem.

13.1. *A sufficient finite-horizon model*

Let X and A be compact metric spaces. At each finite date $j < T$, let $A_j : X \rightrightarrows \mathsf{A}$ have nonempty compact values and be both upper and lower hemicontinuous. Its graph is denoted G_j. Let the innovation space Z_j be compact metric with fixed Borel probability P_j. The original transition $F_j : G_j \times \mathsf{Z}_j \to X$ is jointly continuous and invariant in X. The cost $c_j : G_j \to \mathbb{R}$ and terminal payoff $g : X \to \mathbb{R}$ are continuous. The finite discounts $\beta_j \geq 0$ need not be smaller than one.

Let $\rho_j : L^\infty(P_j) \to \mathbb{R}$ be finite, monotone, cash invariant, and normalized by $\rho_j(0) = 0$. Without normalization, the identical original operator is obtained using $\bar{\rho}_j = \rho_j - \rho_j(0)$ and $\bar{c}_j = c_j + \beta_j \rho_j(0)$; thus the finite deterministic constant remains in the original stage account. For implemented Borel policies we also require the following parameter-measurability property:

$$(s,z) \mapsto w(s,z) \text{ is bounded and Borel} \quad \Longrightarrow \quad s \mapsto \rho_j(w(s,\cdot)) \text{ is Borel} \tag{5}$$

for every standard Borel parameter space and bounded Borel w. This property is verified below for the original expectation and AVaR primitives; it is not deduced from cash invariance alone.

The filtration is generated by the successive original innovations, with their stipulated fixed conditional laws, and the initial state is observed. An admissible history policy is a nonanticipative Borel action rule on this history, feasible at its current original state. Its criterion is the *specified conditional nested recursion*: at a history, apply ρ_j to the next-innovation section of its continuation. This specifies the time-consistent original problem. Static AVaR of total cost and a randomization device with a different observation convention are different criteria and are not inserted into this policy class without being declared.

PROPOSITION S.2—Finite-horizon existence, selection, and validation: *Under these conditions, the original Bellman recursion has one bounded continuous value sequence*

$V_T = g$,

$$V_j(x) = \min_{a \in A_j(x)} \{c_j(x,a) + \beta_j \rho_j(V_{j+1}(F_j(x,a,\cdot)))\}.$$

Each minimum is attained and admits a Borel selector. The resulting Markov policy is optimal among all admissible history policies for the stipulated recursion. Every implemented Borel feasible policy has a well-defined bounded Borel value sequence; its value is at least V_j at every date and original history with state x.

Uniqueness here concerns the value sequence. An optimizer need not be unique. If each action objective has a unique minimizer, that minimizer is continuous in x. In particular, a strongly convex objective on a compact convex feasible action set has a unique minimizer.

For the same primitives, two bounded continuous terminal continuations satisfy

$$\|V_j - \widetilde{V}_j\|_\infty \leq \left(\prod_{k=j}^{T-1} \beta_k \right) \|g - \widetilde{g}\|_\infty. \tag{6}$$

For a fixed implemented policy the same bound holds for bounded Borel terminal continuations, whose recursions are well-defined by (5). Additive stage-cost discrepancies of at most e_k add their dated discounted sum to the right side. Changes in a law, transition, constraints, or preference require their own operator discrepancy; they are not covered by this same-primitives statement.

PROOF: If $\|v - w\|_\infty \leq h$, then $w - h \leq v \leq w + h$. Monotonicity and cash invariance give

$$|\rho_j(v) - \rho_j(w)| \leq h, \qquad |\rho_j(v)| \leq \|v\|_\infty.$$

This also applies to equivalence classes under P_j; a stronger pointwise bound is sufficient. The feasible graph is closed in a compact product and hence compact. For continuous v, joint continuity on $G_j \times \mathsf{Z}_j$ implies uniform continuity. Consequently $(x,a) \mapsto v(F_j(x,a,\cdot))$ is continuous in the supremum norm. The displayed nonexpansiveness proves continuity of its risk evaluation, and thus of the full action objective.

Compactness of $A_j(x)$ gives an attained minimum. To prove value continuity without hiding a selection assumption, let $x_n \to x$. Along a subsequence attain-

ing the lower limit of the values, compactness gives a convergent subsequence of minimizers $a_n \to a$. Upper hemicontinuity gives $a \in A_j(x)$, and continuity of the objective implies that the lower limit is at least $V_j(x)$. Conversely, choose a minimizer a at x. Lower hemicontinuity provides $a_n \in A_j(x_n)$ with $a_n \to a$; evaluating these feasible actions shows that the upper limit is at most $V_j(x)$. Thus the value is continuous. Backward induction from g proves existence and continuity of the complete value sequence; the finite recursion also proves its uniqueness. The norm estimate $\|V_j\|_\infty \le \|c_j\|_\infty + \beta_j \|V_{j+1}\|_\infty$ proves boundedness.

Here is an explicit measurable-selector argument. The argmin graph is closed, because both the objective and value are continuous, and its fibers are nonempty compact. For each n, choose a finite cover of A by closed balls of radius 2^{-n} with fixed centers. In a fixed ordering choose the first ball meeting the argmin fiber, and at each successive level choose the first ball meeting its intersection with the previously chosen balls. All these intersections remain nonempty compact and are nested. The event that the argmin fiber meets a fixed finite intersection of closed balls is closed: it is the projection onto compact X of a closed subset of $X \times \mathsf{A}$. Induction over the finite-valued previous choices therefore makes each new chosen center a Borel function of x. The nested sets have diameters at most 2^{1-n}, so their intersection contains exactly one action. The chosen centers converge to that action; their pointwise limit is Borel and belongs to the argmin fiber. This constructs the required selector. When the argmin is already unique, the same subsequence argument used for value continuity shows continuity of the selector.

For an arbitrary implemented history policy, composition of Borel histories, actions, and the original transition gives a bounded Borel next-history kernel. Property (5) then constructs its bounded Borel value backward from g. Suppose at the next date that the policy continuation is at least V_{j+1} at its next original state. Applying the same conditional risk at the current history preserves this inequality. The resulting current cost is at least the action objective at its feasible chosen action, hence at least V_j. Backward induction proves policy domination, without restricting the comparison to Markov policies. Using the Borel minimizing selector at every date makes every inequality an equality, which proves its optimality.

Finally, the preceding nonexpansiveness yields $\|\mathcal{T}_j v - \mathcal{T}_j w\|_\infty \le \beta_j \|v - w\|_\infty$, by comparing infima over the same actions. A fixed policy has the identical inequality without the infimum. Backward induction proves (6). When the costs differ by at most e_j, the one-step estimate has the extra term e_j; iterating it gives the dated sum. *Q.E.D.*

The exact continuity and measurability obligations. Compact innovations can be replaced by essential-uniform continuity of $v(F_j(x,a,\cdot))$ on the declared continuation class. Pointwise continuity alone is insufficient for an arbitrary monetary risk. For example, let Z have an unbounded exponential law, $F(x,z) = 1 - e^{-xz}$ for $x \in [0,1]$, terminal value $g(y) = y$, and $\rho = \operatorname{ess\,sup}$. All these primitives are continuous and bounded where required, but $\rho(g(F(0,Z))) = 0$ and $\rho(g(F(x,Z))) = 1$ for every $x > 0$. Expectation has stronger dominated-continuity properties, but those cannot be assigned to every monotone cash-invariant risk.

For the original fixed-law AVaR and bounded jointly Borel w, the formula

$$\rho_\alpha(w(s,\cdot)) = \inf_{\eta \in \mathbb{Q}} \left\{ \eta + \frac{1}{1-\alpha} \int (w(s,z) - \eta)_+ \, P(dz) \right\}$$

proves (5). Each integral is Borel: the claim holds on measurable rectangles, extends to indicators of product-Borel sets by a monotone-class argument, and then to nonnegative Borel functions by monotone simple approximation. The infimum is countable and equals the real-threshold infimum because the displayed expression is continuous in η. Expectation is the same measurability argument without an infimum. Thus the needed property is established for both original preference cases, while remaining explicit for a different risk primitive.

13.2. *The original investment domain and unique constrained actions*

PROPOSITION S.3—Original primitive closure and actor regularity: *For the original even-dimensional investment primitives, every feasible state-action-innovation triple satisfies*

$$F(x,a,z) \in [1/32, 27/32]^d \subset [0,1]^d. \tag{7}$$

The capacity correspondence $[0, A(x)]$ *is continuous, compact, and nonempty, with* $1/8 \leq A(x) \leq 1/4$. *Costs are continuous and nonnegative, with valid uniform bounds* $S \leq 49/32$, $g \leq 37/16$, *and* $S + h \leq 103/64$. *Hence the finite-horizon existence and policy theorem applies for the original expectation and nested AVaR criteria.*

For the two-date AVaR model with $\beta = 15/16$, V_1 *is strongly convex in state with modulus* $(4 - 5\beta/4)/d$, *and the original root and last-date action objectives are strongly convex in action with modulus two. They have unique continuous constrained optimal actors. The last-date* V_1 *and the root action objective are continuously differentiable on the relative state interior, with the corresponding one-sided boundary interpretation.*

For either date where Q_j *is continuously differentiable, write* $a_j^*(x)$ *for its constrained minimizer and*

$$\lambda_j(x) = \mathbf{1}_{\{a_j^* = A(x)\}}[-\partial_a Q_j(x, a_j^*)]_+.$$

Then the value gradient is

$$\nabla V_j(x) = \nabla_x Q_j(x, a_j^*(x)) - \lambda_j(x) \nabla A(x). \tag{8}$$

This formula is not a derivative of a rounded deployed actor. Interior actor derivatives require the additional local second derivatives and strict-interiority conditions stated in the retained compiler theorem.

PROOF: For the state part of the jth transition coordinate put $p = x_j$, $q = x_{j+1}$. The function $p/2 + q/8 + p(1 - q)/16$ is nondecreasing in both arguments on $[0,1]^2$, because its two partial derivatives are $1/2 + (1 - q)/16$ and $1/8 - p/16$. Its range lies in $[0, 5/8]$. Since $0 \leq b_j a \leq 1/8$ and $|z| \leq 1/32$, adding the constant $1/16$ proves (7). The affine capacity gives the stated correspondence properties. The inequalities $|x_j - 5/8| \leq 5/8$, $|x_j - x_{j+1}| \leq 1$, and $(1/2 - 2\bar{x})_+ \leq 1/2$ give the stated state-cost bounds. Moreover $h(a) \leq h(1/4) = 5/64$, which gives the full stage bound. Polynomial transitions, squared positive parts, and the fixed compact innovation interval have the required continuity. The previous measurability proof applies to the risk primitive.

For completeness, the inherited joint curvature bound follows directly from the original primitives. On each smooth region, $\nabla^2 S \succeq (4/d) I$, $\nabla^2 g \succeq 0$, and $|\partial_j g| \leq$

$10/d$. The only nonzero second derivative of the transition is the mixed derivative of $-x_j x_{j+1}/16$. In a joint state-action direction (v, w), the Hessian chain rule therefore gives

$$D^2(g \circ F)[(v,w)]^2 \geq -\frac{1}{8}\sum_j \partial_j g(F) v_j v_{j+1} \geq -\frac{5}{4d}\|v\|_2^2.$$

The omitted Jacobian–Hessian–Jacobian term is nonnegative in the full joint direction, including its state-action mixed terms. Together with $h'' \geq 2$, this proves for every fixed shock the lower curvature

$$\frac{4 - 5\beta/4}{d}\|v\|_2^2 + 2w^2.$$

The squared-hinge derivatives match at the junction, so integrating the almost-everywhere second-derivative inequality along a segment extends it across the junction. Integration under any feasible AVaR density preserves the bound. Subtracting its common quadratic term leaves convex functions; their supremum is convex, so the risk supremum preserves the same bound. It follows that the joint objective has the state term $(4 - 5\beta/4)\|x - y\|_2^2/(2d)$ in its strong-convexity inequality. Interpolating two optimal state-action pairs preserves feasibility because the capacity is affine. Taking the value at that feasible interpolation and dropping the nonnegative action-distance term proves strong convexity of V_1 with the announced modulus. This supplies the state convexity used in the active-tail theorem, rather than assuming it for a fitted continuation.

For each fixed root state, the successor is affine in its scalar action. Convexity of V_1, followed by monotonicity and convexity of AVaR, makes $a \mapsto \rho_\alpha(V_1(F(x, a, \cdot)))$ convex. Adding $h'' \geq 2$ proves root-action strong convexity; the same property of the last-date objective follows from the original primitive curvature. The preceding compact theorem now gives existence, uniqueness, and continuity of both optimal actors.

The retained exact last-date formula is C^1 in state and action: the positive-part square and its explicit tail polynomial have matching values and first derivatives at their junctions. Normalize the action by $a = A(x)v$, with $v \in [0, 1]$. The normalized objective has a unique continuous minimizer. Comparing the value increment at the

old optimizer and at the new optimizer, and using uniform first-order expansion on this fixed compact action interval, gives

$$\nabla V_1 = \nabla_x Q_1 + v^* \partial_a Q_1 \nabla A.$$

At a lower endpoint $v^* = 0$, at an interior optimum $\partial_a Q_1 = 0$, and at capacity $v^* = 1$, $\partial_a Q_1 \le 0$. This proves (8) and continuity of the value gradient. In particular the multiplier tends to zero at a switch between an interior and a binding-capacity solution.

For the root objective, invariance puts every successor strictly inside the state cube, where ∇V_1 is continuous and bounded on the compact successor range. Along the innovation, $V_1(F(x,a,z))$ is strongly convex with modulus $181/64$. Every one of its level sets has at most two points, so its uniform-law distribution has no atoms. Continuity of the function on the interval makes its distribution function strictly increasing between its minimum and maximum: the preimage of every nonempty open value interval in that range contains a nonempty open innovation interval. Its quantile is therefore unique at the declared risk levels. The differentiability conditions of the retained AVaR envelope argument apply, with bounded action/state derivatives. Equivalently, the active central interval has a unique continuous minimizer and the value envelope differentiates the integral with that interval held fixed. Thus, writing $E_{x,a}$ for the optimal two-tail set,

$$\partial_a Q_0(x,a) = h'(a) + \frac{\beta}{2R(1-\alpha)} \int_{E_{x,a}} b^\mathsf{T} \nabla V_1(F(x,a,z))\, dz.$$

The analogous state derivative has $D_x F^\mathsf{T} \nabla V_1$ and ∇S. Continuity of the integrands and tail endpoints gives continuity of these derivatives. Applying the same normalized action envelope proves (8) for V_0. Endpoint actions are included; no global differentiability of the actor across active-set changes is asserted. *Q.E.D.*

Observation, feasibility, and terminal-policy transfer. The implemented terminal controller observes a downward 40-bit state $y \le x$ with $\|x - y\|_1 \le d 2^{-40}$ and returns an action feasible at y. Since $A(y) \le A(x)$, it remains feasible at the original state. Its certified action gap at y is at most 2^{-21}. The primitive fixed-action objective

and the optimal value both have state Lipschitz constant at most $15/d$, so its original-state gap is at most

$$2^{-21} + 30\,2^{-40} < 2^{-20}.$$

The fixed-action modulus follows from $\|\nabla S\|_\infty \le 8/d$, the transition column-sum bound $11/16$, $\|\nabla g\|_\infty \le 10/d$, and $8 + \beta(11/16)10 < 15$. The value modulus is the retained capacity-aware bound. The controller's rounding and exact fallback must actually enforce its premise; an optimizer return code alone does not do so. This explains why its allowance remains valid when the new method changes the locations of original-law integration queries while retaining the same deployed controller.

13.3. *Residual validation and the optional infinite-horizon case*

For the finite-horizon one-sided sandwich, suppose $f_j - \mathcal{T}_j f_{j+1} \le u_j$, $\mathcal{T}_j^\pi f_{j+1} - f_j \le v_j$, and $\ell_T \le f_T - g \le u_T$. Monotonicity and cash invariance give separately

$$f_0 - V_0 \le \sum_{j<T} B_j u_j + B_T u_T, \qquad J_0^\pi - f_0 \le \sum_{j<T} B_j v_j - B_T \ell_T.$$

Each follows by backward induction, starting from its corresponding terminal inequality. Adding them proves the retained complete-policy bound, and feasibility supplies $J_0^\pi \ge V_0$. This validates an NBO's own attained continuation without assuming optimal-continuation training labels. Date-specific constant shifts of f_j cancel in the paired residuals and in $u_T - \ell_T$.

For a stationary infinite-horizon model with the same compact and kernel conditions, bounded cost, and $0 \le \beta < 1$, the Bellman map is a β-contraction on the complete space $C(X)$. The finite-horizon continuity proof shows that it maps this space into itself. Thus it has a unique continuous fixed point V. For any Borel feasible stationary policy, the policy map is a β-contraction on bounded Borel functions by (5); it has a unique bounded Borel fixed point J^π. A Borel minimizer of the continuous action objective at V satisfies $T_{\pi^*}V = TV = V$, so uniqueness of its policy fixed point gives $J^{\pi^*} = V$. For another policy, $V \le T_\pi V$; monotone iteration and contraction give $V \le J^\pi$.

The infinite-history criterion is well-defined as the limit of the same finite nested recursions: the omitted cost after n dates is bounded uniformly by $\|c\|_\infty \beta^n/(1-\beta)$. Passing the finite-horizon comparison to this limit proves optimality against the same admissible history-policy class. This argument does not assert that infimizing arbitrary Borel action objectives preserves Borel measurability. The more general policy-iteration theorem in the retained supplement explicitly assumes that additional preservation and selection property when it applies a Bellman infimum to possibly discontinuous policy values.

13.4. *The critic and actor computational graph*

For a smooth parameterized critic and differentiable feasible actor, the derivative must correspond to the objective actually optimized. For example, at a fixed original state write

$$r_\theta = c(x, a_\theta) + \beta\rho(f_{+,\theta}(F(x, a_\theta, Z))) - f_\theta(x).$$

Suppose differentiation is justified by integrable local domination, and either ρ is expectation or it is AVaR at an atomless unique quantile, as in the retained tail-gradient theorem. Let q_θ be its maximizing tail density, with $q_\theta = 1$ for expectation. The chain and risk-envelope rules give

$$\nabla_\theta r_\theta = (D_\theta a_\theta)^\mathsf{T} \left[\nabla_a c + \beta \mathbb{E}_{q_\theta}(D_a F)^\mathsf{T} \nabla_x f_{+,\theta}(F)\right] + \beta \mathbb{E}_{q_\theta} \nabla_\theta f_{+,\theta}(F) - \nabla_\theta f_\theta(x).$$

The last expectation is a partial parameter derivative at a fixed successor. The actor contribution is written separately to avoid double counting. A risk-envelope derivative holds the maximizing density fixed in this formula; its validity comes from the stated regularity assumptions, not from differentiating a sorting permutation. The half-squared-residual objective $\mathcal{L}(\theta) = \frac{1}{2}\mathbb{E}[r_\theta^2]$ has gradient $\mathbb{E}[r_\theta \nabla_\theta r_\theta]$ when its own domination condition holds. Detaching the future target defines a different frozen-target gradient or a semigradient algorithm. It does not establish that this full original coupled gradient was used, and neither formula proves optimizer convergence.

At a projection or rounding discontinuity these differentiability premises may fail. Piecewise training derivatives can be specified as an algorithmic choice, but the global economic certificate still uses the actual feasible action and its original-law inequalities. This is consistent with the own-future constructive theorem, whose validity does not depend on a successful differentiable optimizer.

14. STOPPING BOUNDARIES AND THE CONTINUOUS-TIME RECURSION

The retained Brownian Markov bridge is a separate classical recursive utility verification theorem (Pardoux and Peng, 1990, El Karoui et al., 1997). Its forward coefficients and driver are understood to be jointly Borel in their displayed deterministic arguments, so composition with each admissible progressively measurable control yields progressively measurable coefficients. The forward state-Lipschitz constants and driver (y,z)-Lipschitz constants are uniform in the control. Terminal payoff and the driver at $(y,z)=(0,0)$ are bounded. A feedback must define its own well-posed closed-loop state process. These conditions make the retained existence proof precise; they do not establish existence for a fully coupled FBSDE or an arbitrary discontinuous feedback.

PROPOSITION S.4—Stopped original-BSDE verification: *Let $\tau^a \le T$ be a specified stopping time for each admissible control, such as exit from a declared domain capped at T, with bounded Borel original stopping payoff $G(\tau^a, X_{\tau^a})$. Before stopping use exactly the driver and forward coefficients of the retained BSDE bridge. Let bounded $u \in C^{1,2}$ have the bounded derivatives required for stopped Itô calculus and continuous values on the declared stopping boundary. Suppose*

$$\mathcal{H}^a u \ge -e_-(s) \text{ for every admissible action}, \qquad \mathcal{H}^\pi u \le e_+(s)$$

on the unstopped domain, and at every possible stopping location and at the terminal part of the boundary suppose

$$-\eta_- \le G(s,x) - u(s,x) \le \eta_+.$$

Here $e_\pm$ are nonnegative deterministic integrable functions and $\eta_\pm \geq 0$. The stopped BSDE has a unique solution in $S^2 \times H^2$, and

$$0 \leq Y_0^\pi - \inf_a Y_0^a \leq e^{L_y T}(\eta_- + \eta_+) + \int_0^T e^{L_y s}(e_-(s) + e_+(s))\, ds.$$

The stopping rule and boundary payoff are parts of the original economic criterion. A smaller interior residual does not remove their errors or permit them to be changed.

PROOF: Embed the stopped equation on $[0,T]$ with terminal variable $\xi^a = G(\tau^a, X_{\tau^a})$ and driver $\mathbf{1}_{\{s<\tau^a\}} f(s, X_s, y, z, a_s)$. It is progressively measurable, has the same uniform Lipschitz constants, and has square-integrable terminal data and bounded driver at zero. The weighted Picard argument in the retained proof therefore gives a unique $S^2 \times H^2$ solution. After τ^a it equals ξ^a with $Z = 0$; before stopping it is exactly the declared stopped equation.

Apply stopped Itô calculus to $u(s \wedge \tau^a, X_{s\wedge\tau^a})$. Subtracting gives the same linearized error BSDE as in the retained proof, with its coefficients set to zero after τ^a, terminal error $\xi^a - u(\tau^a, X_{\tau^a})$, and residual driver $\mathbf{1}_{\{s<\tau^a\}} \mathcal{H}^{a_s} u$. Its measurable coefficients satisfy $|A_s| \leq L_y$ and $\|\vartheta_s\| \leq L_z$. The bounded Girsanov coefficient has an exponential density with all finite moments. The same Cauchy–Schwarz and H^1 martingale argument used in the retained verification proof therefore justifies expectation of the stopped integration-factor identity,

$$Y_0^a - u(0,x) = \mathbb{E}^\vartheta \left[e^{\int_0^{\tau^a} A_v dv} (G-u)(\tau^a, X_{\tau^a}) + \int_0^{\tau^a} e^{\int_0^s A_v dv} \mathcal{H}^{a_s} u(s, X_s)\, ds \right].$$

The factor is positive and at most $e^{L_y s}$. Applying the lower residual and boundary bounds gives a lower bound for every admissible control. Applying the upper bounds gives an upper bound for the implemented policy. Since $\tau^a \leq T$, the deterministic integrals and terminal factors are bounded by those displayed in the proposition. Taking the infimum of the lower bounds and subtracting proves the result. *Q.E.D.*

Why the boundary and graph are material. For a deterministic state starting at $1/2$ in $(0,1)$, use either of the two constant controls $a = \pm 1$ in $dX_s = a\, ds$, stop on exit before $T = 1$, and assign boundary costs zero at the left endpoint and one at the

right. With zero driver and $u = 0$, every interior residual is zero, but the right-exit policy has regret one. The stopping-boundary error exactly supplies the missing term. This is a verification counterexample, not a second numerical application.

The BSDE graph is $z_u = \sigma^\top \nabla u$ for the same control and the original driver. Its $C^{1,2}$ requirement is substantive. A piecewise-affine ReLU critic is generally not C^2 and cannot be inserted into this classical formula by declaring its second derivative zero away from its kinks; a different weak, viscosity, or smoothing argument with the required error account would first be needed. The finite-horizon ReLU Bellman construction uses order and Lipschitz arguments and does not rely on this continuous-time smooth verification theorem.

Finally, a general recursive utility is not automatically cash invariant. Even the globally Lipschitz BSDE driver $f(s, y, z) = \lambda y$ gives $Y_0 = e^{\lambda T}\mathbb{E}\xi$, so adding a constant c to the terminal criterion changes Y_0 by $e^{\lambda T}c$, not generally by c. The continuous-time bridge correctly uses its $e^{L_y T}$ stability factors. It does not inherit the monetary discrete-risk contraction or identify an arbitrary NDU/FBSDE with the original investment AVaR objective.

15. ACTIVE-TAIL LOCALIZATION, INFORMATION, AND ORIGINAL-POLICY BOUNDS

15.1. *Proof of Theorem 7*

PROOF: Because f is continuous, the fundamental theorem of calculus gives $H'(u) = f(u + L) - f(u) = D(u)$. For convex f, its increment over a fixed positive length is nondecreasing. One direct proof is to use the almost-everywhere derivative of a Lipschitz convex function:

$$D(v) - D(u) = \int_u^v \{f'(s + L) - f'(s)\}\, ds \geq 0 \quad (u < v).$$

The derivative is monotone wherever defined, and changing its values on a null set does not change the integral. Also,

$$|D(v) - D(u)| \leq |f(v + L) - f(u + L)| + |f(v) - f(u)| \leq 2K|v - u|.$$

Thus H is convex with a $2K$-Lipschitz derivative. Compactness gives a minimizer on I, and the derivative signs give the asserted endpoint rules.

We prove that a minimizing central interval discards precisely a lower α fraction of the outcomes, allowing equality at the threshold. At an interior minimizer u^*, $f(u^*) = f(u^* + L) = \eta$. Convexity gives $f \le \eta$ between these endpoints and $f \ge \eta$ outside. The outside set has original probability t. The feasible density $q = t^{-1}\mathbf{1}_{[-R,u^*]\cup[u^*+L,R]}$ therefore has

$$\int f q \, dP = \eta + t^{-1}\mathbb{E}(f-\eta)_+.$$

The equality is unaffected by any flat portion at level η. For an arbitrary feasible AVaR density $\widetilde{q}$, the constraints $0 \le \widetilde{q} \le t^{-1}$ and $\int \widetilde{q}\, dP = 1$ give

$$\int f\widetilde{q}\, dP = \eta + \int (f-\eta)\widetilde{q}\, dP \le \eta + t^{-1}\mathbb{E}(f-\eta)_+.$$

Consequently this selected q is optimal.

At the left endpoint, $f(-R+L) \ge f(-R)$. Convexity implies that the maximum on $[-R, -R+L]$ is $f(-R+L)$ and that f is nondecreasing to the right of $-R+L$; hence the same threshold argument applies with $\eta = f(-R+L)$. At the right endpoint, the inequality is reversed and its reflected argument applies. These arguments include equal endpoint values and constant functions. They establish (26) without a density for the distribution of $f(Z)$ or a no-atom assumption.

When neither endpoint rule holds, continuity of D gives an interior zero. Sign bisection maintains a bracket containing at least one zero, whose width after m halvings is $2Rt\,2^{-m}$ because $|I| = 2Rt$. Let u^* be a zero in that bracket. Integrating the $2K$-Lipschitz derivative from u^* to its midpoint u_m gives

$$0 \le H(u_m) - H(u^*) \le K|u_m - u^*|^2 \le KR^2t^2\,4^{-m}.$$

The total integral of f is constant in u, so $\mathrm{AVaR}_\alpha(f) - R_{u_m}(f) = [H(u_m) - H(u^*)]/(2Rt)$. This proves (27). An endpoint or exact-zero exit has zero loss, so no smoothness-boundary term is omitted.

For a nonempty cell $[l, r]$ of length h and midpoint z, convexity gives the pointwise secant upper bound and Jensen's lower bound,

$$hf(z) \le \int_l^r f(s)\, ds \le \tfrac{h}{2}\{f(l) + f(r)\}.$$

Their difference is

$$\frac{h^2}{4}\left\{\frac{f(r)-f(z)}{r-z}-\frac{f(z)-f(l)}{z-l}\right\}.$$

The braces are nonnegative. When all cells are ordered from left to right, convexity orders every successive secant slope, including across the omitted central interval. The sum of the bracketed slope differences is therefore at most the difference between the largest and smallest of these slopes, hence at most $2K$. With maximal cell length $h_{\max}$, summation and normalization by $2Rt$ give

$$\overline{R}-\underline{R}\le\frac{Kh_{\max}^2}{4Rt}\le\frac{KRt}{n^2}.$$

This proves the interval and its width bound.

If the selected tail lengths are $b_1,b_2\ge 0$, then $b_1+b_2=2Rt$. Use $\lceil nb_j/(2Rt)\rceil$ cells in each nonempty tail. Since the two unrounded counts sum to the integer n, their ceilings sum to at most $n+1$. Midpoints and distinct endpoints require at most $2(n+1)+2=2n+4$ values, allowing two nonempty components. The two endpoint sign checks require at most four values, and m bisections require at most $2m$ more. Reusing duplicates can only reduce this count. The query bound follows. Putting $m=\lceil\log_2 n\rceil$ gives $4^{-m}\le n^{-2}$.

For completeness, the lower bound preserves its original deterministic adaptive quantifiers. Let $q=-R+2R\alpha$ and $b=R-q=2Rt$, and run the algorithm on

$$f_0(z)=\frac{K}{2b}(z-q)_+^2.$$

This function is convex, nondecreasing, and K-Lipschitz. Insert the N queried locations lying strictly between q and R, together with the two endpoints. Their distinct consecutive interval lengths are $h_1,\dots,h_s$, where $s\le N+1$ and $\sum_i h_i=b$. Define f_1 to equal f_0 to the left of q and its chord on each of these intervals. The chord slopes are nondecreasing in $[0,K]$; thus f_1 belongs to the same class and agrees with every query answer. Induction on the adaptive transcript shows that the algorithm cannot distinguish f_0 and f_1.

Since both functions are nondecreasing, their AVaRs are their averages over $[q,R]$. The integral of the chord error on an interval of length h_i is $Kh_i^3/(12b)$. Jensen's

inequality gives

$$\mathrm{AVaR}_\alpha(f_1) - \mathrm{AVaR}_\alpha(f_0) = \frac{K}{12b^2}\sum_i h_i^3 \geq \frac{Kb}{12(N+1)^2} = \frac{KRt}{6(N+1)^2}.$$

A sound common interval contains both values. This proves the lower bound. With $\alpha = 0$, the same construction uses $q = -R$ and the upper algorithm simply partitions the full interval; no central interval or tail sign search is necessary. For example, for every $N \geq 32$ one may take $n = \lfloor N/4 \rfloor$ in the upper algorithm; its stated query bound is at most N and its width is at most $(75/2)KRt/N^2$. Together with the lower bound this proves the claimed uniform order, without concealing a logarithmic factor in the error rate. *Q.E.D.*

15.2. *Primitive curvature and inexact tail localization*

PROOF: Write $\mu_x = (4 - 5\beta/4)/d$. The inherited primitive curvature bound says that the jointly convex last-date objective satisfies the strong-convexity inequality with terms $\mu_x\|x-y\|_2^2/2$ and $|a-b|^2$. For any x, y, select their optimal actions a_x, a_y. The capacity A is affine, so $(1-\theta)a_x + \theta a_y$ is feasible at $(1-\theta)x + \theta y$ for $0 \leq \theta \leq 1$. Evaluate the strong-convexity inequality at this feasible interpolated action and then take the value infimum. Dropping the nonnegative action-distance term proves

$$V_1((1-\theta)x + \theta y) \leq (1-\theta)V_1(x) + \theta V_1(y) - \tfrac{1}{2}\mu_x\theta(1-\theta)\|x-y\|_2^2.$$

Thus V_1 is μ_x-strongly convex. For fixed original x, a, the successor map is affine in z with direction $s = (1, -1, \ldots, 1, -1)$ and $\|s\|_2^2 = d$. Composition yields $m_z = \mu_x d = 181/64$. The inherited state modulus $15/d$ in the ℓ^1 norm gives $|f_a(z) - f_a(z')| \leq 15|z - z'|$.

For $u < v$, strong convexity of f_a gives monotonicity of $f_a'(z) - m_z z$ at its almost-everywhere derivative points. Therefore

$$D_a(v) - D_a(u) = \int_u^v \{f_a'(s+L) - f_a'(s)\}\, ds \geq m_z L(v-u).$$

This is precisely $m_z L$-strong convexity of H_a. For the true $d = D_a(u)$ it follows that

$$H_a(v) \geq H_a(u) + d(v-u) + \tfrac{1}{2}m_z L(v-u)^2 \quad (v \in I).$$

Taking the infimum proves that the normalized tail deficit is at most the expression in (33) at the true d. The negative of an infimum of affine functions of d is convex; its maximum over $[d_-, d_+]$ is bounded by the larger endpoint value. Completing the square gives the projected minimizer. These facts prove (33) for outward derivative intervals, including both constrained endpoints and $d = 0$. *Q.E.D.*

15.3. *Proof of Theorem 8*

PROOF: The selected density is exactly feasible under the original law. Integrating the midpoint affine supports over their symmetric cells cancels the term proportional to $z - z_i$, yielding $R_u(f_a) \geq r_-$. Convexity of f_a and its endpoint upper values give $R_u(f_a) \leq r_+$. By the tail-defect bound,

$$r_- \leq \mathrm{AVaR}_\alpha(f_a) \leq r_+ + \kappa(u).$$

The complete terminal policy has $V_1 \leq J_1 \leq V_1 + \varepsilon_1$ throughout the original state domain. Monotonicity and cash invariance, for the same original risk primitive, imply

$$\mathrm{AVaR}_\alpha(f_a) \leq \mathrm{AVaR}_\alpha(J_1 \circ F_a) \leq \mathrm{AVaR}_\alpha(f_a) + \varepsilon_1.$$

Adding unchanged stage cost and discounting proves (36). No convexity of J_1 has been used.

For another original feasible action v, the transition satisfies $F(x, v, z) = F(x, a, z) + b(v - a)$. Therefore each global support gives

$$V_1(F(x, v, z)) \geq \ell_i + p_i^\mathsf{T}\{b(v - a) + s(z - z_i)\} \quad (z \in B_i).$$

Integrating over the *same fixed* feasible density removes the last term. The original stage cost satisfies $h(v) \geq h(a) + h'(a)(v - a) + (v - a)^2$ because $h'' \geq 2$. Combining these inequalities yields

$$Q(v) \geq W + s(v - a) + (v - a)^2 \quad (0 \leq v \leq A(x)).$$

Taking the infimum gives (37); the true V_0 is this original optimal-future infimum. For uncertain slope, the constrained quadratic minimum is concave as a function of

its slope, so its minimum over an interval is bounded below by the smaller endpoint minimum. Taking outward endpoints preserves every direction of inequality. A smaller exact feasible action domain can be enlarged when constructing a lower bound, but may never be shrunk to obtain a favorable minimum.

Since $V_0 \le J(a)$, the policy upper minus the optimum lower is a valid nonnegative regret allowance. The same inference holds at $a = 0$ and $a = A(x)$: no first-order condition for an interior actor has been assumed. Tail search need not succeed, a fitted critic need not equal V_1, and a numerical candidate optimizer need not return an optimum, provided the displayed outward witnesses remain valid and the realized action remains feasible. *Q.E.D.*

15.4. *Paid precision and conditional allocation*

PROOF: Qualification at a chosen stage of a finite sequence follows from its own verified inequalities; no monotonicity of earlier reported widths is needed. Summing the actual cost of every attempted stage is the complete cost. If a deterministic ladder has work cn_j per stage with $n_j = 2^j n_0$, its total work through n_J is less than $2cn_J$. Such a bound requires this work model and does not remove fixed costs, descendant solves, refused trials, or rational fallback.

For the allocation formula write $e_b = B4^{-m}$ and $e_q = A/n^2$. At a nonboundary continuous optimum the positive constraint binds, hence $e_q = \epsilon - e_b$ and the objective is

$$F(e_b) = c_q \sqrt{\frac{A}{\epsilon - e_b}} + \frac{c_b}{\log 4} \log \frac{B}{e_b}.$$

Its second derivative is positive for $0 < e_b < \epsilon$. Its derivative tends to negative infinity at zero and positive infinity at ϵ. It therefore has exactly one minimizer, and setting its derivative equal to zero gives (38). If this point violates $n \ge 1$ or $m \ge 0$, the convex problem in (n, m) is minimized on the corresponding boundary or at the remaining feasible stationary point; the constraints $A/n^2 + B4^{-m} \le \epsilon$ form a convex set. Increasing either resolution decreases its allowance, so rounding both upward preserves feasibility and adds less than $c_q + c_b$. All coefficients must be fixed properties of the declared graph and cost model. Estimating them after observ-

ing a favorable outcome does not turn the resulting allocation into a prospective experiment. *Q.E.D.*

15.5. *Recursive interpretation*

Suppose at each date the constructed certificate supplies a uniform original-optimal-continuation action gap $\gamma_t \geq 0$ for a feasible deployed policy. If its next-date policy loss is at most e_{t+1}, the unchanged Bellman preference gives

$$J_t^{\pi} \leq Q_t(V_{t+1}^*; x, \pi_t(x)) + \beta_t e_{t+1} \leq V_t^* + \gamma_t + \beta_t e_{t+1}.$$

Backward induction from the declared terminal discrepancy gives the dated discounted sum. A support of an NBO's own fitted continuation instead requires the retained one-sided or reference-centered continuation-error account. It cannot be silently promoted to a support of V_{t+1}^*. The currently executed original two-date construction obtains the needed V_1 supports from the primitive last-date oracle, and retains its actual terminal-controller allowance.

16. QUALIFICATION AND WORK OF THE BOUNDED PRELIMINARY SERVICE

16.1. *Objects, information, and algorithm*

Fix a state x, its original feasible action interval $[0, A(x)]$, a realized proposal $a \in [0, A(x)]$, and $\alpha \in \{1/2, 3/4\}$. Write

$$R = 1/32, \qquad t = 1 - \alpha, \qquad L = 2R\alpha, \qquad I = [-R, R - L],$$
$$f_a(z) = V_1(F(x, a, z)).$$

The primitive transition is affine in the innovation and the action. The already established original bounds give convexity of f_a, its strong-convexity modulus $m_z = 181/64$, and $V_1 \leq J_1 \leq V_1 + \varepsilon_1$ for the implemented terminal controller, with $\varepsilon_1 = 2^{-20}$. Its complete-policy cost is

$$J(a) = S(x) + h(a) + \beta\, \mathrm{AVaR}_{\alpha}(J_1 \circ F_a),$$

whereas $Q(a)$ uses V_1 in this expression, and $V_0(x) = \inf_{0 \leq v \leq A(x)} Q(v)$.

Two information types are distinguished. A floating terminal value is used only to propose a cut. An outward terminal payload encloses the original V_1 value; when used at a quadrature midpoint it also supplies a valid global affine minorant, with every uncertain slope component retained. These are the same original payloads used by the common terminal compiler. Counting a payload does not assign unit cost to its internal original-risk evaluation, optimization, interval arithmetic, or supporting-witness construction.

The floating proposal compares $f_a(u+L) - f_a(u)$ at the two endpoints of I. A proposed endpoint exit suppresses further floating search for that row. Otherwise six interval-halving updates are performed, after which the midpoint of the final proposed bracket is rounded and clipped to the feasible 2^{-32} grid. The signs need not be correct: only feasibility of the final cut u is used below. The production domain checks enforce this feasibility and the declared risk levels.

At this u, two outward payloads give an interval $[d_-, d_+]$ containing $f_a(u+L) - f_a(u)$. The computed outward upper bound is

$$\widehat{\kappa} \geq \frac{1}{2Rt} \max_{d \in \{d_-, d_+\}} \left[-\min_{v \in I} \left\{ d(v-u) + \frac{m_z L}{2}(v-u)^2 \right\} \right]. \tag{9}$$

The same state, action, risk level, cut, and defect are cached for the two attempted resolutions. Reusing the cache for another state, action, or objective is disallowed.

Let the lengths of the left and right tails be $b_1 = u + R$ and $b_2 = R - u - L$. For $m = 2$ and then, if needed, $m = 4$, partition each tail with $b_j > 0$ into m equal cells. Each cell of that tail has weight $b_j/(m\,2Rt)$. A cell uses one midpoint payload and its two endpoint upper values; endpoints shared by neighboring cells within the same attempted resolution are acquired once. No values are requested solely for a zero-length tail. At each resolution form the original-policy and original-optimum bounds of Theorem 8; accept only if the outward regret allowance is at most $1/256$ and the outward policy width is at most $1/8192$. If neither attempt qualifies, continue with the frozen 128-bin service from the already acquired proposal. Its original conventional repair, further resolutions, and rational fallback are included.

16.2. *Proof of Proposition A.3*

PROOF: We establish feasibility, validity of each possible acceptance, and the full preliminary query count separately.

Exact zero-mass omission. The tail lengths are nonnegative and satisfy

$$b_1 + b_2 = (u + R) + (R - u - L) = 2R - L = 2Rt > 0.$$

Consequently the number p of positive-length tails is either one or two, and the retained weights sum to one. The density relative to the original uniform law is still t^{-1} on these tails and zero on the omitted central interval. Removing a zero-length tail removes a set of zero original probability; it changes none of the exact integrals, risk witnesses, or action supports. This is not an approximation by a small-mass cutoff.

In the implemented domain u is a feasible multiple of 2^{-32}, $R = 2^{-5}$, and L and $2Rt$ are dyadic. With dyadic $m \le 4096$, the tail endpoints, midpoints, and weights used here have exact binary representations with fewer than 53 significant bits. To see this directly, write $u = k2^{-32}$ and $m = 2^q$, with $q \le 12$. The nonnegative tail lengths have integer numerators at most 2^{27} on that grid. Multiplying them by a grid fraction $j2^{-q}$ gives numerators at most 2^{39} on the $2^{-(32+q)}$ grid. Adding a tail endpoint and then taking a midpoint needs at most 41 significant binary digits. Dividing a tail length by $m\,2Rt$ only shifts its binary exponent, because $2Rt$ is either 2^{-5} or 2^{-6}. All these values remain in the normal binary64 exponent range. Thus the production predicate that a weight is positive is precisely a positive-length predicate. The three groups of rows with only a left tail, only a right tail, or both tails partition the admissible batch. Within each group, the retained endpoint columns belong to the same successive cells as the retained midpoint and weight columns. Grouping therefore leaves the preceding exact weighted objects unchanged. Floating summation orders can alter outward interval widths, but must still enclose those objects; bitwise equality of separately implemented certificates is not assumed.

The proposed cut remains certified by the original continuation. Let $H_a(v) = \int_v^{v+L} f_a(z)\,dz$. The original curvature bound implies that H_a is $m_z L$-

strongly convex. For the true $d = f_a(u + L) - f_a(u)$ and every feasible v,

$$H_a(v) \geq H_a(u) + d(v - u) + \frac{m_z L}{2}(v - u)^2.$$

Taking the infimum over the entire original interval I bounds $H_a(u) - \min_I H_a$. The negative of the quadratic minimum in (9) is convex as a function of d. Its value at any $d \in [d_-, d_+]$ is therefore no larger than the larger endpoint value. An outward lower bound on the constrained quadratic minimum, followed by the indicated upper operations, preserves the inequality. Finally,

$$\mathrm{AVaR}_\alpha(f_a) - R_u(f_a) = \frac{H_a(u) - \min_I H_a}{2Rt} \leq \widehat{\kappa}.$$

This derivation uses no floating sign and no assertion that the cut optimizes the original risk. It applies to either endpoint, an incorrect floating endpoint exit, and any imperfect interior cut.

Original policy and all-action validity. For a positive-length cell $B_i = [l_i, r_i]$ with midpoint z_i, write $y_i = F(x, a, z_i)$ and let its valid global affine witness be

$$V_1(y) \geq \ell_i + p_i^{\mathsf{T}}(y - y_i).$$

Integrating over the cell cancels the innovation term proportional to $z - z_i$. If $w_i = (r_i - l_i)/(2Rt)$, this proves $r_- := \sum_i w_i \ell_i \leq R_u(f_a)$. Convexity of f_a and the outward endpoint upper values prove $R_u(f_a) \leq r_+ := \sum_i w_i (u_i^- + u_i^+)/2$. The cut-defect result and the unchanged terminal controller then give

$$S(x) + h(a) + \beta r_- \leq J(a) \leq S(x) + h(a) + \beta\{r_+ + \widehat{\kappa} + \varepsilon_1\}.$$

The last term follows from monotonicity and cash invariance applied to $V_1 \leq J_1 \leq V_1 + \varepsilon_1$. No convexity of the implemented J_1 is needed.

For any other original feasible action v, use the same fixed density and the same global supports. The action-exposure vector b satisfies $F(x, v, z) = F(x, a, z) + b(v -$

a). Integration and $h'' \geq 2$ yield

$$Q(v) \geq W + s(v-a) + (v-a)^2, \qquad W = S(x) + h(a) + \beta r_-,$$
$$s = h'(a) + \beta \sum_i w_i b^\mathsf{T} p_i.$$

Taking the infimum over $[0, A(x)]$ gives a lower bound on $V_0(x)$. If s is interval-valued, the constrained minimum is concave in s, so the smaller minimum at its two enclosing endpoints remains a valid lower bound. This is the original all-action construction, with every arithmetic endpoint directed outward. The policy upper bound minus this optimum lower bound therefore encloses complete-policy regret, and the two policy endpoints enclose policy cost. The qualification test at either attempt preserves precisely the two stated original targets. No monotonicity of successive reported widths is required.

If both tests fail, the preliminary certificates do not overwrite the actor proposal, the economic primitives, the deployed future policy, or the frozen fallback. The continuation branch first constructs its 128-bin original certificate and then runs the unchanged repair and refinement sequence. Its own outward qualification is required before return. A finite prelude of terminating oracle calls therefore preserves the termination and qualification guarantee of the underlying complete service. A computation that fails an enclosure or qualification check must retain its failure evidence and must not return a successful scientific decision record.

All preliminary values are counted. The two floating endpoint differences request four terminal values per proposed action, including any repeated locations. Only rows without a proposed endpoint exit receive further queries. Six such updates request two values each, so their total is at most $4 + 6 \cdot 2 = 16$; a proposed endpoint exit uses four.

The outward defect requests exactly two values once for that state/action/risk identity. A resolution with m cells in each of p nonempty tails uses pm midpoint payloads and $p(m+1)$ endpoint payloads, hence $p(2m+1)$. The calculation allows repeated locations across resolutions, between a midpoint and an earlier endpoint, and between the defect and quadrature. All such repeated calls remain charged. A

first-resolution acceptance thus uses at most $2 + 5p$ outward payloads. If the second resolution is attempted, the complete preliminary total is

$$2 + p\{(2 \cdot 2 + 1) + (2 \cdot 4 + 1)\} = 2 + 14p \leq 30.$$

The cache is why the two defect values are not reacquired at the second resolution. Grouping rows according to their positive tails changes the number of vector calls, not these scalar-payload counts. An earlier failed call cannot enlarge the prescribed subsequent schedule; it terminates with a failed record instead of licensing an unrecorded retry.

Every call made by the frozen continuation is additional to this preliminary count. If its realized continuation transcript has N_F^{outward} outward and N_F^{floating} floating terminal payloads, the corresponding combined counts on a branch that invokes it are bounded by $30 + N_F^{\text{outward}}$ and $16 + N_F^{\text{floating}}$. Actor acquisition and all non-payload operations are outside these two counts and remain in the complete-work ledger. This proves the proposition without assuming a wall-clock cost per payload or an economic ordering of the returned actions. *Q.E.D.*

16.3. *Ablation and prospective scope*

The masked-tail ablation uses the same proposal, cache, exact zero-mass omission, original bounds, and fallback, while attempting all $m \in \{2, 4, 8, 16, 32, 64\}$. The preceding proof applies to each individual acceptance. Its full preliminary outward bound is

$$2 + p \sum_{m \in \{2,4,8,16,32,64\}} (2m + 1) = 2 + 258p \leq 518,$$

with the same floating bound of 16. Its additional attempted resolutions are not free, even if the later fallback is identical.

These two schedules were chosen after the separately preserved Phase A results and are therefore declared as Phase B algorithm development. The five-service comparison reacquires the common training information and executes its conventional controls and original-policy reference enclosures afresh under its own source freeze. Neither the mathematical validity of early acceptance nor the information cap es-

tablishes lower elapsed time, lower policy regret, a calibrated adoption decision, or neural specificity. Those claims require the corresponding current records. In particular, early acceptance may keep an actor proposal that another service repairs; its complete original-policy cost must be evaluated as that returned policy, with the unchanged future controller.

Packing the same original point information. At a fixed attempted resolution, index every positive-weight midpoint request and every endpoint request of a positive-length tail by its original row, column, and request type. Concatenating these indices defines a bijection to a packed list. A packed entry retains the original state x_i, action a_i, and innovation z_i and requests the same original payload at $F(x_i, a_i, z_i)$. Distinct requests at an identical location remain separate paid entries. The inverse indices return the lower values, endpoint upper values, and supporting slopes to precisely their original summands.

Conditional on a given valid collection of original payloads, this operation is only a reindexing of the selected-tail lower sum, secant upper sum, and all-action supporting slope. A zero-length tail has zero weights, so its omitted terms can be represented by zeros without asserting that the unobserved continuation equals zero. Termwise outward operations may enclose an exact zero by a nondegenerate interval; that adds width and does not invalidate the bound. The state/action transition is formed with a singleton innovation axis for each packed entry, preventing cross-products between one entry's state and another entry's shock. Thus the inverse scatter preserves the original probability mass and objective.

The compiler's batch-level fallback may supply different valid outward payloads after a change of batching. In that case the qualification proof applies to the actual returned collection; no bitwise equality of certificates or returned policies is assumed. Proposition A.3's 30/16 preliminary payload bounds are unchanged. The packed implementation makes one original compiler invocation per attempted resolution for its positive-weight midpoint and endpoint requests, in addition to the separately counted cached cut-defect values. Compiler-internal fallback, packing, scattering, zero-term interval arithmetic and the complete decision fallback remain charged

operations. Their effect on elapsed time requires the separately frozen current comparison.

17. COUPLED ACTION COMPARISONS AND SIGNED RECURSIVE GAINS

The common-density comparison below develops a mathematical obligation already frozen in the R111 research protocol. Its atom treatment and the computable active-tail defect complete that interface here. The separate R111 and R112 numerical campaigns retain their original identities and completion status. A common density exploits information shared by two actions; it does not confer a neural-specific capability.

17.1. *Tail witnesses compare the original policy costs*

Let X and Y be two bounded original continuation outcomes on the same innovation space and write $D = Y - X$. For a common convex cash-invariant risk with probability-law dual representation, put

$$k_X(q) = \rho(X) - \{\mathbb{E}_q X - \psi(q)\} \geq 0.$$

For any common feasible q, exact cancellation gives

$$\rho(Y) - \rho(X) = \mathbb{E}_q D + k_Y(q) - k_X(q). \tag{10}$$

If the original risk sets or penalties differ across actions, this particular common-density identity is unavailable unless both differences are included explicitly. A similar-looking objective under a different law is not a comparison in this theorem.

THEOREM S.5—Atom-safe coupled action certificate: *Suppose* $0 \leq k_X(q) \leq s_X$, $0 \leq k_Y(q) \leq s_Y$, *and* $d_- \leq \mathbb{E}_q D \leq d_+$ *are verified for a common feasible density. Then*

$$d_- - s_X \leq \rho(Y) - \rho(X) \leq d_+ + s_Y. \tag{11}$$

For the original AVaR, let η *be an* α*-quantile of* X, $t = 1 - \alpha > 0$, *and choose a measurable* $r \in [0,1]$ *equal to one on* $\{X > \eta\}$ *and zero on* $\{X < \eta\}$, *with* $\mathbb{E}r = t$. *Thus* $q_X = r/t$ *is a maximizing density, including a possible fractional selection on* $\{X = \eta\}$. *For any constant*

c and verified $\|D - c\|_\infty \le \delta$, the exact nonnegative remainder obeys

$$
\begin{aligned}
0 &\le \rho(X + D) - \rho(X) - \mathbb{E}_{q_X} D \\
&\le \frac{1}{t}\mathbb{E}[(X - \eta + D - c)_+ - (X - \eta)_+ - r(D - c)] \qquad (12) \\
&\le \frac{1}{t}\mathbb{E}(\delta - |X - \eta|)_+. \qquad (13)
\end{aligned}
$$

If $P(|X - \eta| \le z) \le Cz^\kappa$ for $0 < z \le \delta$, with $C > 0$ and $\kappa \ge 0$, the last bound is at most $C\delta^{\kappa+1}/\{t(\kappa + 1)\}$. Positive quantile atoms in general preclude an exponent $\kappa > 0$; no differentiability of an actor or risk quantile has been assumed.

For the original investment actions a, b, choose one original-law tail cut u and apply Theorem 8 to bound the two deficits by κ_a, κ_b. If $[d_-, d_+]$ encloses the integral of $f_b - f_a$ under that same density, their optimal-future contrast lies in

$$h(b) - h(a) + \beta[d_- - \kappa_a,\ d_+ + \kappa_b]. \qquad (14)$$

The implemented complete-policy contrast is enclosed after enlarging this interval on each side by $\beta\varepsilon_1$. For exactly the same realized action and same future policy the contrast is exactly zero. A strictly negative upper endpoint, together with the unchanged original qualification predicates, licenses a strict policy improvement from the completed incumbent.

The integral of $f_b - f_a$ must itself be enclosed. A difference of two convex functions need not be convex, so applying convex midpoint–secant rules directly to that difference is invalid. Valid alternatives include separately enclosed integrals, a verified modulus for their paired difference, or a separately implemented exact formula. All acquisitions needed to obtain the tighter paired enclosure belong to the service cost. The centered remainder in (12) completes the R111 hypothesis using the standard AVaR hinge representation (Rockafellar and Uryasev, 2002); it does not retroactively convert an unexecuted tail-aware fitting experiment into new evidence.

The cost difference between two complete policies has its own recursive equation. Let $v_t = J_t^\pi$, $D_t = J_t^\nu - J_t^\pi$, and

$$b_t = T_{\nu,t} v_{t+1} - v_t.$$

At each fixed state and action $\nu_t(x)$, the original risk set and penalty are the same for the two continuations being compared. Define $k_t^v(q)$ by centering that original risk at the successor continuation v_{t+1}. Then

$$D_t = b_t + \beta_t \sup_q \left\{ \mathbb{E}_q D_{t+1} - k_t^v(q) \right\}. \tag{15}$$

This is the signed policy-difference specialization of the retained reference-centered operator. It preserves the reference penalty instead of replacing it immediately by a worst-case physical or adverse law.

PROPOSITION S.6—Only near-maximizing reference tails can affect gain: *At a fixed original action, suppose a verified bracket implies $m \le D \le M$ and $w = M - m > 0$. The risk difference in (15) is unchanged when its supremum is restricted to $\{q : k^v(q) \le w\}$. No attainment assumption is required. If only a stored reference f is available and a verified bound gives* $\operatorname{osc}(v - f) \le r$, *it is sufficient to retain $\{q : k^f(q) \le w + r\}$. On this set, the penalties*

$$\max\{0, k^f(q) - r\} \le k^v(q) \le k^f(q) + r$$

give computable lower and upper risk-difference bounds when combined with valid continuation bounds. For $w = 0$ the difference is the known constant $D = m$, by cash invariance, and no zero-slack optimizer is presumed to exist.

Statewise bounds on b_t and on the continuation contrast can therefore be propagated backward using the retained penalties, with the declared terminal contrast as the boundary condition. This narrows the relevant dual laws when possible gains are small. A finite catalogue does not establish the required statewise premises, and this statement supplies no uncharged original-policy evaluation oracle. Constant

continuation allowances still compose by discounted sums; observed physical-state frequencies cannot replace the original conditional risk operators.

18. COUPLED ACTION AND SIGNED RECURSIVE COMPARISONS

18.1. *Proof of Theorem S.5*

PROOF: Adding and subtracting the same feasible dual value gives

$$\rho(Y)-\rho(X)=\mathbb{E}_q(Y-X)+[\rho(Y)-\mathbb{E}_qY+\psi(q)]-[\rho(X)-\mathbb{E}_qX+\psi(q)].$$

The bracketed quantities are $k_Y(q)$ and $k_X(q)$. Their nonnegativity and their verified upper bounds prove (11). Notice that a feasible witness alone has no zero-slack property; both slacks must be bounded or retained.

For AVaR, an α-quantile satisfies $P(X>\eta)\le t\le P(X\ge\eta)$, so an r with the stated properties exists. On a positive atom one may take a constant fraction; if the atom is null, the mass condition already holds. The identity

$$\mathbb{E}_{q_X}X=\eta+t^{-1}\mathbb{E}(X-\eta)_+$$

shows that q_X is a maximizing feasible density. Feasibility of q_X for $X+D$ therefore gives the nonnegative lower remainder. For its upper bound, use the admissible AVaR threshold $\eta+c$:

$$\rho(X+D)\le\eta+c+t^{-1}\mathbb{E}(X-\eta+D-c)_+.$$

Subtract the preceding exact expression for $\rho(X)$ and $\mathbb{E}_{q_X}D$. The constant c cancels because $\mathbb{E}r=t$, proving (12).

To check the pointwise bound put $s=X-\eta$ and $e=D-c$. If $s>0$, then $r=1$ and the hinge remainder is $(-s-e)_+$, which is at most $(\delta-s)_+$. If $s<0$, then $r=0$ and it is $(s+e)_+$, at most $(\delta+s)_+$. If $s=0$, it is e_+-re, a nonnegative quantity at most $|e|\le\delta$ for every $r\in[0,1]$. These three cases prove (13), including atoms. The layer-cake identity gives

$$\mathbb{E}(\delta-|X-\eta|)_+=\int_0^\delta P(|X-\eta|<z)\,dz\le\frac{C\delta^{\kappa+1}}{\kappa+1}.$$

A quantile atom of positive probability cannot obey the displayed margin bound with $\kappa > 0$ as $z \downarrow 0$.

For investment, the shared u is feasible for both actions because the original innovation law and AVaR cap do not depend on the action. The computed tail defects are precisely upper bounds for its two AVaR optimality slacks. Add the original stage-cost difference and discount (11) to obtain (14). Write the implemented objective as $J(a) = Q(a) + \eta_a$ with $0 \le \eta_a \le \beta\varepsilon_1$. Then $-\beta\varepsilon_1 \le \eta_b - \eta_a \le \beta\varepsilon_1$, which proves the complete-policy enlargement. It is $\beta\varepsilon_1$ on each side, not $2\beta\varepsilon_1$: the error is one-sided for both policies. When $a = b$, the same complete policy is being subtracted from itself, and its contrast is exactly zero despite nonzero numerical widths. A negative upper bound proves strict descent, while the independent qualification inequalities preserve the original regret and width targets. Refusing a candidate preserves the completed incumbent and does not erase the cost of its attempted evaluation. *Q.E.D.*

18.2. *Proof of Proposition S.6*

PROOF: Subtract the two complete-policy Bellman equations, insert the original old-policy continuation at the new action, and use the exact reference-centered identity. This gives (15) without exchanging an action infimum and a risk supremum.

At a fixed action, the risk dual has $k^v(q) \ge 0$ and $\inf_q k^v(q) = 0$. For $D \in [m, M]$, every density with $k^v(q) > w = M - m$ has

$$\mathbb{E}_q D - k^v(q) \le M - k^v(q) < m.$$

On the other hand, densities with slacks tending to zero lie eventually in $\{k^v \le w\}$ and give supremum at least m, because their expectations of D are at least m. Discarding the excluded densities therefore does not change the supremum. This argument uses approximating maximizing densities, not an attained maximum.

Set $e = v - f$. Monotonicity and cash invariance imply $\inf e \le \rho(v) - \rho(f) \le \sup e$. Since every dual law is a probability law, its integral of e lies in the same interval. Therefore

$$|k^v(q) - k^f(q)| = |\rho(v) - \rho(f) - \mathbb{E}_q e| \le \operatorname{osc}(e) \le r.$$

In particular, $k^f(q) > w + r$ implies $k^v(q) > w$, so the larger computable retained set still contains every density needed for the exact supremum. Combining this inequality with the nonnegativity of k^v proves the penalty enclosures. For any verified $L \le D \le U$, they imply

$$\rho(v + D) - \rho(v) \ge \sup_{q:\, k^f(q) \le w+r} \{\mathbb{E}_q L - k^f(q) - r\},$$

$$\rho(v + D) - \rho(v) \le \sup_{q:\, k^f(q) \le w+r} \{\mathbb{E}_q U - \max(0, k^f(q) - r)\}.$$

Both sides use the same original feasible set, with its appropriate statewise cut. For a fixed retained kernel set and fixed penalty enclosures, the bounds are monotone in L, U and nonexpansive in their uniform norm. Selecting that set from a previously verified contrast bracket is a separate construction; no nonexpansiveness of the selection map is asserted. Starting from a valid terminal contrast, applying these inequalities backward with valid signed residual bounds proves the finite-horizon account. When $w = 0$, $D = m$ is constant; cash invariance gives risk difference m exactly and avoids an empty zero-slack set when the original supremum is unattained. A loose bracket with positive width can also be used in that case, at the price of a weaker bound. *Q.E.D.*

19. INDEPENDENT FINITE ENCLOSURES FOR THE ORIGINAL POLICY CATALOGUE

This section supplies the mathematical justification for validation D. It is a finite verification procedure for the original investment objective, using classical convexity and the AVaR threshold representation (Rockafellar and Uryasev, 2002, Boyd and Vandenberghe, 2004). It is not a new optimization theorem, an independent population experiment, or a claim of neural exclusivity. The implementation reconstructs the original economic primitives separately from the production terminal compiler. Its mathematical guarantee still uses the original domain, continuation-convexity, and implemented-terminal-policy bounds; software independence does not remove these assumptions.

19.1. *Original objective and precise validation statement*

Fix $d = 2$, $X = [0,1]^2$, $R = 1/32$, $\beta = 15/16$, and $\alpha \in \{1/2, 3/4\}$. Put $t = 1 - \alpha$ and let P be the uniform law on $[-R, R]$. Indices below are cyclic. The original primitives are

$$A(x) = \frac{1}{8} + \frac{x_1 + x_2}{16}, \qquad h(a) = a^2 + 4a^4,$$

$$F_j(x,a,z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1 - x_{j+1})}{16} + b_j a + s_j z,$$

$$b = (1/2, 1/4), \quad s = (1, -1),$$

$$S(x) = \sum_{j=1}^{2} (x_j - 5/8)^2 + \frac{1}{8} \sum_{j=1}^{2} (x_j - x_{j+1})^2 + 2(1/2 - x_1 - x_2)_+^2,$$

$$S_T(x) = 2 \sum_{j=1}^{2} (x_j - 5/8)^2 + \frac{1}{8} \sum_{j=1}^{2} (x_j - x_{j+1})^2 + 2(1/2 - x_1 - x_2)_+^2.$$

For bounded w, write

$$\rho_\alpha(w) = \inf_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{t} \mathbb{E}_P (w - \eta)_+ \right\}.$$

The last-date optimal value and the two root fixed-action objectives are

$$G_y(v) = S(y) + h(v) + \beta \rho_\alpha(S_T(F(y, v, \cdot))),$$

$$V_1(y) = \min_{0 \le v \le A(y)} G_y(v),$$

$$Q_x(a) = S(x) + h(a) + \beta \rho_\alpha(V_1(F(x, a, \cdot))),$$

$$J_x(a) = S(x) + h(a) + \beta \rho_\alpha(J_1(F(x, a, \cdot))).$$

Here J_1 is the unchanged implemented terminal policy, not a newly optimized or independently reconstructed controller. The original results in Proposition S.3 estab-

lish domain invariance, existence, and strong state convexity of V_1 with modulus $(4-5\beta/4)/2$. The implemented terminal guarantee is

$$V_1(y) \le J_1(y) \le V_1(y) + \varepsilon_1, \qquad y \in X, \qquad \varepsilon_1 = 2^{-20}. \tag{16}$$

The validation below relies on this proved uniform guarantee and makes no convexity assumption about J_1.

PROPOSITION S.7—Finite original-policy validation enclosure: *Fix an original state $x \in X$ and feasible action $a \in [0, A(x)]$. Let $M \ge 1$ be an integer and put*

$$z_k = -R + \frac{Rk}{M}, \qquad y_k = F(x, a, z_k), \qquad k = 0, \dots, 2M.$$

At each y_k, choose any feasible v_k and an exactly maximizing AVaR density q_k for $S_T(F(y_k, v_k, \cdot))$. Define

$$\begin{aligned} g_k &= h'(v_k) + \beta \mathbb{E}_P[q_k \, \partial_v S_T(F(y_k, v_k, Z))], \\ \delta_k &= \operatorname{proj}_{[-v_k, A(y_k)-v_k]}(-g_k/2), \\ \ell_k &= G_{y_k}(v_k) + g_k \delta_k + \delta_k^2, \qquad\qquad u_k = G_{y_k}(v_k). \end{aligned}$$

These are valid bounds $\ell_k \le V_1(y_k) \le u_k$, including boundary actions. For an n-vector w, let $\rho_\alpha^{[n]}(w)$ denote AVaR under equal atom weights $1/n$. Form the vectors

$$\ell^{\mathrm{mid}} = (\ell_1, \ell_3, \dots, \ell_{2M-1}), \qquad u^{\mathrm{end}} = (u_0, u_2, u_2, u_4, \dots, u_{2M-2}, u_{2M}).$$

Thus each cell contributes both endpoints to u^{end}; its interior endpoints occur twice. With $c = S(x) + h(a)$, define

$$L_Q = c + \beta \rho_\alpha^{[M]}(\ell^{\mathrm{mid}}), \qquad U_Q = c + \beta \rho_\alpha^{[2M]}(u^{\mathrm{end}}). \tag{17}$$

Then the original objectives satisfy

$$L_Q \le Q_x(a) \le U_Q, \qquad L_Q \le J_x(a) \le U_Q + \beta\varepsilon_1. \tag{18}$$

Every quantity in the construction is rational when the original state, action, and primitive-node actions are rational. A finite terminal search affects the widths, not the validity, of these inequalities.

19.2. *Proof: exact original primitives and boundary-safe node bounds*

PROOF: We first verify that all evaluations concern the original feasible economy. For $p, q \in [0,1]$, the state part $p/2 + q/8 + p(1-q)/16$ is increasing in both arguments and lies in $[0, 5/8]$. Since $A(x) \in [1/8, 1/4]$, $0 \le b_j a \le 1/8$, and $|s_j z| \le 1/32$, every successor belongs to $[1/32, 27/32]^2 \subset X$. This applies both to a root action and to each feasible terminal action. The continuous terminal objective attains its minimum on the compact feasible interval.

Fix a terminal state y and feasible v. Write $F(y, v, z) = (w_1 + z, w_2 - z)$. The aggregate-state squared hinge in S_T is constant in z because the two shock loadings sum to zero. The first two squared deviations contribute $4z^2$ and the cyclic coupling term contributes z^2. Consequently

$$p(z) := S_T(F(y, v, z)) = p_0 + p_1 z + 5z^2.$$

This is a primitive identity. Its evaluation at $-R, 0, R$ determines the polynomial exactly; three-point interpolation would not identify a general nonlinear function. At rational inputs, all three coefficients are rational.

Here is an explicit exact maximizing density. Set $L = 2R\alpha$ and

$$u = \mathrm{proj}_{[-R, R-L]}\left(-\frac{p_1}{2p_2} - \frac{L}{2}\right), \quad p_2 = 5, \qquad q(z) = t^{-1}\mathbf{1}_{[-R,u] \cup [u+L,R]}(z).$$

The central interval has length L, so $0 \le q \le t^{-1}$ and $\mathbb{E}_P q = 1$. To verify maximality directly, let $\eta = \max\{p(u), p(u+L)\}$. An interior cut is centered on the quadratic's vertex, so its endpoint values coincide and $p \le \eta$ inside the central interval and $p \ge \eta$ on its two tails. At the left boundary the clipped vertex lies weakly to the left of the central midpoint, $p(u+L) \ge p(u)$, and the only nonempty tail is to the right of $u + L$; it again has $p \ge \eta$. The right-boundary case is symmetric. Thus, up to endpoints of probability zero,

$$\eta + t^{-1}\mathbb{E}_P(p - \eta)_+ = \mathbb{E}_P(qp).$$

For any feasible density $\widetilde{q}$ and any threshold e,

$$\mathbb{E}_P(\widetilde{q}p) = e + \mathbb{E}_P[\widetilde{q}(p - e)] \le e + t^{-1}\mathbb{E}_P(p - e)_+.$$

Taking an infimum in e proves weak duality; the displayed equality proves that q attains the original AVaR. Integrating the quadratic over its two rational tail intervals gives the exact rational risk. This proof includes clipped cuts and does not differentiate a moving cut.

The function S_T is convex and continuously differentiable, including at the squared-hinge junction. For any other feasible terminal action w, its supporting inequality along the affine action direction gives

$$S_T(F(y,w,z)) \geq S_T(F(y,v,z)) + \partial_v S_T(F(y,v,z))(w-v).$$

Use the same fixed feasible density q at every w. Weak duality and its equality at v, followed by integration of this support, show

$$\rho_\alpha(S_T(F(y,w,\cdot))) \geq \rho_\alpha(S_T(F(y,v,\cdot))) + \mathbb{E}_P[q\,\partial_v S_T(F(y,v,Z))](w-v).$$

Since $h'' \geq 2$, adding the original stage term proves the global action minorant

$$G_y(w) \geq G_y(v) + g(w-v) + (w-v)^2, \qquad 0 \leq w \leq A(y). \tag{19}$$

The action derivative inside the expectation is affine in z: the quadratic parts of S_T have affine gradients, while the squared-hinge gradient is constant in z along the zero-aggregate shock. Evaluations at $0, R$ therefore reconstruct this affine derivative exactly. No derivative of the optimal tail endpoints is used.

The minimum of $g\delta + \delta^2$ over the full feasible displacement interval is attained at $\delta = \mathrm{proj}_{[-v,A(y)-v]}(-g/2)$. Minimizing the minorant gives the stated lower bound; evaluating the feasible action v gives the upper bound. This reasoning applies if $v = 0$, $v = A(y)$, or an imperfect interior search iterate. In particular, a nonnegative support slope at zero certifies that endpoint and a nonpositive support slope at capacity certifies that endpoint. At an interior point, a negative support slope rules out a minimizer to its left, and a positive support slope rules out one to its right. Thus rational bisection is legitimate, but the support bound remains sound after any finite number of steps. The separately recorded check $u_k - \ell_k \leq 2^{-20}$ is a width check; it is not the justification of the lower bound.

The continuous-risk outer enclosure. The original strong state convexity of V_1 implies convexity of $f(z) = V_1(F(x,a,z))$, because the root transition is affine in z. For every real threshold η, the function $\phi_\eta(z) = (f(z) - \eta)_+$ is convex. On the cell $[z_{2j}, z_{2j+2}]$, Jensen's inequality and convex secant domination give, with cell length $\Delta = 2R/M$,

$$(\ell_{2j+1} - \eta)_+ \le \phi_\eta(z_{2j+1}) \le \frac{1}{\Delta}\int_{z_{2j}}^{z_{2j+2}} \phi_\eta(z)\,dz,$$

$$\frac{1}{\Delta}\int_{z_{2j}}^{z_{2j+2}} \phi_\eta(z)\,dz \le \frac{\phi_\eta(z_{2j}) + \phi_\eta(z_{2j+2})}{2} \le \frac{(u_{2j} - \eta)_+ + (u_{2j+2} - \eta)_+}{2}.$$

Summing cells with weights $1/M$ and adding the threshold yields

$$\eta + \frac{1}{Mt}\sum_{j=0}^{M-1} (\ell_{2j+1} - \eta)_+ \le \eta + t^{-1}\mathbb{E}_P(f - \eta)_+,$$

$$\eta + t^{-1}\mathbb{E}_P(f - \eta)_+ \le \eta + \frac{1}{2Mt}\sum_{j=0}^{M-1} \{(u_{2j} - \eta)_+ + (u_{2j+2} - \eta)_+\}.$$

These hold at every common threshold. Taking infima over all real thresholds preserves each inequality and proves

$$\rho_\alpha^{[M]}(\ell^{\mathrm{mid}}) \le \rho_\alpha(f) \le \rho_\alpha^{[2M]}(u^{\mathrm{end}}).$$

Neither the lower nor the upper node array has to be convex. The required convexity is that of the original optimal continuation along the innovation. A bound only on the ordinary integral of f would not prove this risk enclosure. Adding c and multiplying by the positive β gives (17).

Finally, the threshold formula is monotone in its random argument, and replacing its threshold by $\eta - c_0$ proves $\rho_\alpha(w + c_0) = \rho_\alpha(w) + c_0$ for every constant c_0. Apply these facts to (16) at the original successors to obtain

$$Q_x(a) \le J_x(a) \le Q_x(a) + \beta\varepsilon_1.$$

Combining with the Q bounds proves (18). The node-optimization uncertainty is already contained in ℓ_k, u_k. The distinct implemented-future allowance $\beta\varepsilon_1$ is added exactly once at the root. No convexity or differentiability of the rounded deployed future controller is asserted. *Q.E.D.*

19.3. *Exact atom handling and finite policy comparisons*

For completeness, the equal-mass risk in the proposition can be computed without floating cutoffs. Sort $w_{(1)} \geq \cdots \geq w_{(n)}$, put $r = n(1-\alpha) > 0$, $k = \lfloor r \rfloor$, and $\theta = r - k$. Then

$$\rho_\alpha^{[n]}(w) = \frac{\sum_{i=1}^{k} w_{(i)} + \theta w_{(k+1)}}{r}, \tag{20}$$

where the final term is omitted if $\theta = 0$, including $k = n$. Indeed, assign masses $\lambda_i \in [0,1]$ with $\sum_i \lambda_i = r$. The threshold inequality in the proof gives $r^{-1} \sum_i \lambda_i w_i \leq \rho_\alpha^{[n]}(w)$. The assignment of unit mass to the k largest values and mass θ to the next value attains the threshold expression at the cutoff value. This proves the formula, including ties and split atoms. At $\alpha = 0$ it reduces to the ordinary mean; $\alpha = 1$ is outside this definition. At the two frozen risks and the frozen $M = 64$, both atom counts are integral. The implementation nevertheless uses exact rational masses and preserves the repeated endpoints.

Suppose a fixed catalogue contains n_0 original states and each complete policy r has a valid interval $[L_{jr}, U_{jr}]$ at state j. Summing its pointwise inequalities gives the deterministic mean bound

$$\frac{1}{n_0} \sum_j L_{jr} \leq \frac{1}{n_0} \sum_j J_{jr} \leq \frac{1}{n_0} \sum_j U_{jr}.$$

For two policies r, s, use the interval $[L_{jr} - U_{js}, U_{jr} - L_{js}]$ for their difference at state j. If they have the same exact action, the same original state and risk, and the identical future controller, the two original defining objectives are equal, and that state's difference is exactly zero. Replacing its subtraction interval by $[0,0]$ is then an identity, not an independence assumption. Summing these statewise difference intervals proves the reported mean-contrast interval. A strictly negative upper endpoint certifies a lower cost for the left policy; a strictly positive lower endpoint certifies a higher cost. An interval containing zero remains unresolved, except that $[0,0]$ certifies equality. No statistical or population inference follows from these deterministic sums.

In D, exact stored binary64 identities are converted to their precise rational values. The input binding checks the common state across every action and service and the unchanged future-controller identity. The frozen catalogue has 144 states per risk, 983 distinct state/action/risk keys, and 13,824 primary policy records. The prescribed output includes all 96 arm/service/risk means, all 56 unordered arm contrasts with the fixed compiled-service labels, and all 240 within-arm service-pair identity cells. These are declared coverage requirements, not assertions that execution succeeded. No favorable sign is a pass condition.

19.4. *What interval compatibility does and does not establish*

Let I be an independently justified interval for an original policy value and P_0 a recorded service or fine-reference interval. If they are disjoint, both cannot enclose the same value: at least one enclosure argument, arithmetic step, input identity, or model premise requires investigation. The case must be retained. If they intersect, their intersection alone does not establish correctness of P_0; for example, $I = [0,2]$ and $P_0 = [1,3]$ overlap although a true value $1/2$ lies outside P_0. The stronger inclusion $I \subseteq P_0$ would transfer coverage from I to P_0, conditional on validity of I. The reverse inclusion does not. D records overlap and both inclusion directions separately and uses overlap only as a compatibility diagnostic. It does not thereby verify every endpoint of the old 1,024-cell reference, which shares the production terminal compiler.

The lower bound L_Q in (17) concerns the specified action. It does not by itself provide a valid lower bound on $\inf_a Q_x(a)$ and therefore creates no new global regret or action-distance certificate. The procedure leaves all original A/B/C policies, observations, timing records, negative results, and qualification proofs unchanged. Its full actual work is additional author-side validation cost. The subsequent execution receipt, rather than this mathematical proposition, determines whether every fixed case was actually completed and passed.

20. ORIGINAL-POLICY ACTION COMPARISONS AND THEIR INFORMATION COST

This section proves the R120 comparison construction under the original investment primitives. The completed-incumbent safeguard, three projected support

proposals, and original all-action witnesses retain their R108–R110 ownership. The common-risk identity retains its R111/R119 ownership. The new ingredients proved here are an original AVaR comparison modulus, the associated mixed integration interface and its restricted information bound, and quantitative control of common-tail mismatch. The statistical transfer and strong-convexity safeguards use classical arguments and are proved to make their assumptions explicit. No statement supplies a neural representation separation.

20.1. *Proof of the original comparison modulus*

PROOF OF THEOREM 9: Throughout this proof $d = 2$. Let $\boldsymbol{b} = (1/2, 1/4)^\top$ and $\boldsymbol{s} = (1, -1)^\top$. For a terminal state y, normalize its feasible action by $a = A(y)v$, where $v \in [0,1]$ is held fixed when differentiating in y. Put

$$m(y,v) = F(y, A(y)v, 0), \qquad k(y,v) = 5\{m_1(y,v) - m_2(y,v)\}.$$

The original domain argument in Proposition S.3 puts m, and every $m + \boldsymbol{s}z$ used here, in the state cube. The zero-sum shock leaves the aggregate squared hinge unchanged. Direct expansion of the original terminal cost therefore gives

$$g(m + \boldsymbol{s}z) = g(m) + kz + 5z^2.$$

By the exact original tail formula in Theorem A.1, the normalized last-date objective is

$$\widetilde{Q}(y,v) = S(y) + h(A(y)v) + \beta\{g(m(y,v)) + \mathcal{R}_\alpha(k(y,v))\}.$$

The function $\mathcal{R}_\alpha$ is continuously differentiable and

$$\mathcal{R}'_\alpha(k) = \operatorname{clip}\left(\frac{\alpha k}{10t}, -\alpha R, \alpha R\right).$$

Its derivative has Lipschitz constant $\alpha/(10t)$, including the two clipping junctions.

We establish a uniform upper state-curvature bound for $\widetilde{Q}(\cdot, v)$. For $d = 2$, the target, cyclic, and squared-hinge terms in S have Hessian operator norms bounded by 2, 1, and 8, respectively. Thus $\nabla^2 S \preceq 11I$ wherever the second derivative is defined.

The same decomposition for g gives $0 \preceq \nabla^2 g \preceq 13I$. Its gradient coordinates satisfy

$$|\partial_j g(y)| \le 4|y_j - 5/8| + \tfrac{1}{2}|y_j - y_{j+1}| + 4(1/2 - y_1 - y_2)_+ \le 5.$$

The state Jacobian of m is nonnegative on the cube. Its row sums are at most $3/4$, and its column sums at most $47/64$. Indeed the unnormalized transition has both sums at most $11/16$; the term $\boldsymbol{b}v\nabla A^\top$ adds at most $1/16$ to a row and $3/64$ to a column. Hence

$$\|D_y m\|_2 \le \sqrt{(3/4)(47/64)} \le 3/4.$$

The only second state derivatives of m_j are the mixed derivatives $-1/16$ of $-y_1y_2/16$; the normalized capacity term is affine. In a state direction w,

$$\left|\sum_{j=1}^{2} \partial_j g(m)\, D_y^2 m_j[w,w]\right| \le \frac{10}{8}|w_1 w_2| \le \frac{5}{8}\|w\|_2^2.$$

The chain rule consequently gives the uniform upper bound

$$D_y^2(g \circ m)[w,w] \le \left\{13(3/4)^2 + 5/8\right\}\|w\|_2^2 = \frac{127}{16}\|w\|_2^2.$$

Also $h''(a) = 2 + 48a^2 \le 5$ on $[0, 1/4]$ and $\|\nabla A\|_2^2 = 1/128$. The stage upper bound is therefore $11 + 5/128 = 1413/128$.

For the risk-motion term, the bilinear transition terms cancel in the difference of the two coordinates:

$$k(y,v) = \frac{35}{16}(y_1 - y_2) + \frac{5}{4}A(y)v.$$

This is affine in y for fixed v, and

$$\nabla_y k = (35/16 + 5v/64,\ -35/16 + 5v/64)^\top,$$

$$\|\nabla_y k\|_2^2 = 2(35/16)^2 + 2(5v/64)^2 \le \frac{19625}{2048}.$$

It follows that the upper Hessian contribution of $\mathcal{R}_\alpha(k(y,v))$ is at most $19625\alpha/(20480t)$. Combining these estimates gives $D_y^2\widetilde{Q}[w,w] \le M_\alpha\|w\|_2^2$. Squared hinges and the tail formula have matching first derivatives at their junctions. Their

restrictions to a line have absolutely continuous first derivatives, so integrating this almost-everywhere second-derivative bound along a segment proves that

$$y \longmapsto \widetilde{Q}(y, v) - \tfrac{1}{2} M_\alpha \|y\|_2^2$$

is concave on the cube, uniformly over $v \in [0, 1]$.

An infimum of concave functions is concave: its value at a convex combination is at least the same combination of the two infima. Since the normalized feasible interval is fixed, this proves semiconcavity of $V_1(y) - M_\alpha \|y\|_2^2/2$. The retained original joint curvature and the affine capacity graph give convexity and a unique continuous minimizing $v(y)$. For clarity, continuous differentiability does not require differentiability of that minimizer. Evaluate a value increment at the old minimizer for one inequality and at the new minimizer for the opposite inequality. Uniform first-order expansion of the C^1 objective on the compact normalized interval and continuity of $v(y)$ give $\nabla V_1(y) = \partial_y \widetilde{Q}(y, v(y))$ in the interior. This formula extends continuously to the boundary, and incorporates the capacity term already displayed in (8).

Convexity together with M_α-semiconcavity now implies the claimed gradient modulus. One explicit justification is to mollify V_1 on any smaller interior cube. Convolution preserves convexity and semiconcavity, so the smooth Hessian has every eigenvalue between 0 and M_α. Integrating the Hessian along a segment bounds the difference of the mollified gradients by M_α times its length. The gradients converge locally uniformly because V_1 is C^1. Pass to the limit and then use continuity at the boundary.

Finally the two successor points at a fixed innovation differ by $\boldsymbol{b}(b - a)$. Differentiation in the innovation gives

$$\begin{aligned} \left| \frac{d}{dz} \{ f_b(z) - f_a(z) \} \right| &\le \|\boldsymbol{s}\|_2 \|\nabla V_1(F(x, b, z)) - \nabla V_1(F(x, a, z))\|_2 \\ &\le M_\alpha \|\boldsymbol{s}\|_2 \|\boldsymbol{b}\|_2 |b - a| \\ &= \frac{\sqrt{10}}{4} M_\alpha |b - a| \le \frac{4}{5} M_\alpha |b - a|. \end{aligned}$$

All points and joining segments are in the invariant convex cube. Integration in z proves (41). *Q.E.D.*

20.2. *Valid paired integration and the policy contrast*

PROOF OF THEOREM 10: Write $D = f_b - f_a$. A Lipschitz function of constant E_{ab} satisfies on a cell with midpoint z_i and length h_i

$$\left| \int_{B_i} D(z)\, dz - h_i D(z_i) \right| \le E_{ab} \int_{B_i} |z - z_i|\, dz = E_{ab} h_i^2 / 4.$$

Summing and dividing by $B = 2Rt$ proves the second interval in (43), including arbitrary valid midpoint value errors. Separately, convexity gives for each f_a

$$h_i f_a(z_i) \le \int_{B_i} f_a(z)\, dz \le \frac{h_i}{2} \{ f_a(l_i) + f_a(r_i) \}.$$

Valid lower midpoint and upper endpoint values thus give $L_a \le \mathbb{E}_P q_u f_a \le U_a$, and likewise for b. Subtracting these two valid intervals proves the first interval. Both contain the same integral, so their exact intersection is nonempty and valid. Empty outward intersections signal a failed numerical construction, not a license to choose a favorable endpoint. Zero-length tails contribute zero measure and are omitted.

For completeness, each exact convex midpoint–secant width is bounded by KRt/n^2 under $h_i \le B/n$. On a cell its unnormalized gap is $h_i^2/4$ times the difference of the right and left half-cell secant slopes. These differences are nonnegative; ordering all cells, including across the omitted center, bounds their sum by $2K$. Thus the normalized width is at most $K h_{\max}^2/(2B) \le KRt/n^2$. Two such widths sum to $2KRt/n^2$. The paired Lipschitz interval has width

$$\frac{E_{ab}}{2B} \sum_i h_i^2 \le E_{ab} h_{\max}/2 \le E_{ab} Rt/n.$$

Intersection proves (45). The midpoint–secant inequality is applied only to each convex continuation, never to D.

For the original continuations, Theorem 9 also gives a Lipschitz constant $2M_\alpha$ for f_a': the innovation direction has squared norm two. The two Taylor upper bounds

at the endpoints of a cell therefore imply

$$\frac{f_a(l_i) + f_a(r_i)}{2} - f_a(z_i) \le \frac{2M_\alpha h_i^2}{8}.$$

The opposite linear terms cancel. Multiply by h_i/B and sum over the cells. The midpoint–secant width for one function is at most $M_\alpha h_{\max}^2/4$, and the difference of the two separately enclosed integrals has width at most $M_\alpha h_{\max}^2/2 \le 2M_\alpha R^2 t^2/n^2$. Intersecting this analytic bound with the previous ones proves (46). The numerical algorithm already computes these same midpoint and endpoint values; this consequence changes neither its inputs nor its prescribed decisions.

For the same feasible original density, define $k_a = \rho_\alpha(f_a) - \mathbb{E}_P q_u f_a \ge 0$ and similarly k_b. Direct cancellation yields

$$\rho_\alpha(f_b) - \rho_\alpha(f_a) = \mathbb{E}_P q_u D + k_b - k_a.$$

Consequently the optimal-future root contrast lies in $h(b) - h(a) + \beta[d_- - \kappa_a, d_+ + \kappa_b]$. By (39), the difference of the two actual future allowances lies in $[-\delta, \delta]$, not $[-2\delta, 2\delta]$. Adding this interval proves (44). This inference uses the same implemented future policy at both actions and makes no convexity assertion about its value. When the two realized actions agree, their complete-policy values and their continuation functions agree exactly. Their difference is zero before any interval widening or new acquisition. *Q.E.D.*

Explicit node-precision allowance. Suppose every acquired scalar value interval has width at most ξ and the sums are computed exactly. The paired midpoint interval gains width at most 2ξ from its node intervals. The separate convex difference gains width at most 4ξ. Thus a valid bound replacing (45) is

$$\min\left\{\frac{2KRt}{n^2} + 4\xi, \frac{ERt}{n} + 2\xi\right\}. \tag{21}$$

To prove this, write each midpoint lower error and each endpoint upper error separately. Their nonnegative normalized weights sum to one, so each such weighted error is at most ξ. For the paired midpoint difference the sum of the two interval widths is at most 2ξ. The remaining terms are the exact quadrature gaps just proved.

Outward accumulation adds its separately bounded arithmetic enlargement. In an implementation, the directly retained interval endpoints remain authoritative; neither ξ nor the future-policy allowance is silently sent to zero.

20.3. *A matching bound for the fixed-tail information interface*

PROPOSITION S.8—Paid paired point information: *Fix $R, K > 0$, $t = 1 - \alpha > 0$, $\alpha \in [0, 1)$, and $E \geq 0$. The unknown pair consists of convex K-Lipschitz functions on $[-R, R]$ whose difference is E-Lipschitz. A known feasible two-tail density and these class restrictions are supplied. The information oracle returns an exact value of one chosen function at one chosen point. A reported interval must contain the fixed-density integral of the difference for every pair consistent with its transcript.*

For every integer $n \geq 1$, there is a nonadaptive quadrature algorithm using at most $4n + 8$ scalar values whose width is at most $\min\{2KRt/n^2, ERt/n\}$. Conversely, for the known one-sided density $q = t^{-1}\mathbf{1}_{[-R+2R\alpha, R]}$, every deterministic adaptive algorithm using at most N scalar values has an input requiring width at least (47). For $N \geq 32$ the upper width can in particular be bounded by

$$\min\{128KRt/N^2,\ 8ERt/N\}.$$

These are interval-width bounds, not point-estimation error bounds.

PROOF: For the upper bound, divide a nonempty tail of length b_j into $\lceil nb_j/(2Rt) \rceil$ equal cells. The two tail lengths sum to $2Rt$, so at most $n + 1$ nonempty cells suffice. Each function requires at most $2n + 4$ midpoint and distinct endpoint values. Paying for both functions gives $4n + 8$; Theorem 10 proves its width. At $\alpha = 0$ this is just integration over the whole interval and requires no tail location. If $N \geq 32$, take $n = \lfloor (N - 8)/4 \rfloor \geq N/8$. The displayed upper bound follows.

For the lower bound put $q_0 = -R + 2R\alpha$ and $B = R - q_0 = 2Rt$. Run the proposed algorithm on the pair (f_0, f_0), where

$$f_0(z) = \frac{K}{2B}(z - q_0)_+^2.$$

It is convex, nondecreasing, and K-Lipschitz. Partition its right tail into $2(N + 1)$ equal cells of length $h = B/\{2(N + 1)\}$. At most N cells contain a queried location

in their interior, irrespective of which function label was queried. Choose $N+1$ cells containing no such point. On each selected cell replace f_0 by the convex combination

$$f_1 = (1-\lambda) f_0 + \lambda\,(\text{chord of } f_0), \qquad \lambda = \min\{1, 4E(N+1)/K\},$$

and put $f_1 = f_0$ elsewhere. Every queried value remains unchanged, including queries at selected cell endpoints. Induction on the adaptive transcript therefore gives the same queries and answers on (f_0, f_1).

The chord slope is between the two endpoint derivatives of f_0. At entry to a selected cell the derivative of f_1 jumps upwards by $\lambda Kh/(2B)$; at its exit it jumps upwards by the same amount. Interior derivatives are nondecreasing, and all slopes lie in $[0, K]$. Adjacent selected cells have the same ordering. Hence f_1 is convex and K-Lipschitz. On a selected cell

$$|(f_1 - f_0)'| \le \lambda Kh/(2B) = \lambda K/\{4(N+1)\} \le E.$$

The difference is continuous and zero at every selected endpoint and elsewhere, so it is globally E-Lipschitz. The construction thus preserves every information restriction.

The integral of the chord excess of a quadratic with second derivative K/B on a cell of length h is $Kh^3/(12B)$. Summing $N+1$ selected cells and dividing by B gives

$$\begin{aligned} \mathbb{E}_{Pq}(f_1 - f_0) &\ge \frac{\lambda KB}{96(N+1)^2} \\ &= \min\left\{\frac{KRt}{48(N+1)^2}, \frac{ERt}{12(N+1)}\right\}. \end{aligned}$$

The first pair has contrast zero. A sound interval for their common transcript must contain both contrasts and hence has at least this width. Both functions are nondecreasing, so the known right tail is also exactly maximizing for both AVaRs. This supplies a subclass with no localization slack; it does not eliminate localization costs from the upper algorithm on a general pair. *Q.E.D.*

The proof uses the standard indistinguishable-input approach to numerical information (Novak, 2015). Its upper bound is conditional on a known fixed tail, and

its lower bound allows arbitrary convex functions with the stated moduli. It does not claim that the original finite-parameter investment primitives realize this entire adversarial class. Formula access, derivative access, additional separate smoothness restrictions, a shared exact symbolic representation, and free terminal solves are different information conventions. Providing one of them changes the comparison for every method, including splines, polynomials, hinges, and neural networks. In particular (46) uses an additional proved property of the original continuation. No matching lower bound for that full original-primitives information class is inferred from the nonsmooth construction above.

20.4. *Transferring the defect of a common tail*

THEOREM S.9—Action-scaled transfer between constrained tail witnesses: *Let* $0 < \alpha < 1$, $L = 2R\alpha$, $B = 2Rt$, *and* $I = [-R, R - L]$. *Suppose* f_a, f_b *are* m_z*-strongly convex on the innovation interval, with* $m_z > 0$, *and their difference is E-Lipschitz. Define* $H_j(u) = \int_u^{u+L} f_j(z)\,dz$ *and* $k_j(q_u) = \{H_j(u) - \min_I H_j\}/B$. *For any feasible* u *and any verified* $\kappa_b \geq k_b(q_u)$,

$$k_a(q_u) \leq \left\{ \sqrt{\kappa_b} + E\sqrt{\frac{\alpha}{2m_z t}} \right\}^2.$$

The statement includes an endpoint optimizer or a one-sided tail. For the original investment pair, $m_z = 181/64$ *and* $E = E_{ab}$ *are valid. At* $\alpha = 0$ *the only relevant density is the full original law and both deficits are zero.*

PROOF: The original tail representation, or its direct proof in Theorem 7, gives the stated expression for the risk deficit. Strong convexity implies that H_a and H_b are λ-strongly convex on I, where $\lambda = m_z L$. Indeed $H_j'(u) = f_j(u + L) - f_j(u)$, and its increment between two cuts is at least $m_z L$ times their distance. Let u_b minimize H_b on I. The constrained first-order inequality, followed by strong convexity, gives

$$H_b(v) \geq H_b(u_b) + \tfrac{1}{2}\lambda|v - u_b|^2 \quad (v \in I).$$

There is no assumption that $H_b'(u_b) = 0$ at a boundary. Write $G = H_b - H_a$. Since $G'(v) = (f_b - f_a)(v + L) - (f_b - f_a)(v)$, G has Lipschitz constant EL.

Put $e = H_b(u) - H_b(u_b) = Bk_b(q_u) \le B\kappa_b$. The preceding inequality gives $|u - u_b| \le \sqrt{2e/\lambda}$. For any $v \in I$, the triangle inequality now yields

$$\begin{aligned} H_a(u) - H_a(v) &\le e - \tfrac{1}{2}\lambda|v - u_b|^2 + EL|v - u| \\ &\le e + EL\sqrt{2e/\lambda} - \tfrac{1}{2}\lambda|v - u_b|^2 + EL|v - u_b| \\ &\le e + EL\sqrt{2e/\lambda} + \frac{E^2L^2}{2\lambda}. \end{aligned}$$

The last step maximizes a scalar quadratic over a larger nonnegative-distance domain, so it is valid at either boundary. Take the supremum in v, divide by B, and use $L/B = \alpha/t$. The result is $k_a(q_u) \le \{\sqrt{k_b(q_u)} + E\sqrt{\alpha/(2m_z t)}\}^2$. Monotonicity in k_b proves the asserted bound. *Q.E.D.*

Outward square roots in the transferred bound. The implementation intersects the directly acquired incumbent deficit with an outward version of the transfer bound. For an exact nonnegative rational $z = p/q$ with $q > 0$, let m be the integer square root of $\lfloor 2^{80}p/q \rfloor$, and increase m by one if $m^2 q < 2^{80}p$. The resulting $m/2^{40}$ is at least $\sqrt{z}$: before a possible increment, m is the floor of $2^{40}\sqrt{z}$, and the exact square comparison detects whether that floor was exact. A binary64 conversion is moved upwards when its exact rational value is smaller than the dyadic bound, and its squared exact rational value is checked against z. Applying this procedure to upper endpoints, followed by outward additions, products, and squaring, gives a valid transferred deficit. Taking the smaller of two valid upper bounds remains valid. Both direct endpoint payloads and all integer operations are still paid. Because this is an incumbent-deficit bound, its tightening directly affects the lower contrast and the certified-refusal predicate; it does not itself tighten the candidate-deficit term in the upper acceptance endpoint.

20.5. *Uniform safeguards, finite refinement, and an honest stopping floor*

PROOF OF PROPOSITION 11: At any accepted step the previous theorem gives $J_x(b) - J_x(a) \le \overline{D} < 0$, and hence the original cost decreases by at least $-\overline{D} > 0$. The replacement is explicitly feasible and qualified. A refused trial changes no action or future policy. Induct over the finite sequence, and telescope the original cost

differences. This gives the lower gain account in (48). The initial completed policy satisfies $J_x(a_{\text{initial}}) - V_0(x) \le \tau$, while every final feasible policy has cost at least $V_0(x)$. Subtraction gives the upper bound τ.

This proof is statewise and representation independent. Finite Borel proposal, arithmetic, ordering with fixed tie rules, and acceptance maps compose into a Borel returned-action rule, provided the completed incumbent service is defined statewise. The original finite-horizon policy theorem then defines its policy value. Exhaustion of a new comparison returns the already completed incumbent. A failed scientific execution does not license omitting that requested state from the sample or presenting an incomplete service as a completed policy. *Q.E.D.*

PROPOSITION S.10—Finite precision above the full comparison margin: *Fix two distinct feasible actions and one acquired feasible cut. Let $g_Q = Q_x(a) - Q_x(b)$ and suppose κ_b is a valid, fixed upper bound on its candidate deficit. Put*

$$\gamma = E_{ab}\sqrt{\frac{\alpha}{2m_z t}}, \qquad F_q = \kappa_b + (\sqrt{\kappa_b} + \gamma)^2.$$

Assume refinement at positive integer resolutions n produces fixed-density contrast intervals of width

$$w_n \le \eta_{\text{num}} + \min\{A_2/n^2, A_1/n\}, \quad A_2 = \min\{2KRt, 2M_\alpha R^2 t^2\}, \quad A_1 = E_{ab}Rt,$$

where $\eta_{\text{num}} \ge 0$ includes the stated terminal-node and accumulation errors. Suppose

$$\epsilon = (g_Q - \delta)/\beta - F_q - \eta_{\text{num}} > 0.$$

Then the upper contrast gate is strictly negative whenever

$$n > \min\{\sqrt{A_2/\epsilon},\, A_1/\epsilon\}. \tag{22}$$

Acceptance also requires the separate original candidate qualification; if it is established by a resolution n_{qual} and remains valid at the considered round, the two conditions together suffice.

If the actual work of each doubling round is at most $cn + c_0$, with $c, c_0 \ge 0$ and a positive integer starting resolution n_0, the total through a final round n_J is at most $c(2n_J - n_0) +$

$c_0(J+1) \le 2cn_J + c_0(J+1)$, apart from separately paid setup. A first doubling round meeting an integer target $n_* \ge n_0$ satisfies $n_J < 2n_*$ and $J \le \lceil \log_2(n_*/n_0) \rceil$. This cost statement is conditional on the declared work model. Actual terminal solves, proposals, refused rounds, recording, and qualification costs are not replaced by a point-query count.

PROOF: Let $d = \mathbb{E}_P q_u(f_b - f_a)$ and denote the true deficits by k_a, k_b. If d_+ is the reported upper integral bound, $0 \le d_+ - d \le w_n$. The complete-policy upper contrast used by the gate is $U = h(b) - h(a) + \beta(d_+ + \kappa_b) + \delta$. The exact common-risk identity gives

$$\begin{aligned} U &= -g_Q + \delta + \beta\{(d_+ - d) + (\kappa_b - k_b) + k_a\} \\ &\le -g_Q + \delta + \beta\{w_n + F_q\}, \end{aligned}$$

where Theorem S.9 bounds k_a and $k_b \ge 0$ was retained in the first line and dropped only for the upper estimate. Under the stated positive margin, (22) makes the right side strictly negative. No inference about candidate qualification follows from that sign alone; its separate verified predicates are part of the assertion.

For $n_j = 2^j n_0$, $\sum_{j=0}^{J} n_j = 2n_J - n_0 < 2n_J$. Sum the per-round work bound. Minimality of the first doubling round meeting n_* gives the two displayed bounds on n_J and J. For the finite executed ladder $2, 4, 8, 16, 32, 64$, this guarantee applies only if a qualifying round also lies within that declared cap. *Q.E.D.*

The fixed-cut theorem does not assume that increasing the cell count also improves its tail localization. If the cut and its deficit remain fixed, their allowances remain in F_q. The actual implementation retains both direct original deficits and intersects the incumbent bound with the transferred one. A failed sufficient margin can therefore be conservative; it is not proof that a particular candidate is economically inferior.

PROPOSITION S.11—Sharp residual uncertainty of the unchanged-future interface: *Suppose the only information about the actual-policy correction at two distinct actions is $0 \le \eta_x \le \delta$ in (39), and their two Q_x values are known exactly. The interval $Q_x(b) - Q_x(a) + [-\delta, \delta]$ is then sharp over the compatible corrections. Thus $g_Q > \delta$ is*

necessary and sufficient for this information alone to certify strict improvement uniformly. Further refinement of the V_1 integration alone cannot remove that floor.

PROOF: The difference $\eta_x(b) - \eta_x(a)$ is in $[-\delta, \delta]$. The upper endpoint is attained in the interface by setting $\eta_x(a) = 0$, $\eta_x(b) = \delta$; reversing those values attains the lower endpoint. If continuity were additionally required without any quantitative modulus, the two prescribed values could still be joined by a continuous bounded interpolation between these distinct actions. Consequently a strict negative upper contrast is equivalent to $g_Q > \delta$. The claim is about compatibility with the declared information: it does not assert that both extremal corrections are generated by the actual fixed terminal policy. That policy's further verified structure would be additional information.

For exactly the same action, both values of η_x are the same and cancellation is exact; the two-endpoint construction does not apply. If only the magnitude of the actual gain g_J is specified, then $g_Q \geq g_J - \delta$. The corresponding sufficient worst-case condition for the exact-Q gate is $g_J > 2\delta$, rather than $g_J > \delta$. This distinction prevents a known-continuation gain and an actual-policy gain from being assigned the same resolution claim. *Q.E.D.*

21. EXACT PRIMITIVE VALIDATION OF ORIGINAL-OPTIMUM REGRET

The R119 independent catalogue enclosure bounded $Q_x(a)$ and $J_x(a)$ at fixed actions. The following additional construction also provides an independent lower bound on $\inf_a Q_x(a)$. It uses the retained original curvature proposition and explicit primitive gradients; it is a validation strengthening, not a new claim that a scalar fixed-action lower value already bounded the all-action optimum.

PROPOSITION S.12—An independent rational all-action witness: *Fix the original $d = 2$ model and a rational original state and root action. At a terminal anchor y, choose any feasible rational v and any feasible original tail density q. Let W be the exact terminal objective under this fixed density, and let (p_y, p_v) be its state and action gradients. For*

arbitrary rational $\lambda, \nu \geq 0$, define

$$e = p_v + \lambda - \nu, \qquad c = W + \lambda\{v - A(y)\} - \nu v - e^2/4,$$
$$r = p_y - \frac{\lambda}{8d}\mathbf{1}. \tag{23}$$

Then $V_1(y') \geq c + r^\top (y' - y)$ for every original state y'. At a boundary one may choose $\lambda = \mathbf{1}_{\{v=A(y)\}}(-p_v)_+$ and $\nu = \mathbf{1}_{\{v=0\}}(p_v)_+$; no exact stationarity is required.

Partition the root innovation interval into n equal cells, with midpoints z_i and successors $y_i = F(x, a, z_i)$. Acquire such supports (c_i, r_i), and choose rational masses $q_i \in [0, 1/(nt)]$ summing to one. Put

$$W_0 = S(x) + h(a) + \beta \sum_i q_i c_i,$$
$$p_0 = h'(a) + \beta \sum_i q_i r_i^\top \boldsymbol{b}, \qquad a_* = \mathrm{proj}_{[0,A(x)]}(a - p_0/2),$$
$$B_0 = W_0 + p_0(a_* - a) + (a_* - a)^2. \tag{24}$$

Then $B_0 \leq V_0(x)$. Together with any separately valid original complete-policy upper bound $U_J(a)$, this proves

$$0 \leq J_x(a) - V_0(x) \leq U_J(a) - B_0, \qquad |a - a_0^*(x)|^2 \leq U_J(a) - B_0.$$

The maximum of finitely many such lower bounds acquired at the same original state is also valid. No support or optimum bound is transferred across distinct states without an additional justified state comparison.

PROOF: For a fixed feasible original density, the joint terminal state-action objective W_q has the lower curvature $m_x \|dy\|_2^2 + 2\,dv^2$, where $m_x = (4 - 5\beta/4)/d > 0$. This follows directly from $\nabla^2 S \succeq (4/d) I$, $h'' \geq 2$, $\nabla^2 g \succeq 0$, and $|\partial_j g| \leq 10/d$: the only potentially negative transition-chain term is $-\sum_j \partial_j g(F)\, dy_j dy_{j+1}/8$, bounded below by $-5\|dy\|_2^2/(4d)$. The other chain Hessian is positive semidefinite even in a joint state-action direction. Integration under the fixed nonnegative density preserves this bound. Matching first derivatives extend it across squared-hinge

junctions. Dropping the positive state-curvature term gives

$$Q_1(y',v') \geq W_q(y',v') \geq W + p_y^\top (y'-y) + p_v(v'-v) + (v'-v)^2.$$

Replace p_v by $e - \lambda + \nu$. For a feasible $v' \in [0, A(y')]$, $-\lambda v' + \nu v' \geq -\lambda A(y')$. The affine capacity and the inequality $e(v'-v) + (v'-v)^2 \geq -e^2/4$ therefore give $Q_1(y',v') \geq c + r^\top (y'-y)$. Taking the infimum over the original feasible v' proves (23). In particular the intercept must be this valid global-support intercept; a larger scalar lower bound obtained by clipping a quadratic at only the anchor state cannot be attached to r without a further proof.

Here are explicit gradients suitable for an independent rational implementation. Put $m = F(y,v,0)$. The original terminal polynomial in the innovation is $g(m) + kz + 5z^2$, whose coefficients can be reconstructed by evaluating the primitives at $-R, 0, R$. For its exact clipped quadratic tail with left cut u, let

$$L = 2R\alpha, \quad B = 2Rt, \quad \mu_1 = -L(2u+L)/(2B), \qquad G_j = \partial_j g(m) + (10/d)s_j\mu_1.$$

The squared hinge is constant along the zero-sum shock, and direct differentiation of the remaining quadratics gives $\partial_j g(m + \boldsymbol{s}z) = \partial_j g(m) + (10/d)s_j z$. Thus G_j is the exact fixed-density mean of that derivative. With cyclic indices,

$$\begin{aligned} p_v &= 2v + 16v^3 + \beta \sum_j b_j G_j, \\ p_{y,j} &= \partial_j S(y) + \beta \left[G_j \left\{ \frac{1}{2} + \frac{1 - y_{j+1}}{16} \right\} + G_{j-1} \left\{ \frac{1}{8} - \frac{y_{j-1}}{16} \right\} \right]. \end{aligned} \tag{25}$$

These formulas differentiate the fixed density, not its optimizing cut. The density supplied by the independent quadratic threshold construction is exactly feasible and maximizing at the anchor, but feasibility alone suffices for the global support proof. All quantities are rational at the stated inputs.

For the root, the masses q_i define a piecewise constant feasible AVaR density with value nq_i on cell i. For another feasible root action w and any z in cell i,

$$F(x,w,z) - y_i = \boldsymbol{b}(w-a) + \boldsymbol{s}(z - z_i).$$

Integrating its affine support over the symmetric cell removes the last term exactly. The same fixed feasible density consequently yields

$$Q_x(w) \geq S(x) + h(w) + \beta \sum_i q_i \{c_i + r_i^\top \boldsymbol{b}(w-a)\} \geq W_0 + p_0(w-a) + (w-a)^2.$$

Its quadratic minimum over the full original capacity interval is (24). Taking the infimum of Q_x proves the optimum lower bound. For equal cells with integral nt, selecting the largest nt values of c_i with fixed ties and assigning mass $1/(nt)$ to them is feasible and maximizes this finite linear witness. This sorting changes the chosen original density, not the original continuous innovation law.

Every implemented feasible policy costs at least V_0, so $J_x(a) \leq U_J(a)$ gives the regret inequality. The original Q_x is strongly convex in its action with modulus two. Its constrained minimizer obeys the boundary variational inequality. Hence $Q_x(a) - V_0(x) \geq |a - a_0^*(x)|^2$. Use $Q_x(a) \leq J_x(a) \leq U_J(a)$ and $B_0 \leq V_0(x)$ to prove the action-distance bound. A maximum of valid lower bounds on the same scalar optimum remains a valid lower bound. This completes the proof. *Q.E.D.*

The independent fixed-action upper value is still constructed by the original midpoint/duplicated-endpoint convex-order argument in Proposition S.7, with $\beta\varepsilon_1$ added once for the actual future policy. The new all-action witness does not assert containment of a fine production interval inside an arbitrarily selected coarse one, nor equality of any fine endpoint to the truth. It supplies its own exact, reproducible bounds on the original optimum, regret, and action error.

22. POPULATION GAIN WITH FROZEN DECISIONS AND FINITE NUMERICAL INTERVALS

PROOF OF PROPOSITION 12: Let $\mathcal{F}$ contain the frozen actors, fitting data, proposal and refinement protocol, representation choices, initial-state sampling law, and statewise execution rule. Condition on $\mathcal{F}$. For each of the 48 fixed groups, require that its completed incumbent and final action be measurable functions of its own initial state, with the uniformly safe guarantee in Proposition 11. In particular they are not fitted, selected, or changed after observing the validation states. Apply each complete service to a single original state; batching innovations within that state does not violate this condition. Batch logic that lets another sampled state's

failed certificate alter the current state's action would require a different probability argument.

For the declared finite dyadic law one may take

$$\mathcal{X}_{40} = \{0, 1, \ldots, 2^{40} - 1\}^2 / 2^{40}.$$

Independent uniform coordinate draws from this set make $X_1, \ldots, X_{64}$ iid under the finite uniform population. A fixed sequence of 640 independently uniform bytes can encode these 128 coordinates in blocks of five bytes. Their later retention is reproducibility evidence. Independence and uniformity of the generating entropy, and its independence of $\mathcal{F}$, remain the statistical assumption. The conclusion is for this declared law, not automatically for a continuous uniform population or an empirical economic state distribution.

Write $G_j(x)$ for the actual complete-policy gain over the corresponding completed incumbent. The uniform safeguard gives $0 \le G_j \le \tau$ before the sample is drawn. Let $\ell_{ji} \le G_j(X_i) \le u_{ji}$ be the saved valid gain bounds, intersected with $[0, \tau]$. For example, accepted-step lower gains can be summed outwardly, and the two separately valid complete-policy value intervals can provide another gain interval to intersect. Numerical interval widths are retained, not replaced by midpoints. Exact equal-policy identities give $G_j = 0$ without independent interval subtraction. A missing independent numerical validation can conservatively use $[0, \tau]$ and must not be dropped. The numerical enclosure procedure may depend on already observed validation information, provided its intervals remain valid and it does not change the frozen returned policy rule.

For a fixed group and threshold define the latent events $B_i = \mathbf{1}_{\{G_j(X_i) \ge \theta_k\}}$ and their probability p_{jk}. These are iid Bernoulli variables conditional on $\mathcal{F}$. The observable certified count satisfies

$$K_{\text{cert}} = \sum_i \mathbf{1}_{\{\ell_{ji} \ge \theta_k\}} \le \sum_i B_i =: K_{\text{true}}$$

pointwise. The certified count itself need not be binomial if the numerical validation is incomplete or adaptive. Its domination by the latent binomial count is sufficient.

The parameters may be dependent across groups; no cross-group independence is required.

Here is the exact inversion used in the statement. For an observed count $k \in \{1, \dots, n\}$, $n = 64$, define

$$T_k(p) = \sum_{r=k}^{n} \binom{n}{r} p^r (1-p)^{n-r}.$$

It is continuous and strictly increasing on $(0,1)$, with endpoint values zero and one. Differentiating the finite sum and telescoping proves $T_k'(p) = n\binom{n-1}{k-1} p^{k-1}(1-p)^{n-k} > 0$. There is therefore a unique solution of $T_k(p) = 1/7680$ in $(0,1)$; call it $p_*(k)$. For $k = 0$ set $p_*(0) = 0$. The map $k \mapsto p_*(k)$ is nondecreasing because $T_{k+1}(p) \le T_k(p)$. For the true parameter p, the event $p_*(K) > p$ is either empty or of the form $K \ge k_0$, where $T_{k_0}(p) < 1/7680$. Its probability is therefore at most $1/7680$.

An exactly rational implementation need not evaluate a beta quantile. Start with the bracket $[0,1]$ and bisect it 48 times, evaluating the finite binomial polynomial as an exact rational. Move its lower endpoint to the midpoint only when T_k there is at most $1/7680$. The returned lower endpoint p^- satisfies $p^- \le p_*(k)$ and $p_*(k) - p^- \le 2^{-48}$. Thus it retains the same conservative coverage when applied to K_{true}. Monotonicity of $p_*(k)$ and $K_{\text{cert}} \le K_{\text{true}}$ imply $p^-(K_{\text{cert}}) \le p_*(K_{\text{true}})$. Consequently the same coverage holds for the actually reported certified count, including any conservative missing-validation intervals. There are $48 \cdot 4 = 192$ prespecified event bounds; their union failure probability is at most $192/7680 = 1/40$.

On the simultaneous coverage event, $p_{jk}^- \le p_{jk}$. Because the thresholds are increasing and $\theta_0 = 0$, the following pointwise inequality holds:

$$\sum_{k=1}^{4} (\theta_k - \theta_{k-1}) \mathbf{1}_{\{G_j(x) \ge \theta_k\}} \le G_j(x).$$

If the largest passed threshold is θ_s, the left side is exactly θ_s; otherwise it is zero. Taking expectations proves (50). Each individual lower bound $\theta_k p_{jk}^-$ is also valid. The staircase statement uses all fixed thresholds under the same simultaneous coverage event.

For the mean interval, a short proof of the needed bounded-variable inequality is useful. If $W \in [0,\tau]$, the logarithmic moment generating function $\psi(\lambda) = \log \mathbb{E}e^{\lambda W}$ has $\psi''(\lambda) = \mathrm{Var}_\lambda(W) \le \tau^2/4$ under its exponentially tilted law. This variance bound follows from $\mathbb{E}_\lambda(W - \tau/2)^2 \le \tau^2/4$ and the minimizing property of the mean. Integrating twice from zero gives

$$\mathbb{E}e^{\lambda(W-\mathbb{E}W)} \le e^{\lambda^2\tau^2/8}.$$

Independence, exponential Markov inequality, and minimization over λ yield for either tail

$$P\left(\left|\overline{W} - \mathbb{E}W\right| \text{ exceeds } r \text{ in that tail}\right) \le e^{-2nr^2/\tau^2}.$$

Since $\overline{\ell}_j \le \overline{G}_j \le \overline{u}_j$, the lower failure $\mathbb{E}G_j < \overline{\ell}_j - r$ and upper failure $\mathbb{E}G_j > \overline{u}_j + r$ are contained in the corresponding two deviation events for the unobserved true gain average. Thus no numerical value error has been ignored.

For the prespecified rational radius $r = 33\tau/128$,

$$2nr^2/\tau^2 = 1089/128 =: x.$$

The strict inequality $e^x > 3840$ can itself be checked without a floating logarithm or exponential. Write $S_m(z) = \sum_{k=0}^{m} z^k/k!$. Since $x > 17/2$, positivity of coefficients gives

$$S_{20}(x) \ge S_{12}(17/2) \ge S_{11}(8)(1 + 1/2) > 2644(3/2) = 3966 > 3840.$$

The middle inequality follows by expanding $(8 + 1/2)^k$: every term of $S_{11}(8)S_1(1/2)$ occurs with nonnegative coefficient in $S_{12}(8 + 1/2)$. For the last lower bound, integer lower bounds for the successive terms of $S_{11}(8)$ are $1, 8, 32, 85, 170, 273, 364, 416, 416, 369, 295, 215$; their sum is 2644, and several inequalities are strict. Since $e^x \ge S_{20}(x)$, each one-sided mean failure is strictly less than $1/3840$. The 48 two-sided mean intervals therefore fail with total probability less than $96/3840 = 1/40$.

A final union bound over the event and mean families gives failure probability at most $1/20$. Intersecting valid bounds with the known population range $[0,\tau]$

proves (51) and the simultaneous 0.95 assertion. No independence between the two inference families is used. These are ordinary probability bounds (Clopper and Pearson, 1934, Hoeffding, 1963), applied after the original-objective safeguard makes their gain range and events mathematically valid. *Q.E.D.*

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